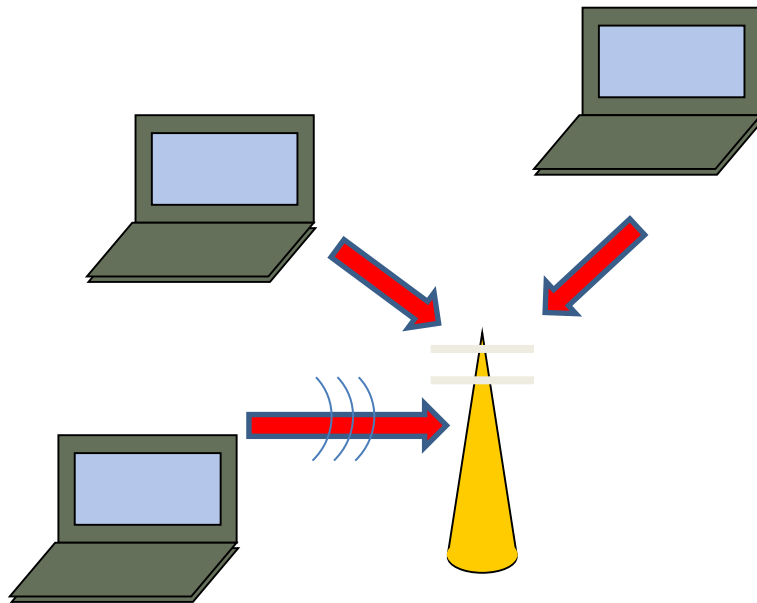


Mobile Internet

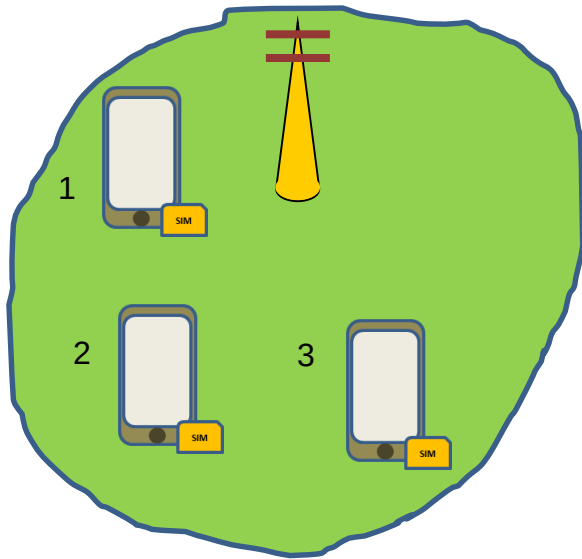
4.- Cellular Networks:
2G (GSM & GPRS), 3G(UMTS &
HSPA), 4G (LTE), 5G

Multiple access protocols

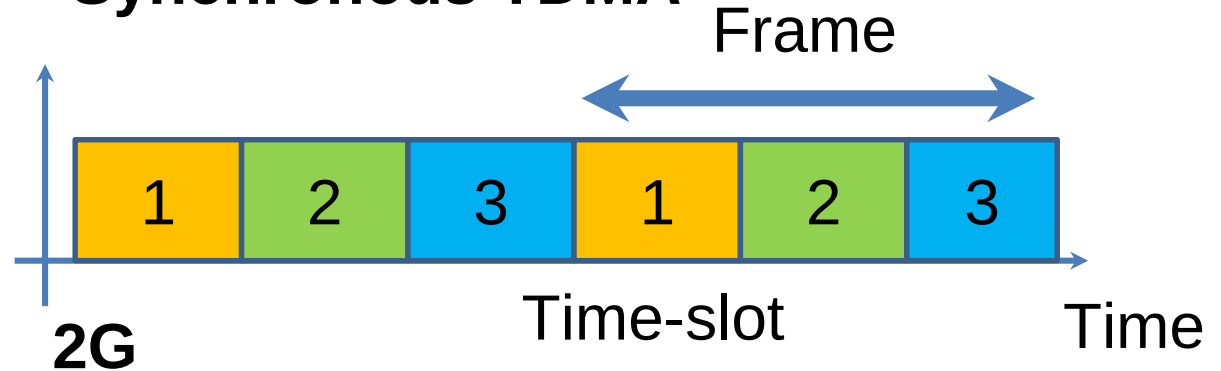


Simultaneous transmission at the same time, frequency and code, can lead to a collision

Time Division Multiple Access (TDMA)

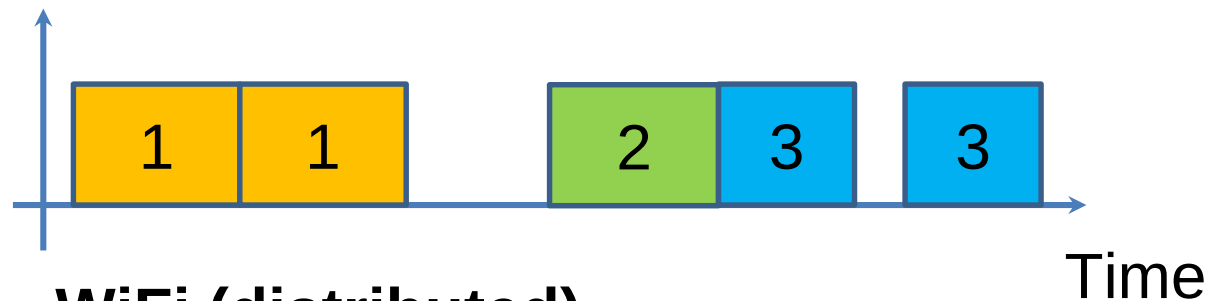


Synchronous TDMA



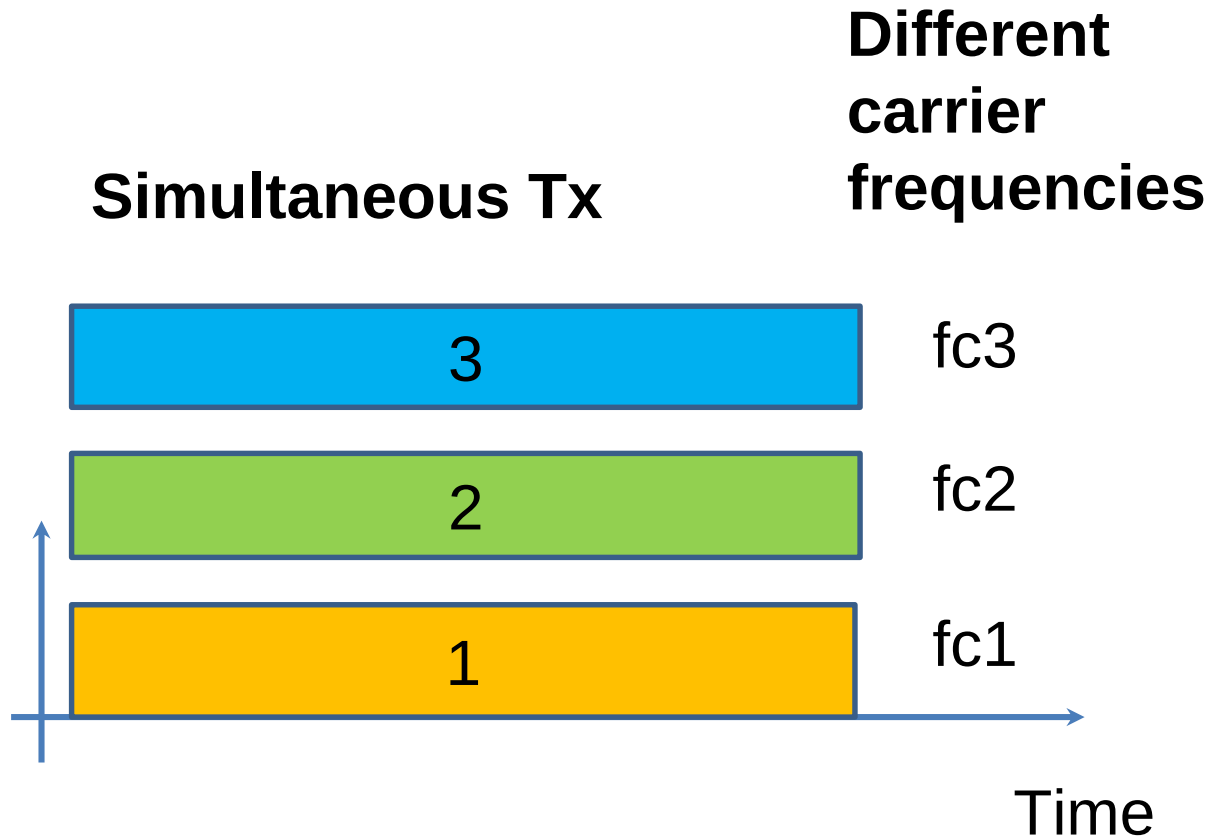
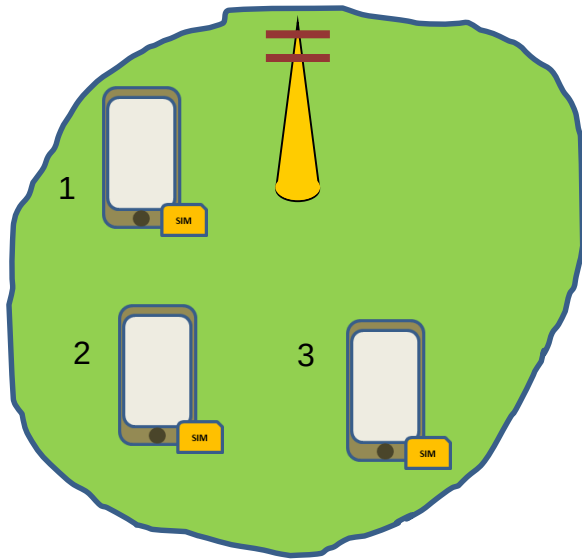
Statistical TDMA

Same carrier frequency



**WiFi (distributed),
2.5G (centralized)**

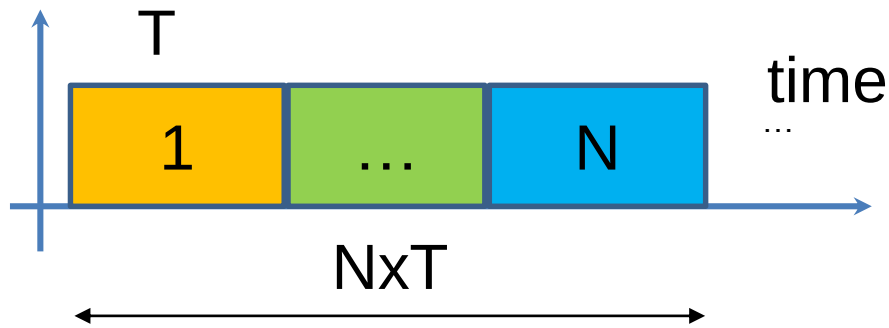
Frequency Division Multiple Access (FDMA)



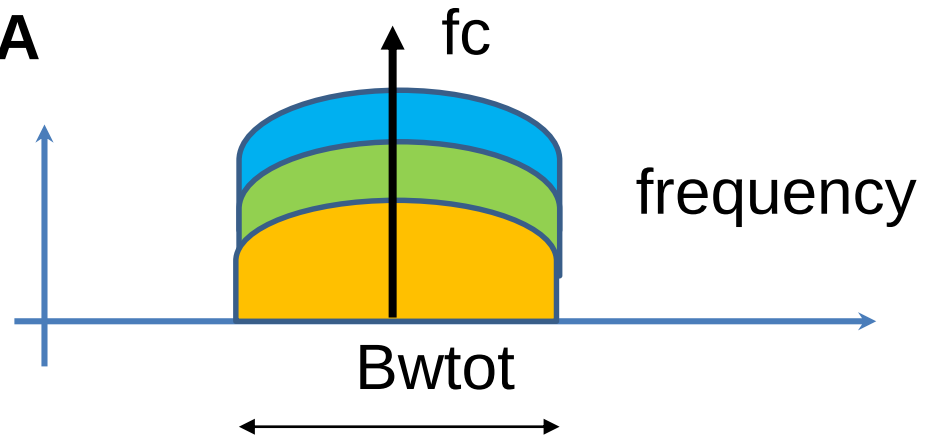
2G, 2.5G: FDMA+TDMA

4G: OFDMA, SC-FDMA

TDMA

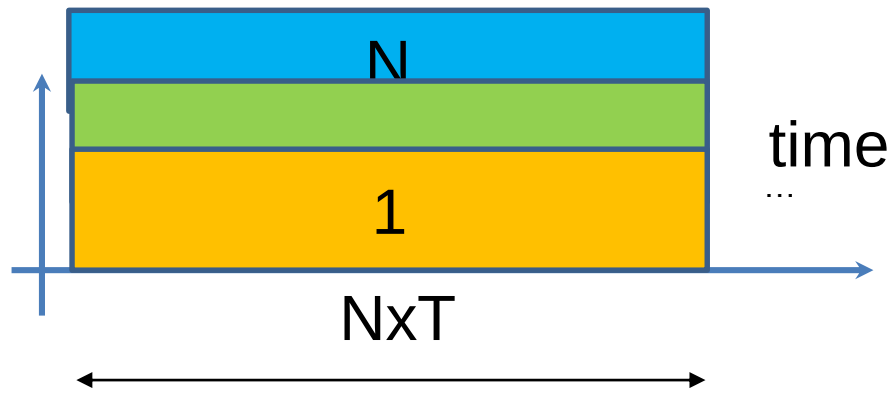


$$R_{\text{tot}} = N \times R = N(1/N \times T) = 1/T$$

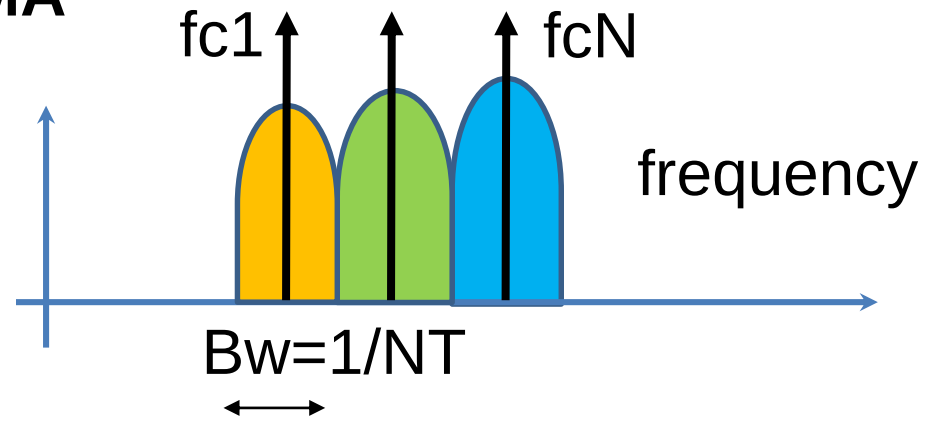


$$B_{\text{wtot}} = 1/T$$

FDMA

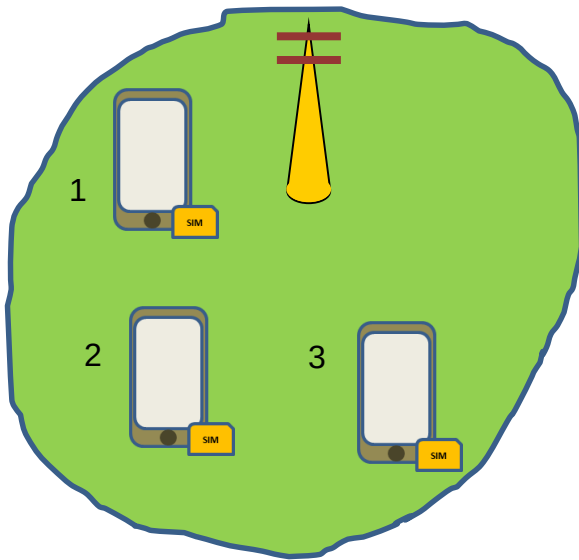


$$R_{\text{tot}} = N \times R = N(1/N \times T) = 1/T$$



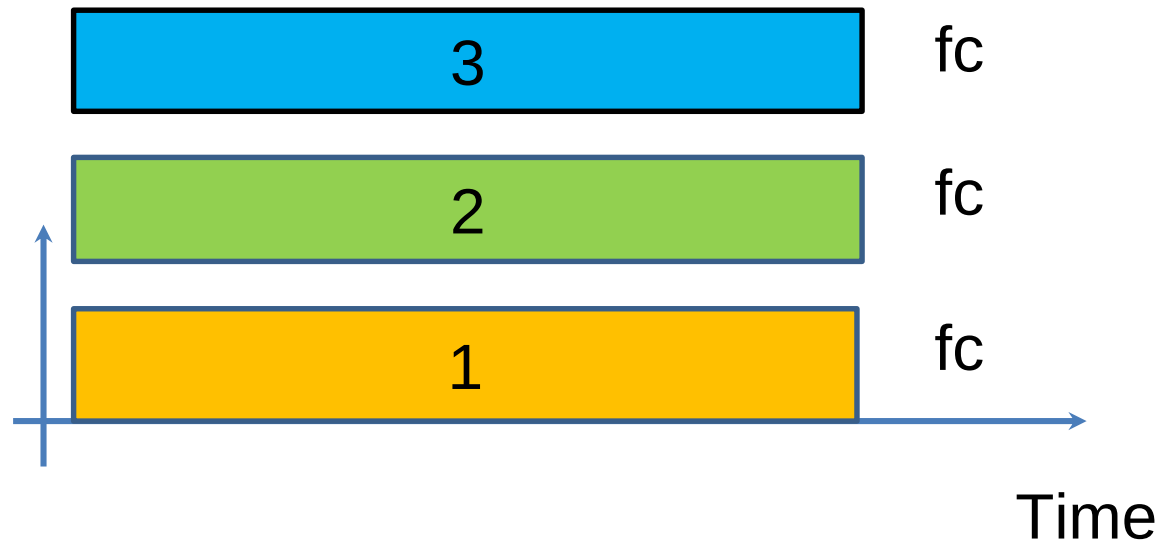
$$B_{\text{wtot}} = N \times (1/NT) = 1/T$$

Code Division Multiple Access (CDMA)



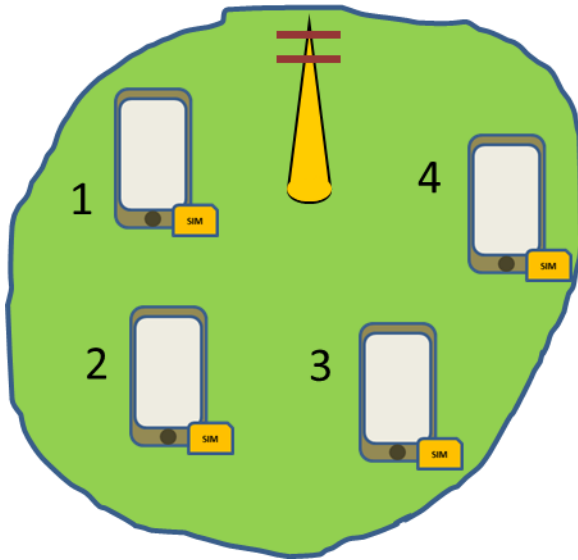
Orthogonal codes

Same
carrier
frequencies



3G:WCDMA

**CDMA+Aloha: Random access
channel 4G**



$C1 = (1, 1, 1, 1)$
 $C2 = (1, 1, -1, -1)$
 $C3 = (1, -1, 1, -1)$
 $C4 = (1, -1, -1, 1)$

$$R = (2, 2, 2, -2)$$

$$R = s_1 c_1 + s_2 c_2 + s_3 c_3 + s_4 c_4$$

$$\langle R, c_3 \rangle = s_1 \langle c_1, c_3 \rangle + s_2 \langle c_2, c_3 \rangle + s_3 \langle c_3, c_3 \rangle + s_4 \langle c_4, c_3 \rangle$$

$$\langle R, c_3 \rangle = 2 - 2 + 2 + 2 = 4$$

$$\langle c_1, c_3 \rangle = 0 \quad \langle c_2, c_3 \rangle = 0 \quad \langle c_3, c_3 \rangle = 4 \quad \langle c_4, c_3 \rangle = 0$$

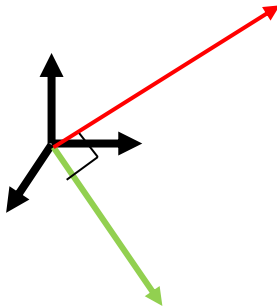
$$4 = 0 + 0 + s_3 4 + 0 \Rightarrow s_3 = 1$$

CDMA: scalar product

- Reminder

Assume $\mathbf{x}$ and $\mathbf{y}$ are two vectors of N components $\mathbf{x}=(x_1, x_2, \dots, x_N)$ and $\mathbf{y}=(y_1, y_2, \dots, y_N)$

Scalar product: $\langle \mathbf{x}, \mathbf{y} \rangle = x_1 y_1 + x_2 y_2 + \dots + x_N y_N$



Fundamental properties:

- $\langle \mathbf{x}, \mathbf{y} + \mathbf{z} \rangle = \langle \mathbf{x}, \mathbf{y} \rangle + \langle \mathbf{x}, \mathbf{z} \rangle$
- $\langle \mathbf{x}, a\mathbf{y} \rangle = \langle a\mathbf{x}, \mathbf{y} \rangle = a\langle \mathbf{x}, \mathbf{y} \rangle$
- If $\mathbf{x}$ and $\mathbf{y}$ are orthogonal, $\langle \mathbf{x}, \mathbf{y} \rangle = 0$

CDMA: scalar product

- E.g. Check that these 3 vectors are mutually orthogonal (we will call them “**codes**”)

c1=<1,1,1,1>, **c2**=<1,-1,1,-1>, **c3**=<1,1,-1,-1>

- Try to find **c4** which is orthogonal to **c1**, **c2**, and **c3**
(Note, as $N=4$, we cannot find more than N mutually orthogonal codes)

Scalar product, trick

- Check that $\mathbf{c}_4=(1,-1,-1,1)$ is a solution of the question in previous slide
- Think on 4 numbers, a,b,c,d (restrict to values $+1$ or -1 , although the trick will work in general)
- Find now $\mathbf{x}=\mathbf{a}\mathbf{c}_1+\mathbf{b}\mathbf{c}_2+\mathbf{c}\mathbf{c}_3+\mathbf{d}\mathbf{c}_4$
- For instance, if we choose $a=1, b=-1, c=1, d=-1$, we have:
 $\mathbf{x} = 1\mathbf{c}_1+(-1)\mathbf{c}_2+1\mathbf{c}_3+(-1)\mathbf{c}_4 = \mathbf{c}_1-\mathbf{c}_2+\mathbf{c}_3-\mathbf{c}_4 = (0,4,0,0)$
- Question: how to find a,b,c and d if you only know $\mathbf{x}$?
- E.g. find a,b,c , and d if $\mathbf{x}=(0,0,0,-4)$

Answer

If you can find $a, b, c,$ and d solving a 4×4 linear system of equation, but we can use a trick that makes things much easier:
find the scalar product between one of the codes and the vector x :

$$x = ac_1 + bc_2 + cc_3 + dc_4$$

$$\begin{aligned} \langle c_1, x \rangle &= a\langle c_1, c_1 \rangle + b\langle c_1, c_2 \rangle + c\langle c_1, c_3 \rangle + d\langle c_1, c_4 \rangle = \\ &a\langle c_1, c_1 \rangle + 0 + 0 + 0 = a\langle c_1, c_1 \rangle \end{aligned}$$

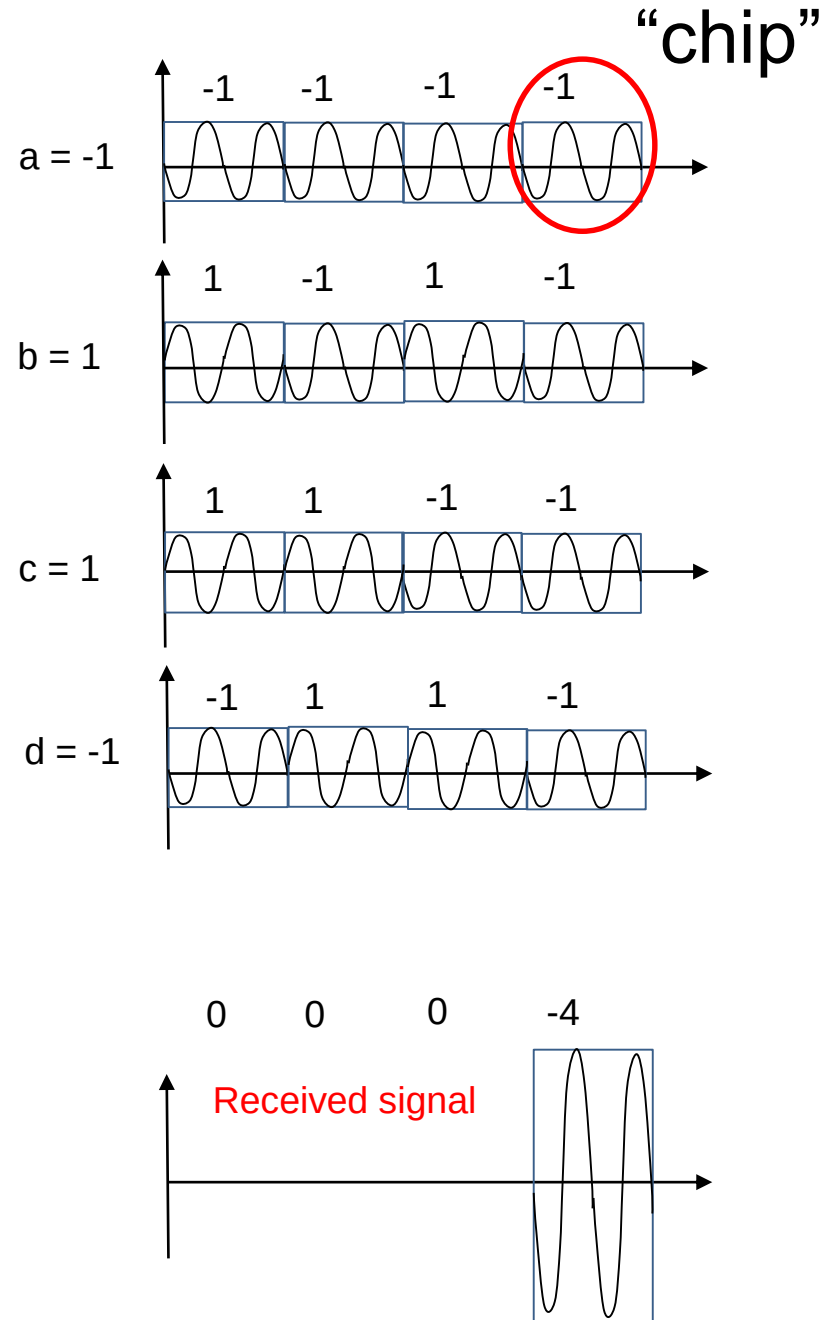
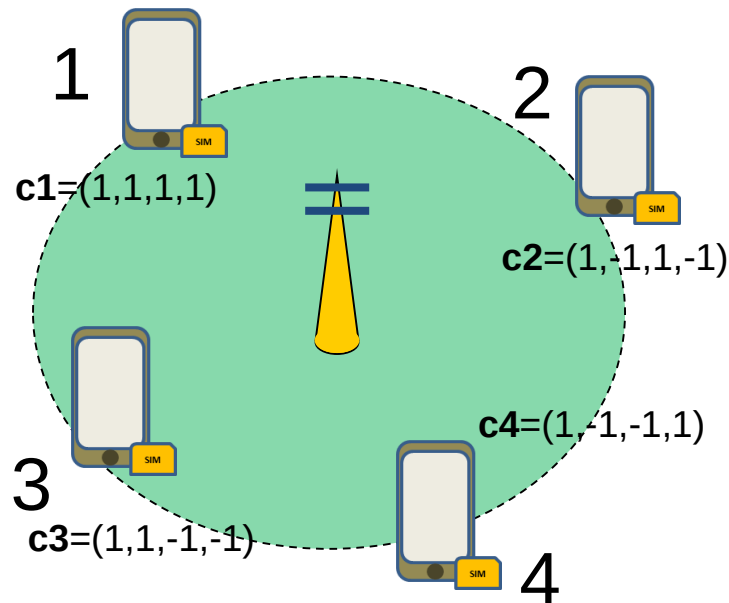
$$\text{e.g. } x = (0, 0, 0, -4) \Rightarrow \langle c_1, x \rangle = -4, \langle c_1, c_1 \rangle = 4 \Rightarrow a = -1$$

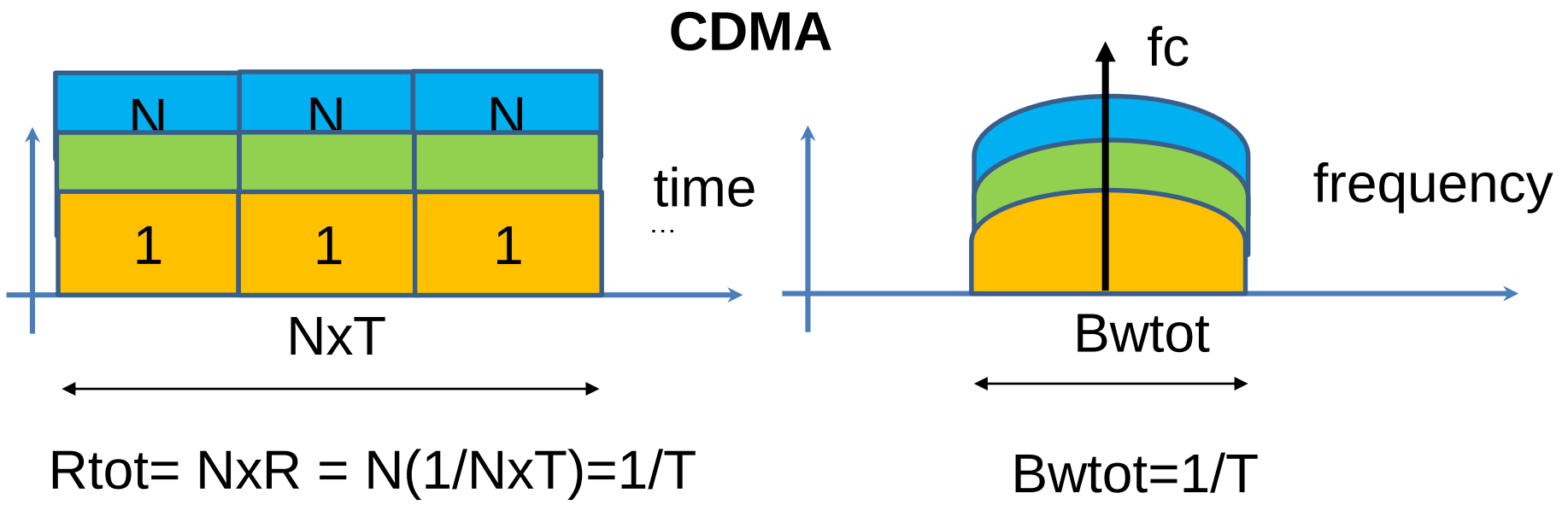
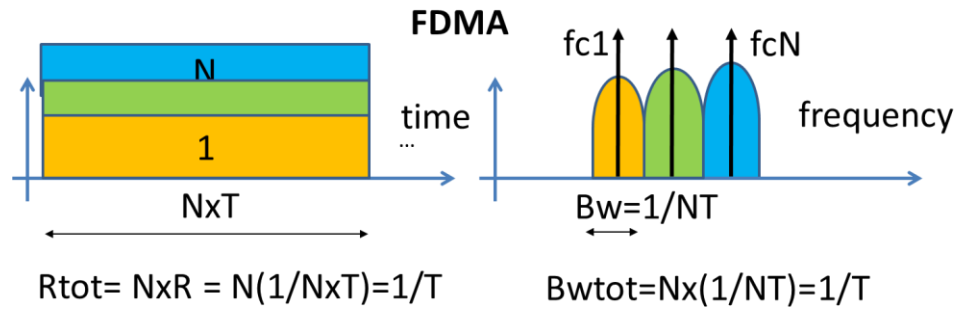
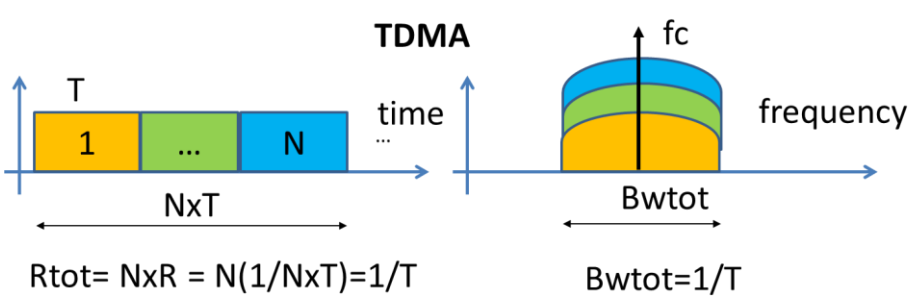
Correct! The same trick can be used to find b, c and d if we multiply x by c_2, c_3 and c_4 . Check this, you should obtain $b=1, c=1, d=-1$

$$R_c \text{ (Chip rate)} = \text{number of chips/second} = N \times R$$

CDMA

This trick can be used by a base station in a multiaccess channel, assuming it knows the codes used by each user:



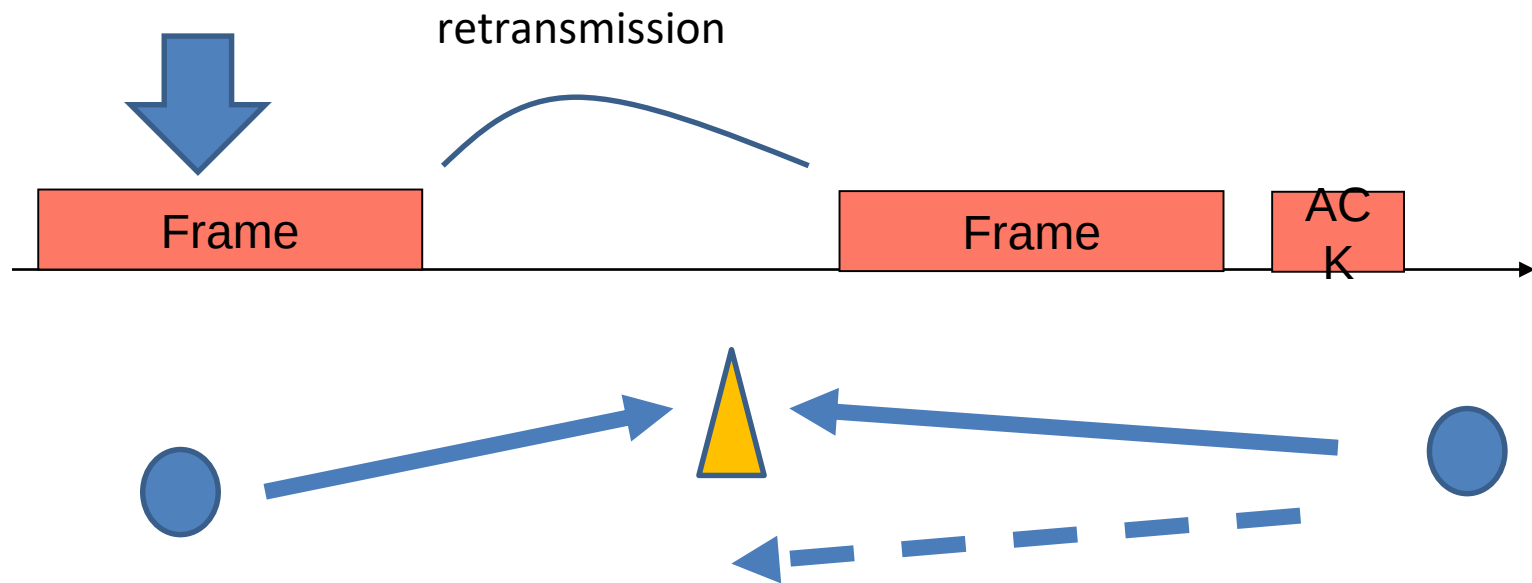


Access protocols Aloha

(e.g. LoraWAN class A device, RFID, RACH
4G RACH (Aloha+CDMA))

The node sends a packet and wait for an ACK. IN case no ACK is received, it waits a random time and retransmits the frame

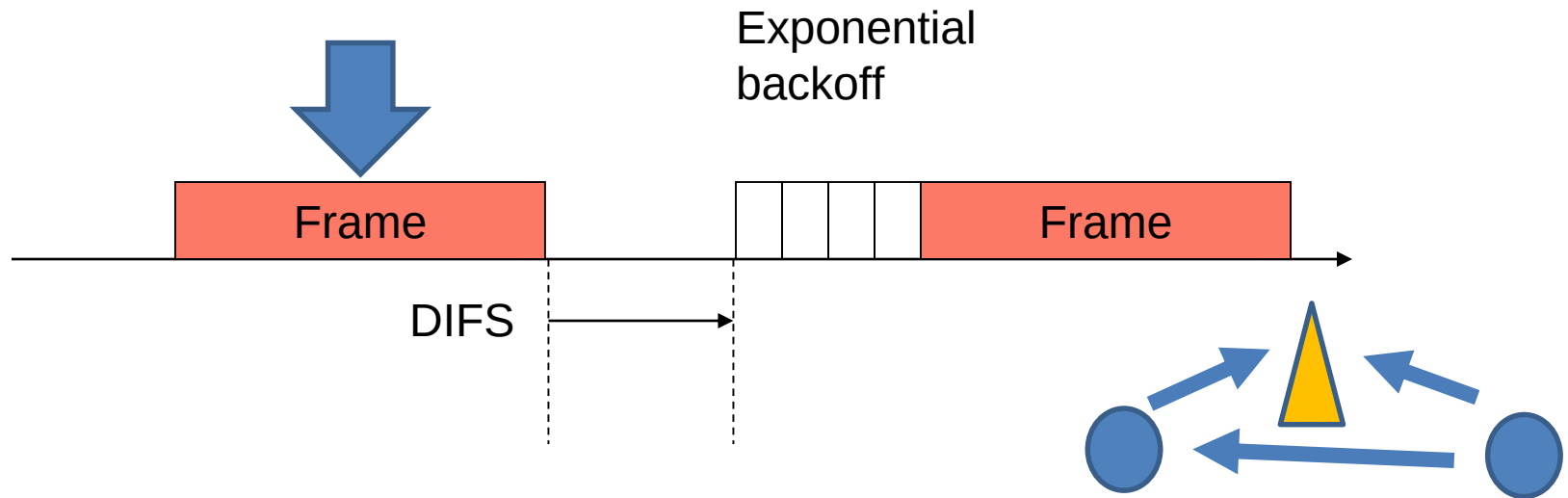
NO! Carrier sense (CSMA), simple!



Access protocols CSMA/CA (e.g. 802.11, 802.15.4)

If the medium is busy, the station must wait until it becomes idle for longer than a DIFS, and then it performs an exponential backoff...

Carrier sense (CSMA)

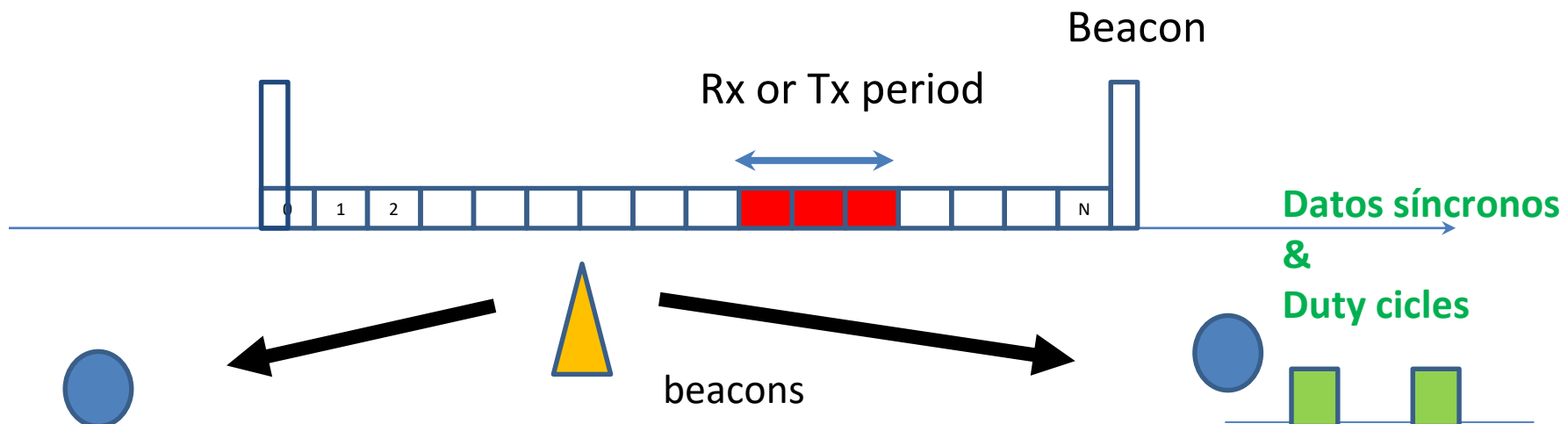


In the first attempt, the Contention Window (CW) measures CW_{min} (e.g. 7 slots). In case of collision, in attempt n -th, the CW is set to $CW_{min} \times 2^{n-1} - 1$, until reaching a maximum value of CW_{max} ...

Scheduled transmission (Beacons)

2G, LoraWan Class B, 802.15.4

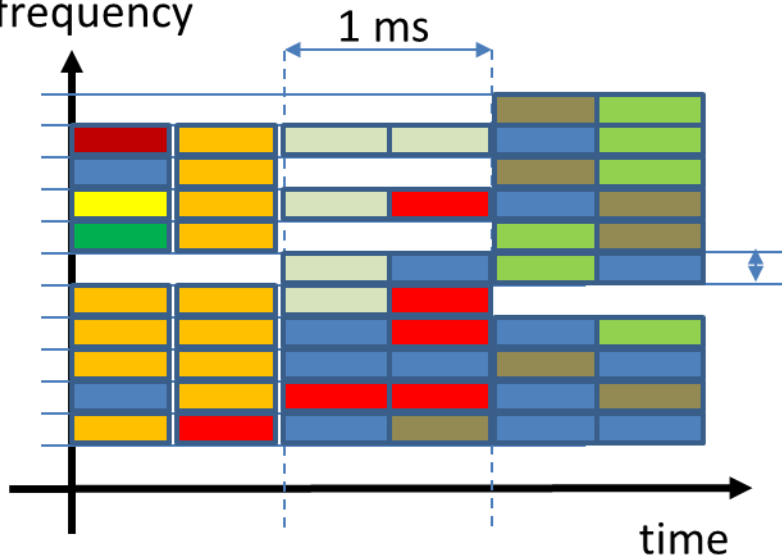
- A dedicated network coordinator transmits beacons in predetermined intervals
- All nodes switch awake/sleep periods (net nodes need to receive periodic beacons)



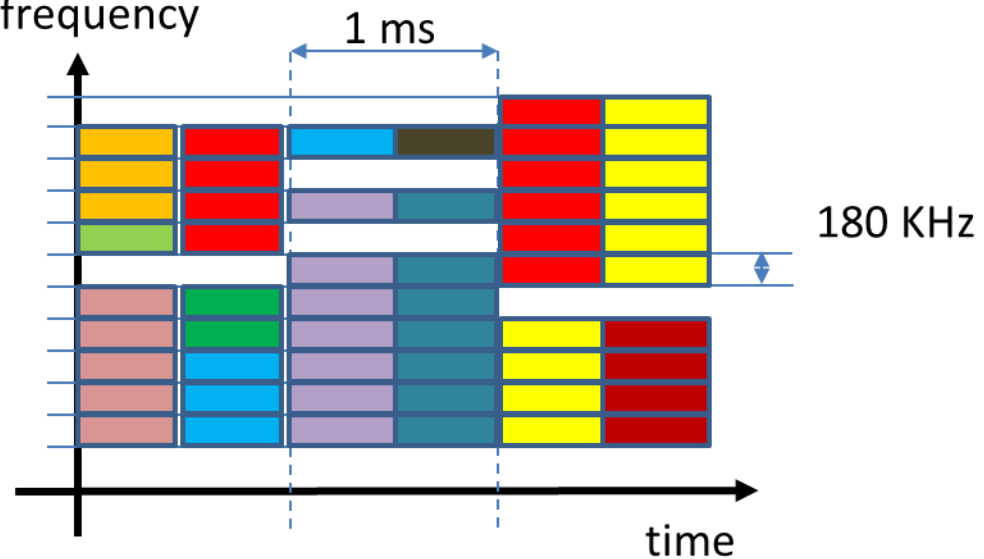
Frequency diversity frames (2G, 4G, 15.4 TSCH)

4G

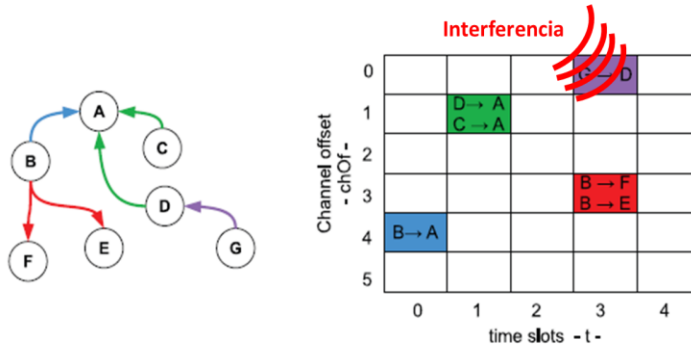
Downlink
frequency



Uplink
frequency

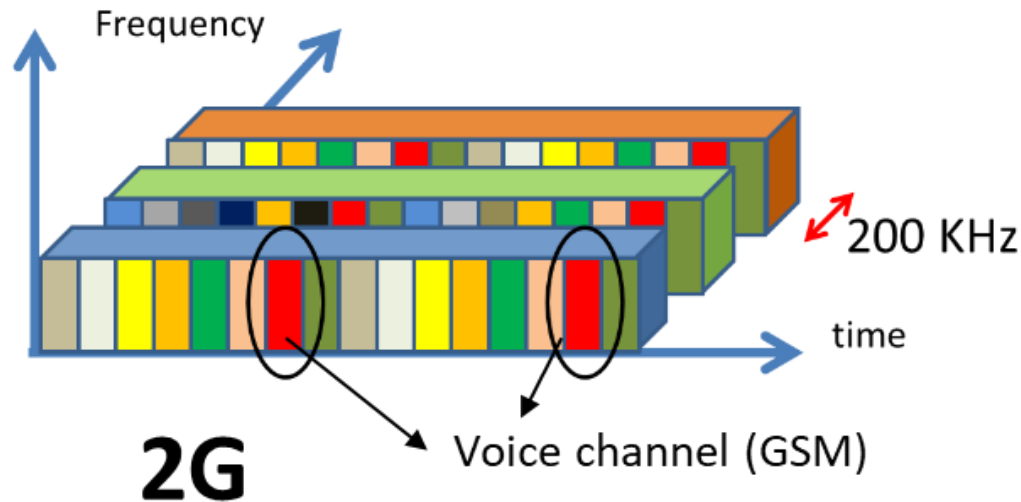
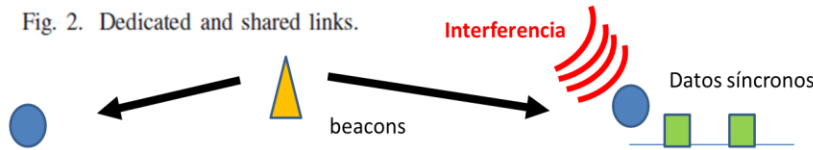


Frequency diversity frames (2G, 4G, 15.4 TSCH)

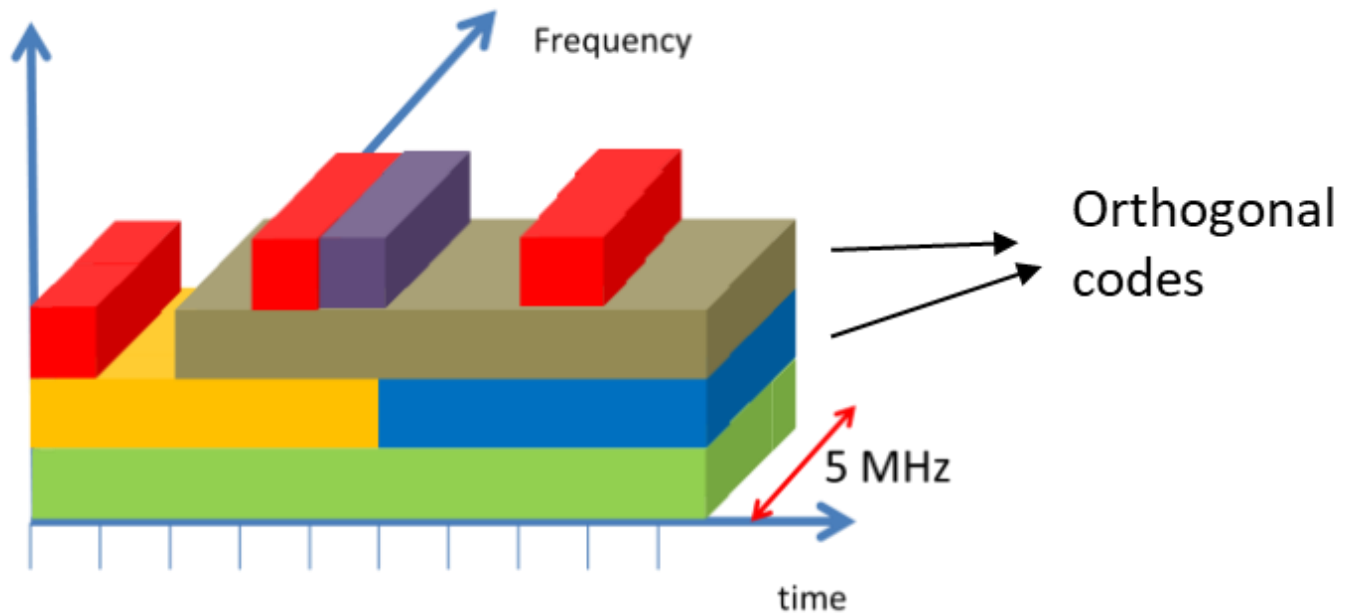


IEEE 802.15.4 TSCH

Fig. 2. Dedicated and shared links.

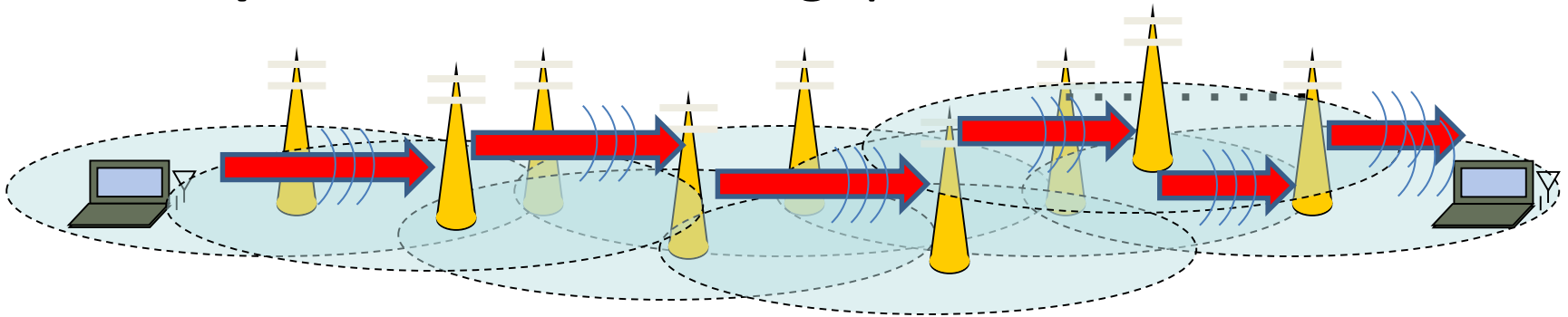


CDMA (3G)

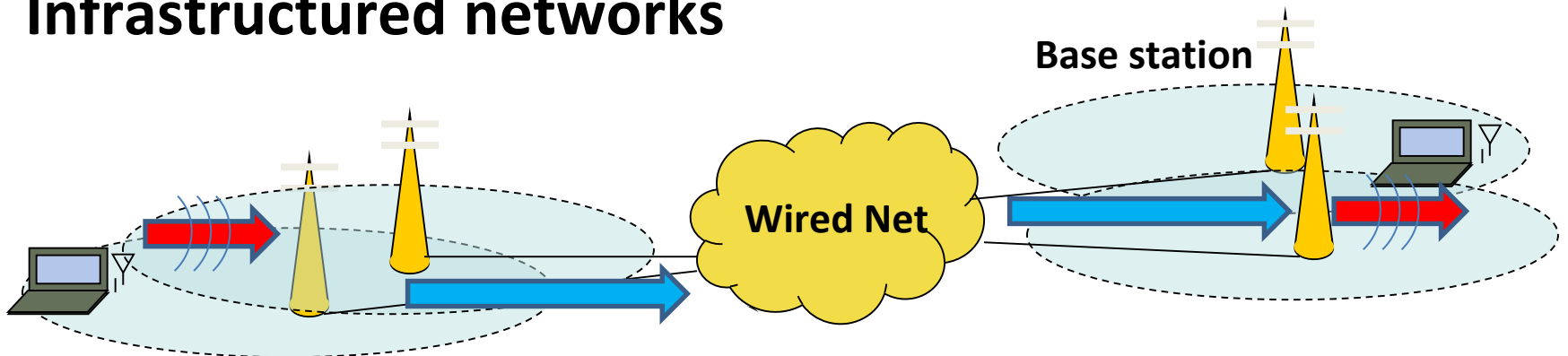


Wireless Nets

Multihop: Small & low throughput nets

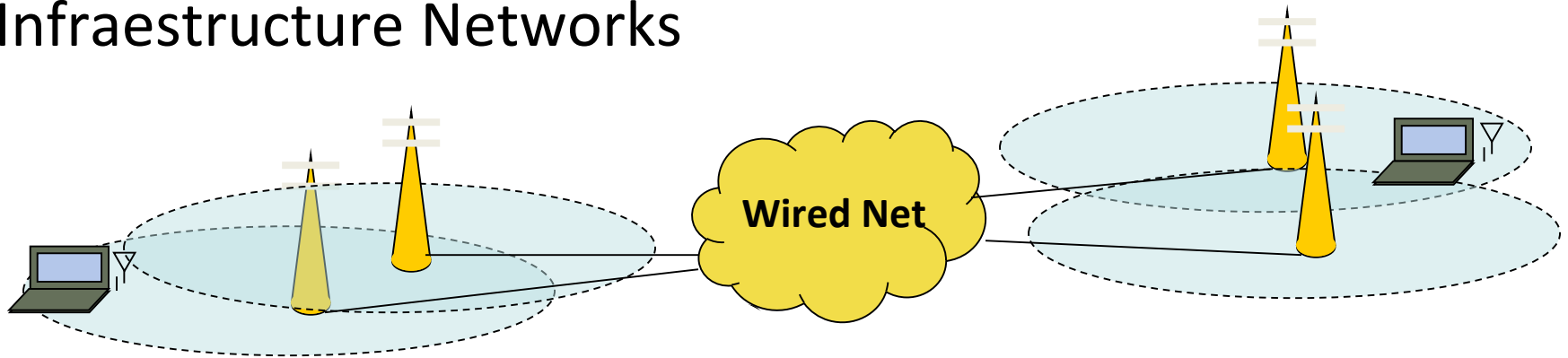


Infrastructured networks



Wireless Nets

Infraestructure Networks



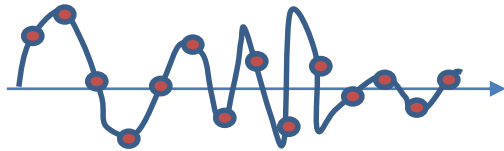
GSM (2G), GPRS (2.5G), UMTS (3G), HSPA (3.5G), LTE (4G)

LoRa/LoRaWAN

WiFi Hotspots

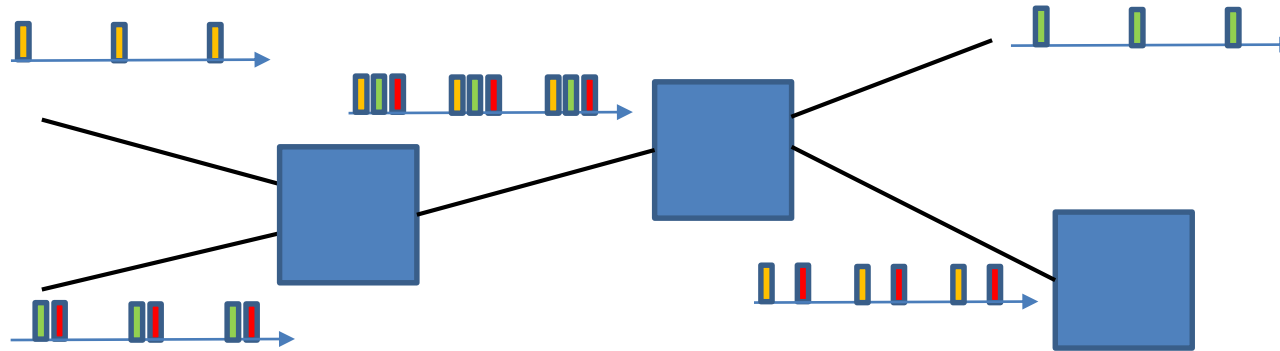
Circuit & packet switching

Voice,
Sampling

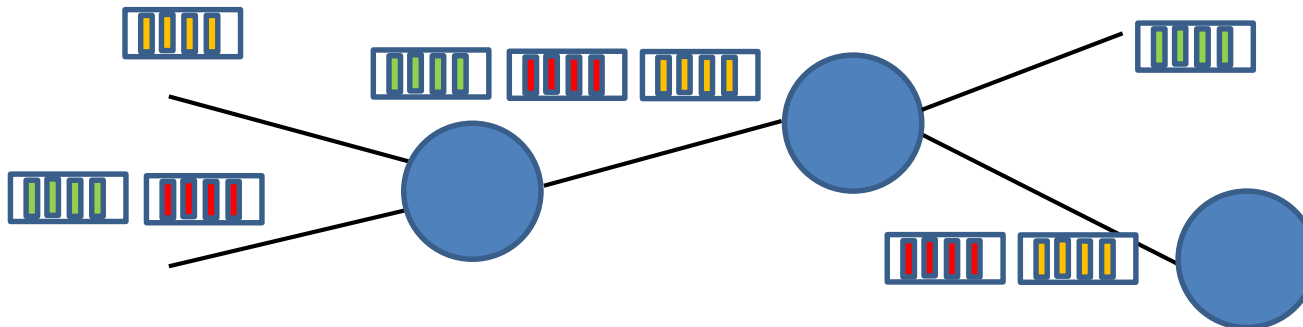


0	0	1	1	0	0	1	0	0	1	0	0
1	1	0	0	1	1	1	1	0	1	0	1
1	1	0	1	1	1	1	1	0	1	1	1
0	1	0	0	0	1	0	0	1	0	0	1

Quantization

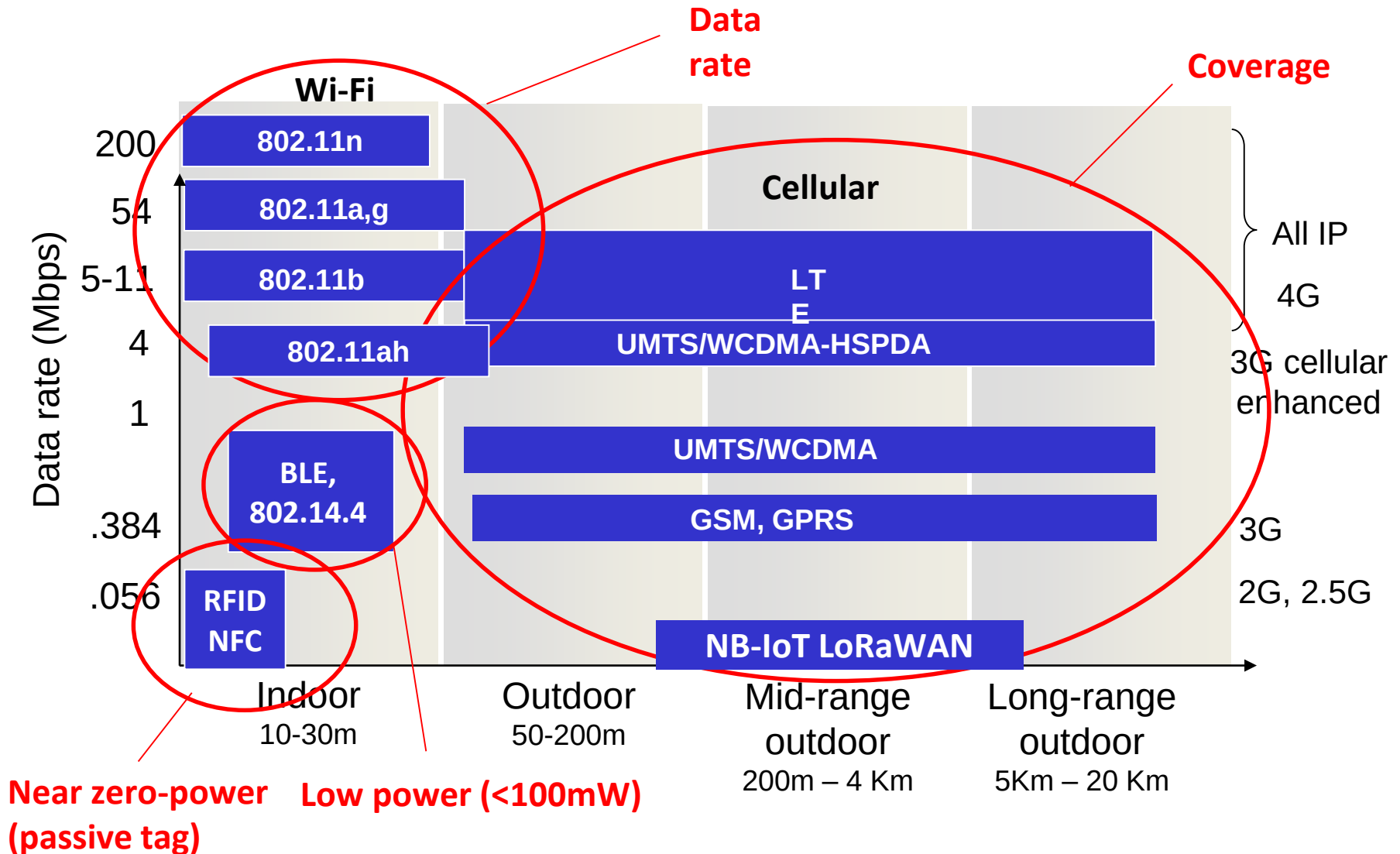


Circuit
switching



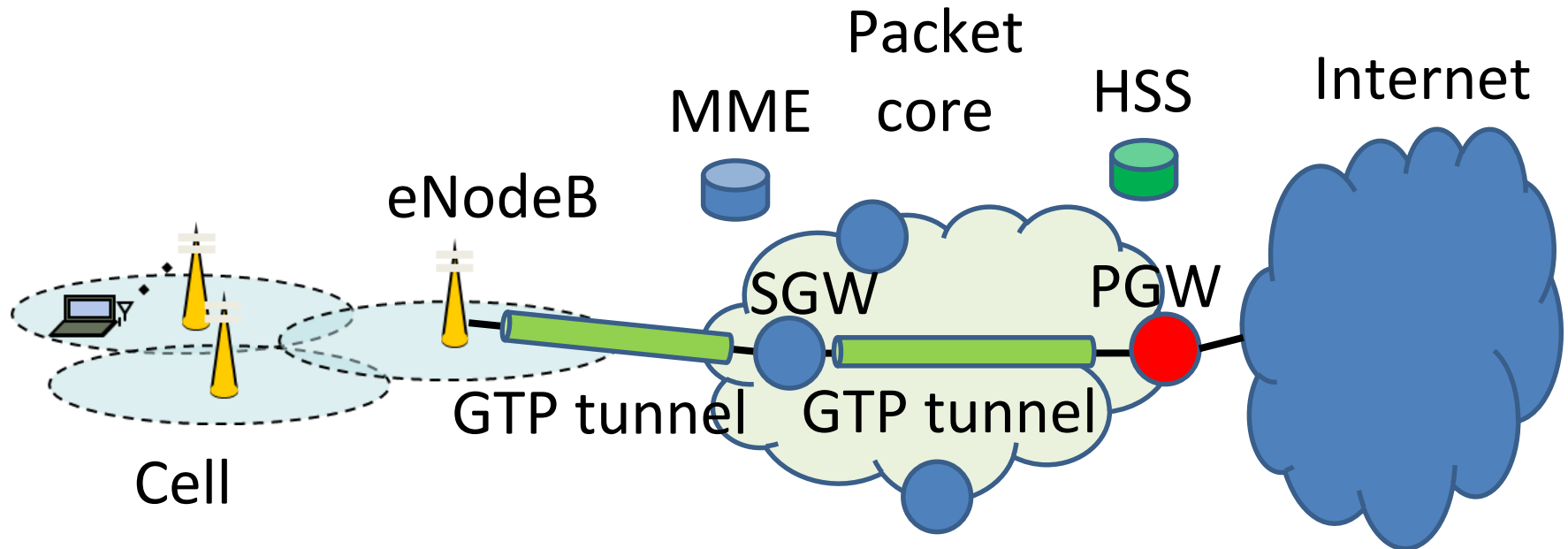
Packet
switching

Comm: Wireless technologies



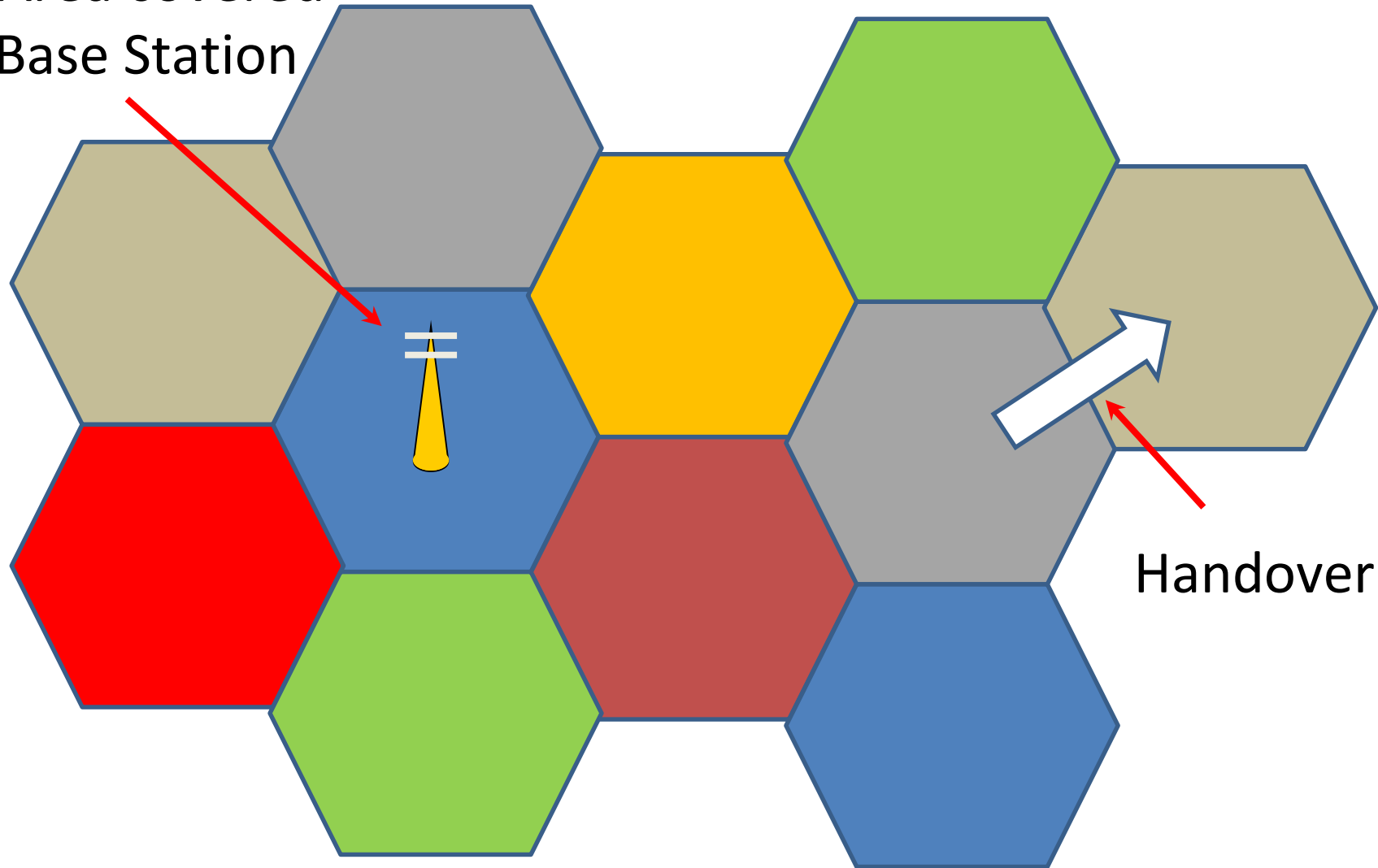
Cellular Networks

Cellular Networks



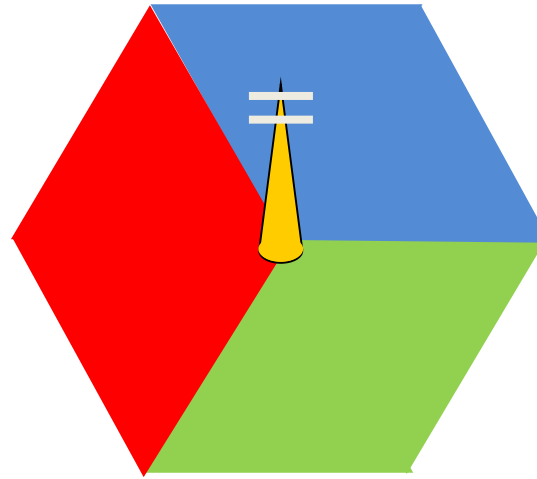
Cellular networks

Cell: Area covered
by a Base Station



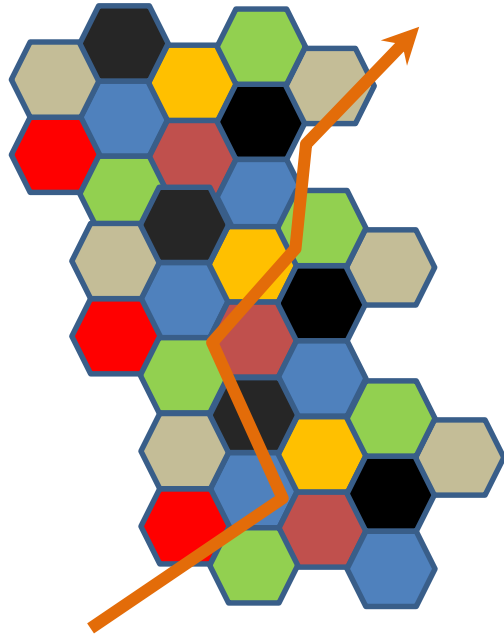
Real cells are not hexagons

- Real shapes can be very complicated
- Different sizes:
 - Urban (x500 m)
 - Suburban
 - Rural (x 5 Km)
- Sectorization



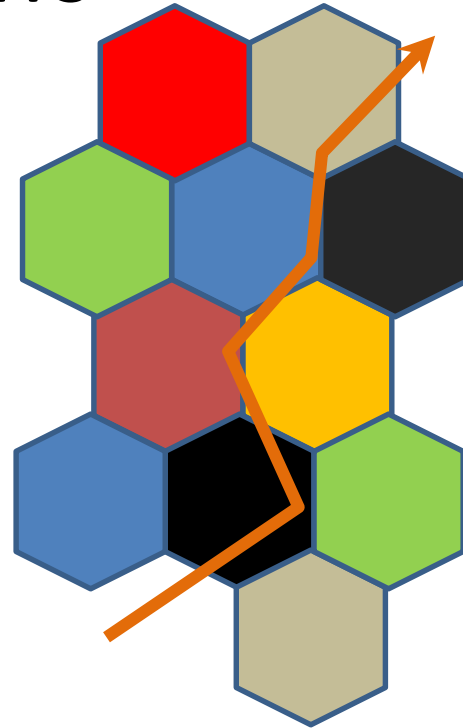
e.g. 120 degrees sectors

Small cells =>
Higher capacity



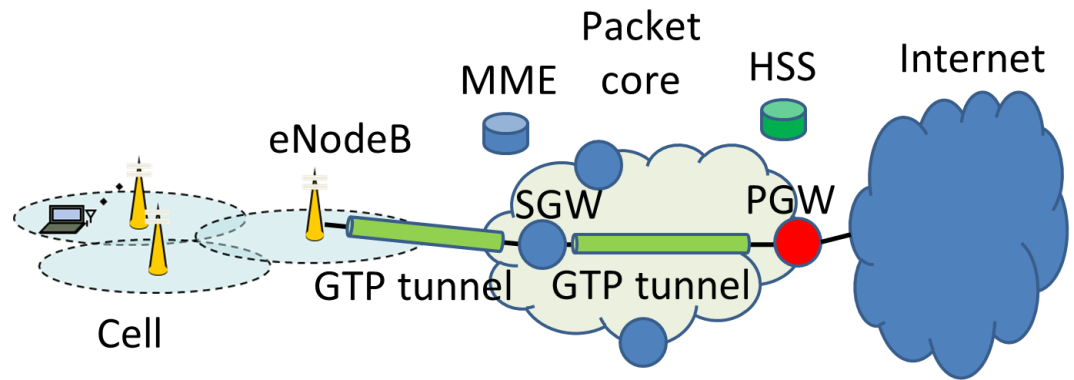
Cell size tradeoffs

Large cells =>
Cheaper coverage



Small cells => More users per km², but more handovers,
more towers and more connections to the backbone
Network (more expensive to install and maintain)

Mobile Packet Core



- **Session management**

Establishing and managing information flow of a particular service or application

- **Mobility management**

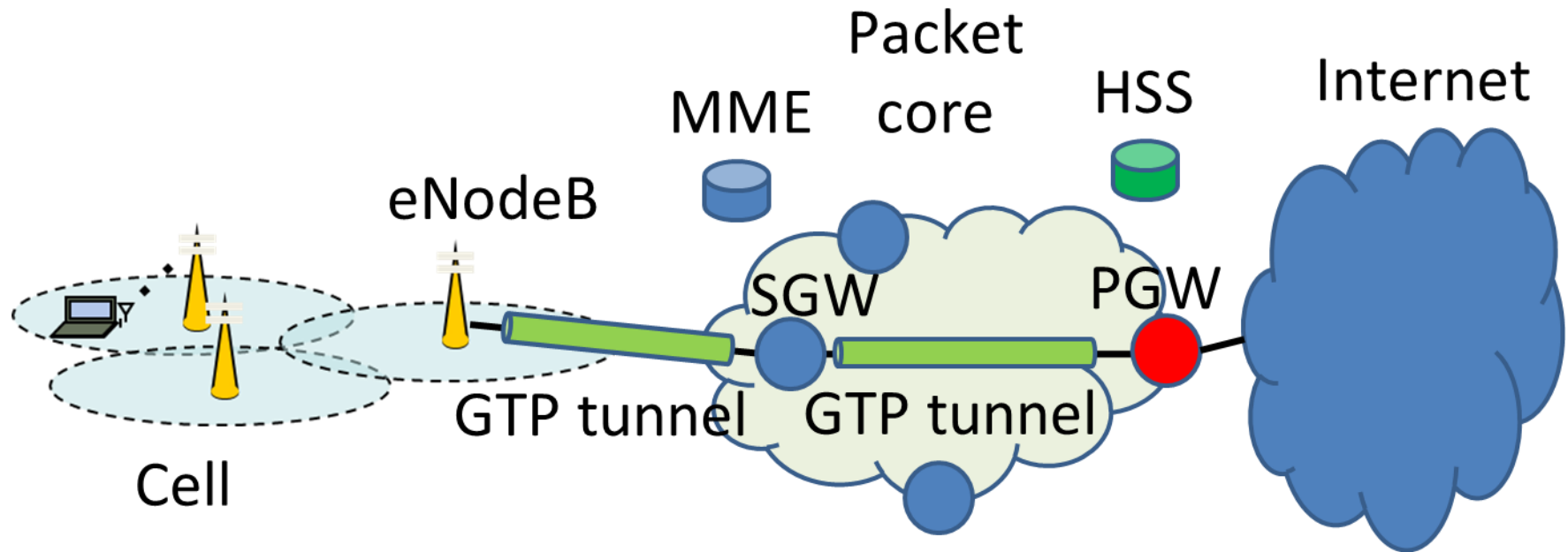
Track users as they move between cell site and route traffic accordingly

- **Security & privacy**

Authentication, encryption and user privacy.

- **Policy & charging**

LTE (Evolved Packet Core)



IP network: Data and VoLTE

Serving Gateway (SGW) (3g *SGSN*) **(Voice over LTE)**

Packet Data Network Gateway (PGW), (3g *GGSN*)

Mobility Management Entity (MME) (3g *VLR*)

Home Subscriber Server (HSS) (3g *HLR, AuC*)

Cellular net evolution

2G 3G 4G 5G 6G

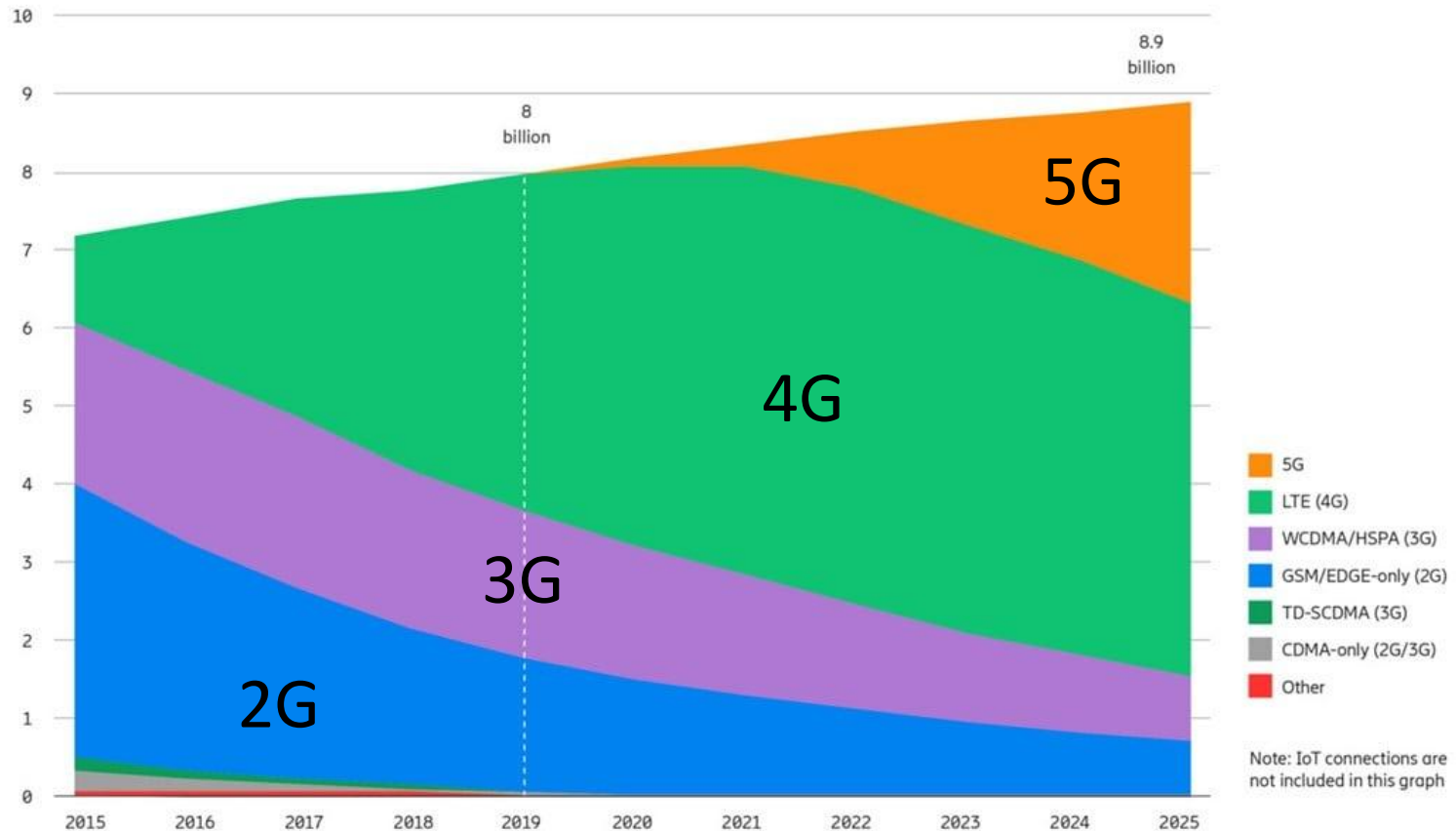
Peak throughput · bandwidth · frequency bands (EU) · latency

GEN	STANDARD	PEAK DOWNLINK	PEAK UPLINK	CHANNEL BW	EU FREQUENCY BANDS	LATENCY
2G	GSM / GPRS / EDGE	~384 Kbps (GPRS ~114 Kbps)	~384 Kbps	200 kHz	900 MHz · 1800 MHz	300–1000 ms
3G	UMTS / HSPA / HSPA+	~42 Mbps (HSPA+ max)	~11 Mbps (HSUPA max)	5 MHz	900 MHz · 2100 MHz	50–200 ms
4G	LTE / LTE-A / LTE-A Pro	~1 Gbps (typ. 20–150 Mbps)	~500 Mbps (typ. 10–50 Mbps)	1.4–20 MHz (up to 100 MHz CA)	700 / 800 / 1800 / 2100 / 2600 MHz	10–50 ms (~30 ms typ.)
5G	NR (New Radio) SA/NSA	~20 Gbps (sub-6 GHz: 100–900 Mbps)	~10 Gbps (sub-6 GHz: 50–200 Mbps)	Up to 400 MHz (sub-6: ≤100 MHz)	700 / 1500 / 2100 / 3.5 GHz (n78) mmWave: 26 GHz (n258)	<10 ms (URLLC target: <1 ms)
6G	IMT-2030 (proposed ~2030)	~1 Tbps (theoretical)	~1 Tbps (theoretical)	Multi-GHz (THz bands)	Sub-6 GHz + mmWave + THz (0.3–3 THz, not yet allocated)	<0.1 ms (target, not ratified)

Peak figures are theoretical maxima under ideal conditions. Real-world performance is typically 10–30× lower.

Forecast global subscriptions

Mobile subscriptions by technology (billion)



Source: <https://www.ericsson.com/en/mobility-report/reports/november-2019/mobile-subscriptions-outlook>

Cellular net evolution

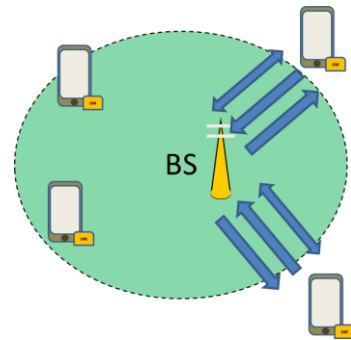
- To increase the overall capacity of cellular networks, there are three main possibilities:
 - **Networks with small cells**
 - Use wide bandwidth available **at higher frequencies**
 - Use **multiantenna transmissions** or other techniques that increase spectral efficiency

Source: David López-Pérez, Ming Ding, Holger Claussen, Amir H. Jafari, “Towards 1 Gbps/UE in Cellular Systems: Understanding Ultra-Dense Small Cell Deployments”

Cellular networks:

Cellular networks standards

Channels

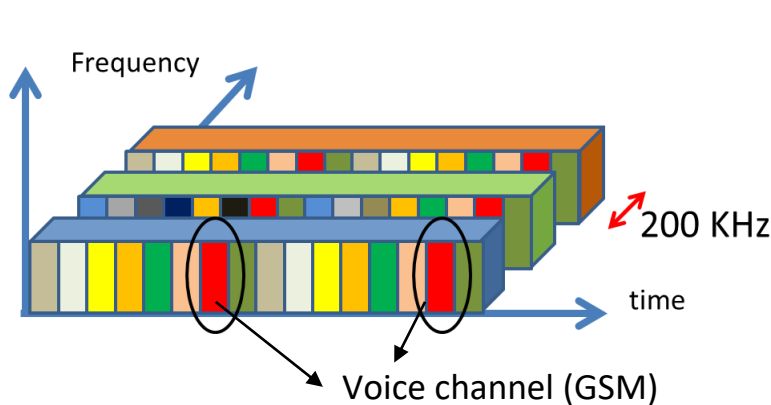


- **Traffic channel** (TCH, UD): user info
- **Random Access channel** (RACH, U), the user request use of channel, (ALOHA)
- **Access Grant channel** (AGCH, D): The network assigns channel to user
- **Paging channel** (PCH, D): The network signals the user about a new call
- **Broadcast Control Channel** (BCCH, D, broadcast): Identifies cellular network, Network, BS id, etc

Radio access

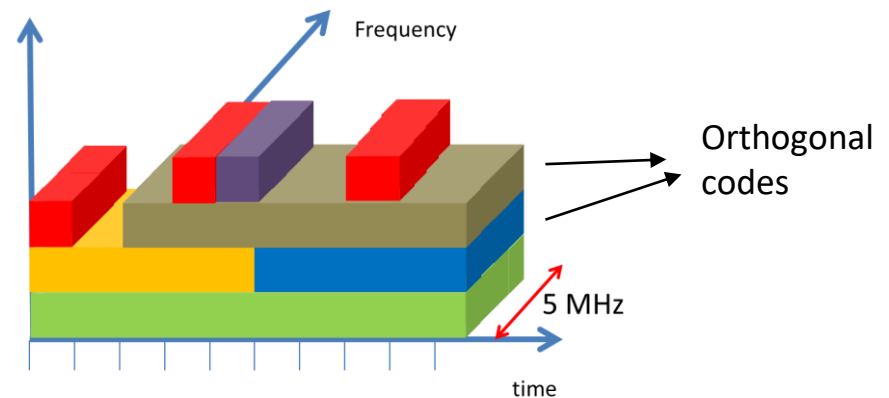
2G (GSM), 2.5 G (GPRS)

- FDMA + TDMA
- Voice: Frames of 8 time slots/200 KHz
- GSM: Voice (13 Kbps)
- GPRS: Data (< 170 Kbps)



3G (UMTS), 3.5 G (HSPA)

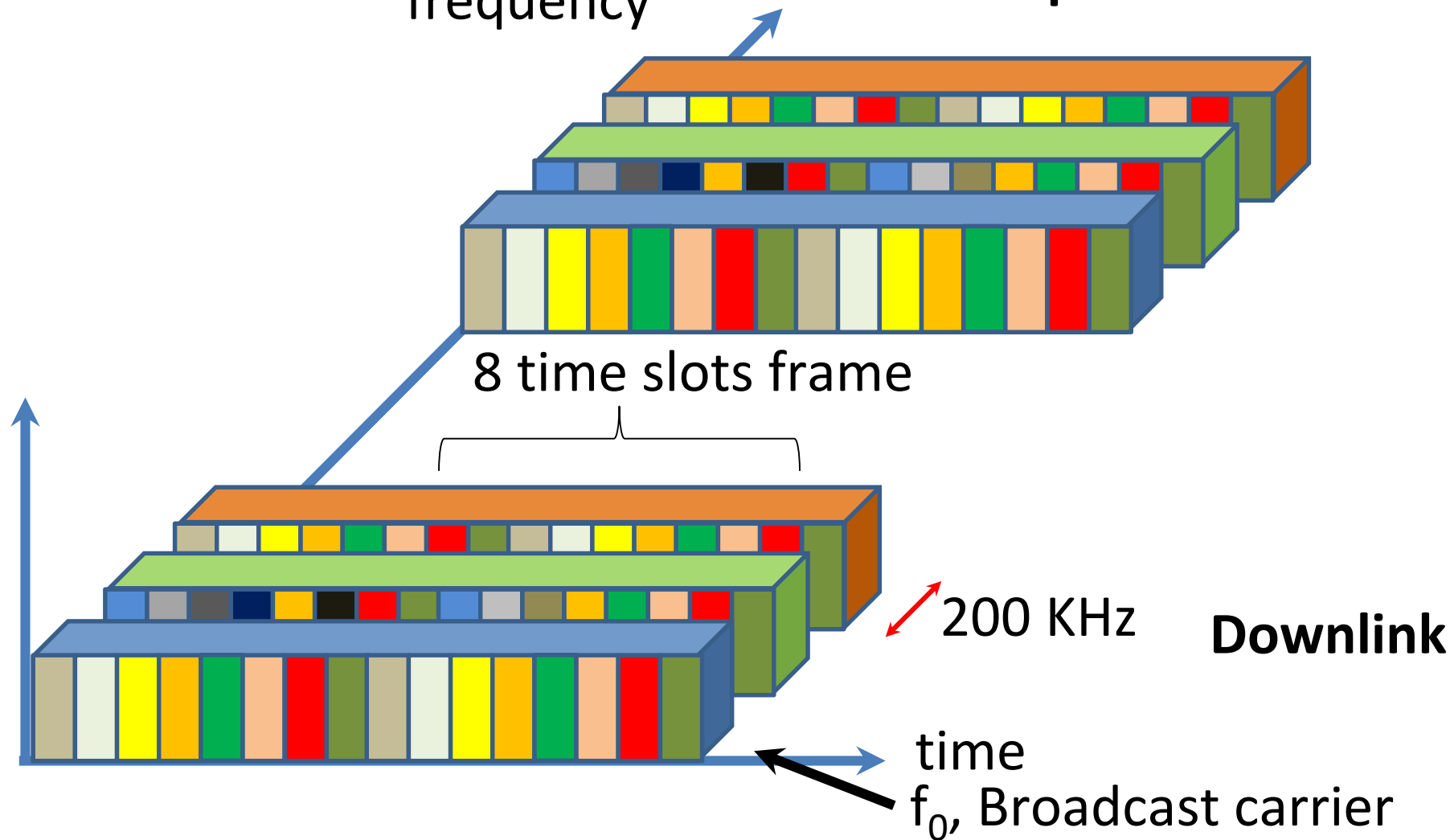
- WCDMA
- Bw= 5 Mhz, Rchip =3.84 Mcps
- UMTS: $R < 2/0.3$ Mbps,
- HSDPA/HSUPA, $R < 10/2.4$ Mbps



These networks (in special 3G) will probably be turned off during the next years to free spectrum for 4G and 5G

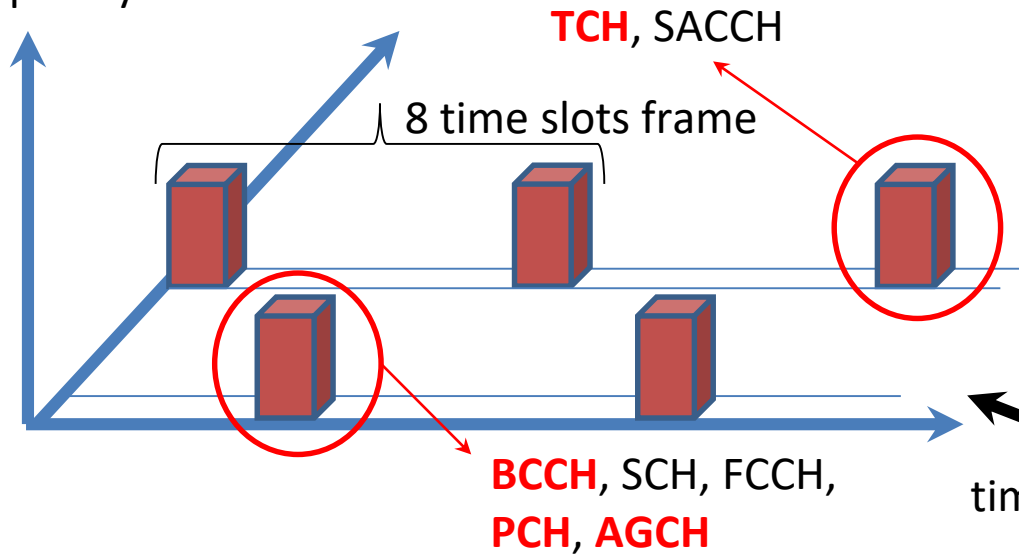
2G (GSM)

frequency **Uplink**

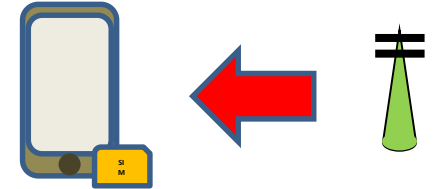


2G (GSM)

Frequency

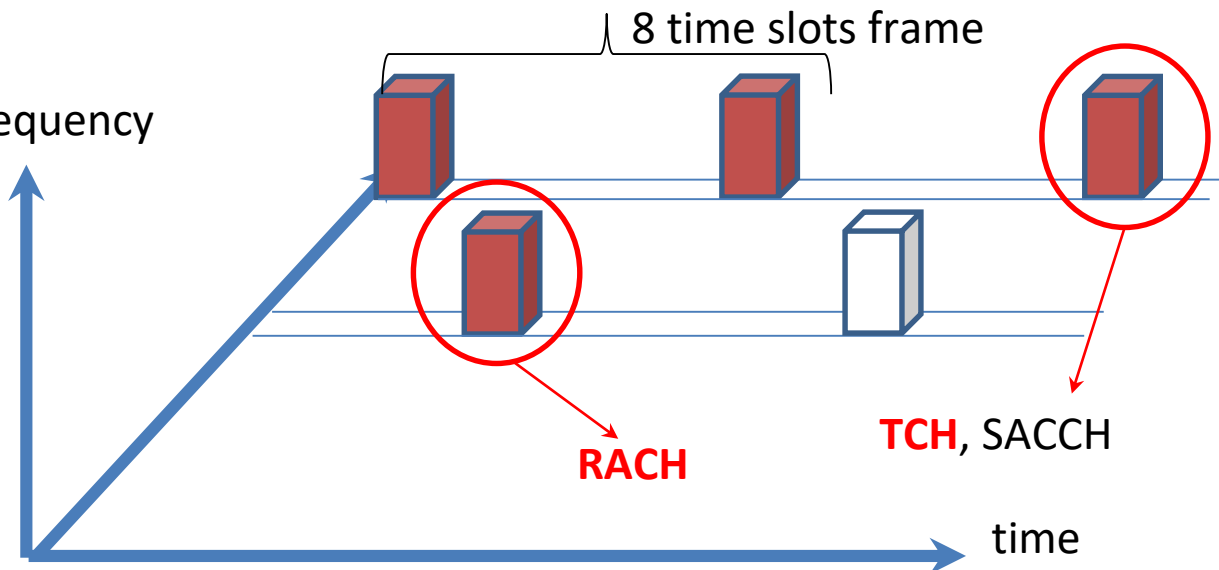


Downlink

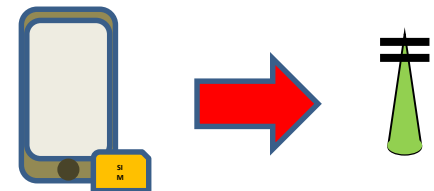


f_0 , Broadcast carrier
time

Frequency

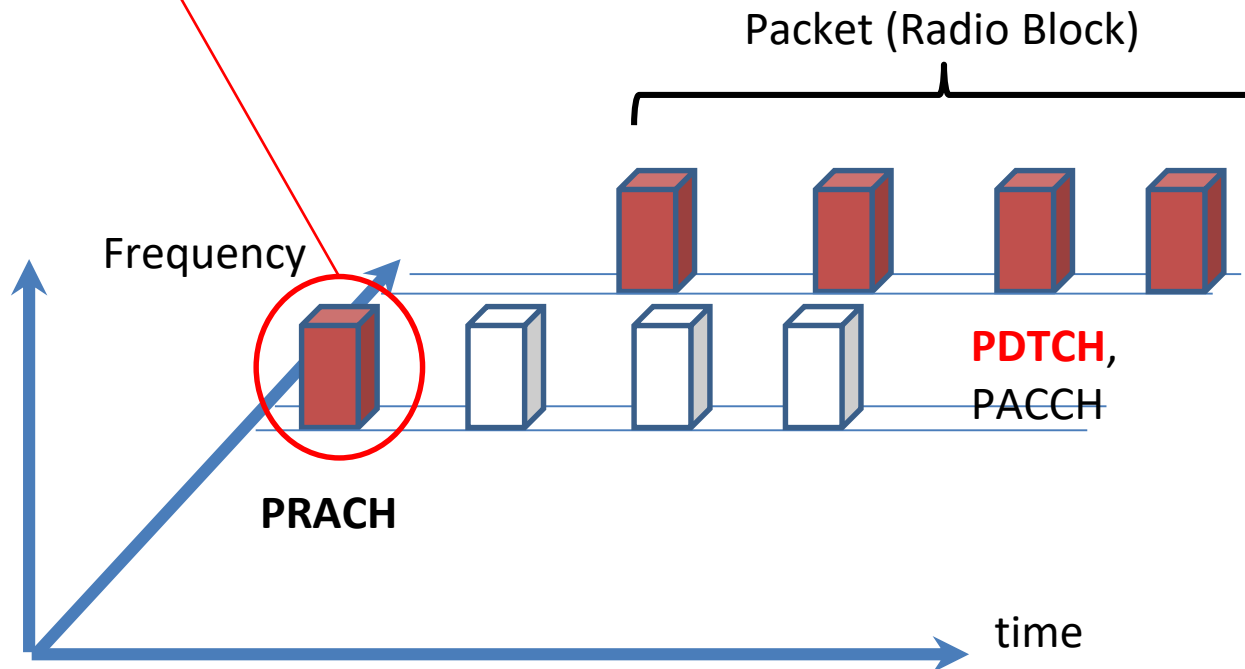
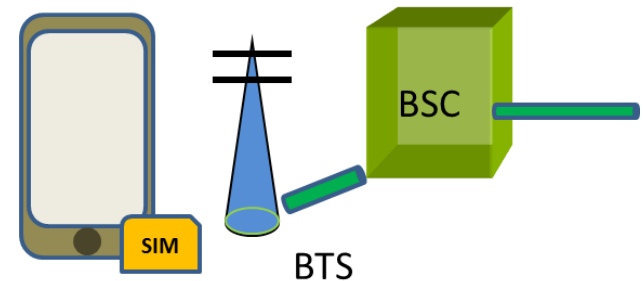


Uplink



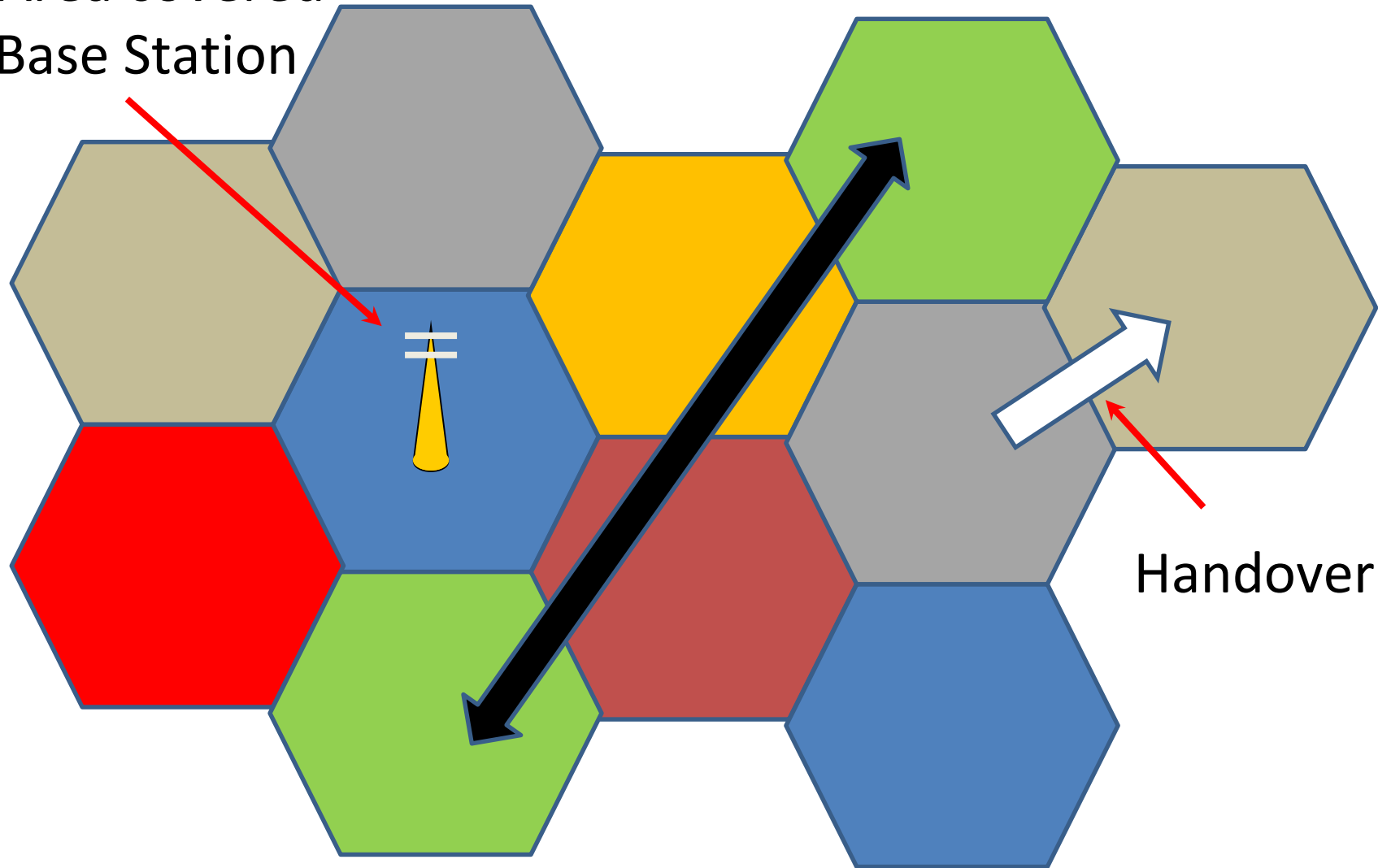
2,5G (GPRS)

Uses Slotted ALOHA in the PRACH channel in order to obtain access to a PDTCH and send one or more data packets



2G/2.5G: Frequency reuse

Cell: Area covered
by a Base Station

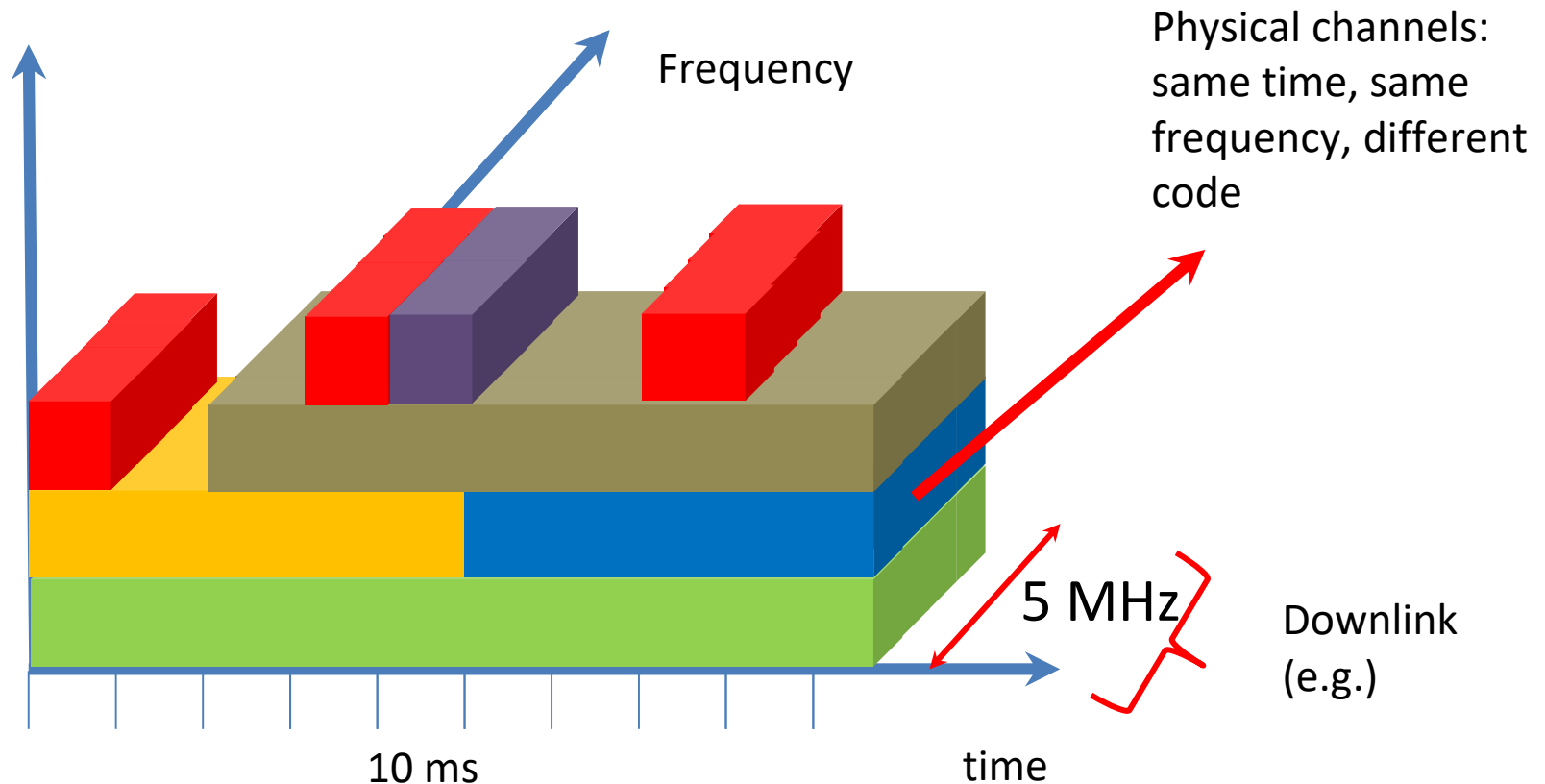


3G (UMTS)

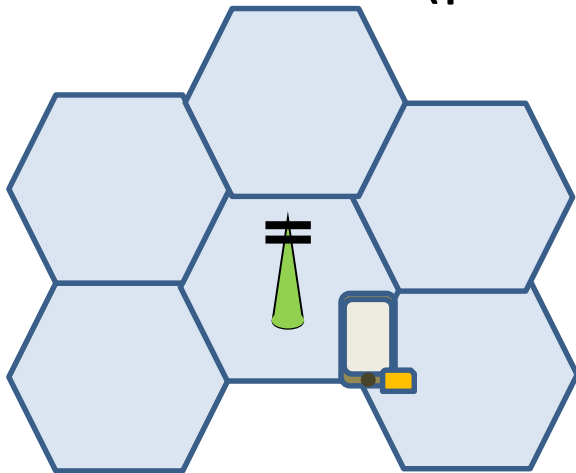
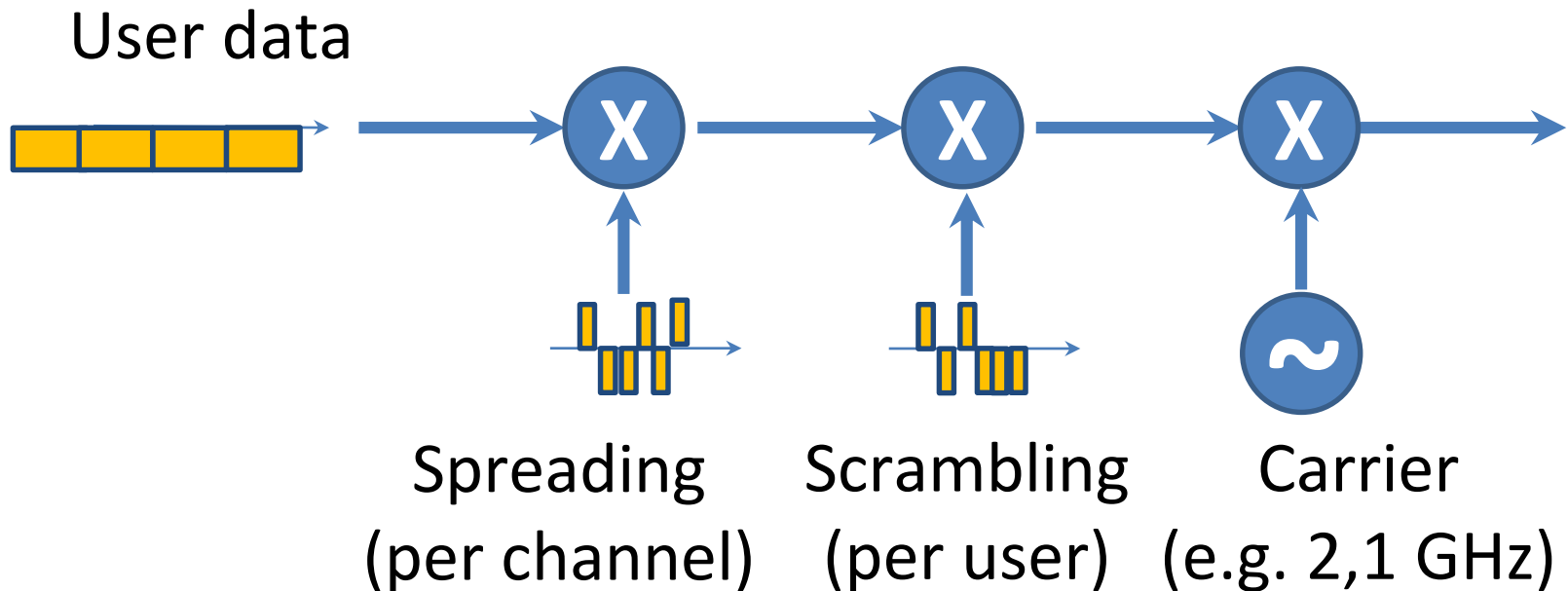
- Europe: WCDMA (UMTS FDD)
- 1900-1980, 2100-2170 and 2110-2170 MHz
- $B_w = 5$ Mhz
- Chip Rate 3.84 Mcps
- Data rates: $\leq 2/0.3$ Mbps

3G (UMTS)

- Uses CDMA (same frequency & time, spreading code)
- Uplink and Downlink in different frequency bands



3G, spreading and scrambling codes



No frequency reuse!, orthogonal scrambling codes
The available number of SCs is **512**; however, they are used to identify thousands of sectors over the network. Therefore code planning is essential in the downlink direction

3.5G (HSPA)

- Physical channel for higher data rate (10/2.4 Mbps)
 - Higher order modulation (16-QAM)
 - Fast scheduling (users with more favourable channel conditions)
 - Improved ARQ (soft and hybrid ARQ)
- HSDPA: Enhanced Downlink
- HSUPA: Enhanced Uplink

2G/3G architecture

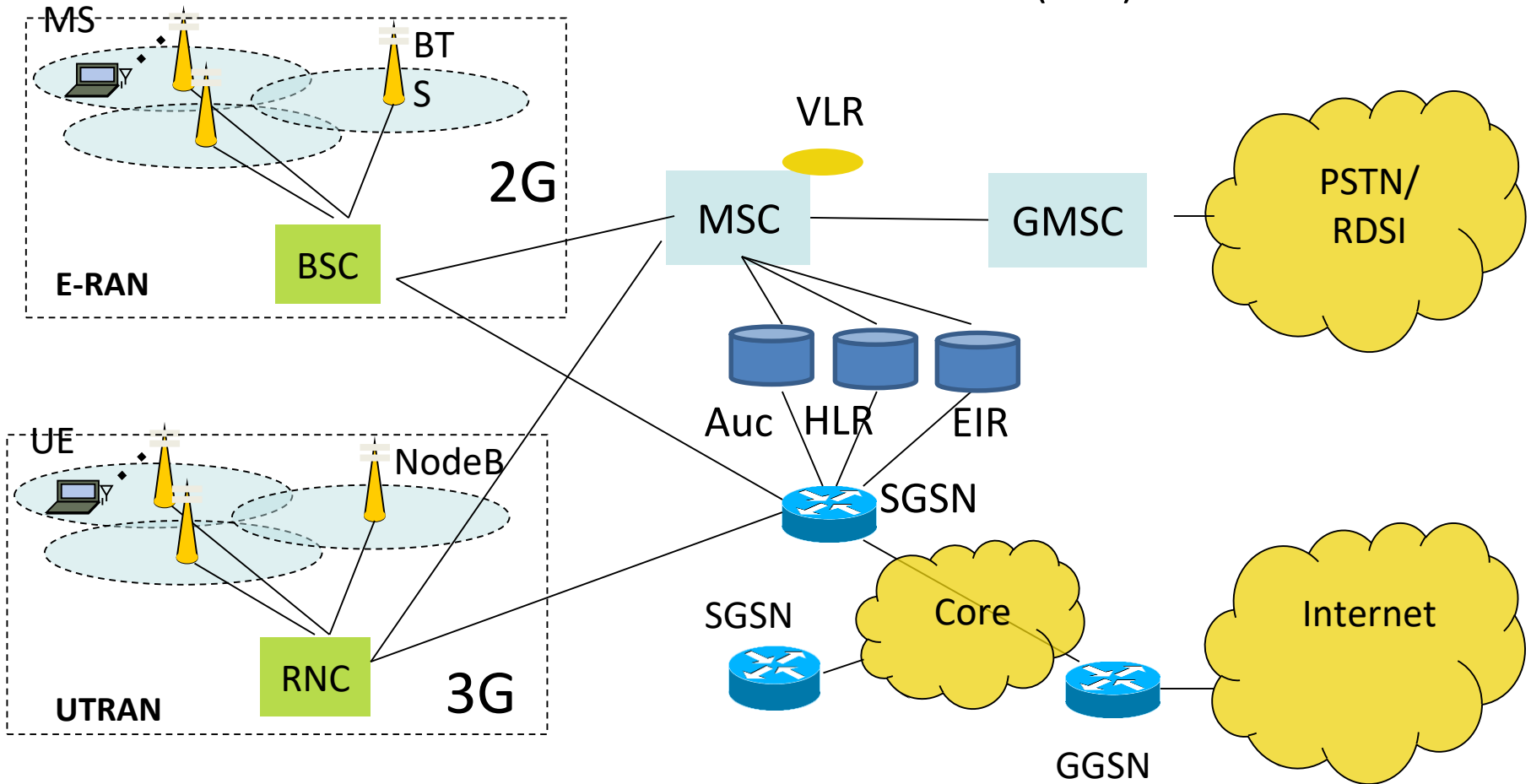
Mobile Switching Center (MSC)

Serving/Gateway GPRS Support Node (SGSN/GGSN)

Home Location Register (HLR)

Visitor Location Register (VLR)

Authentication Center (AuC)



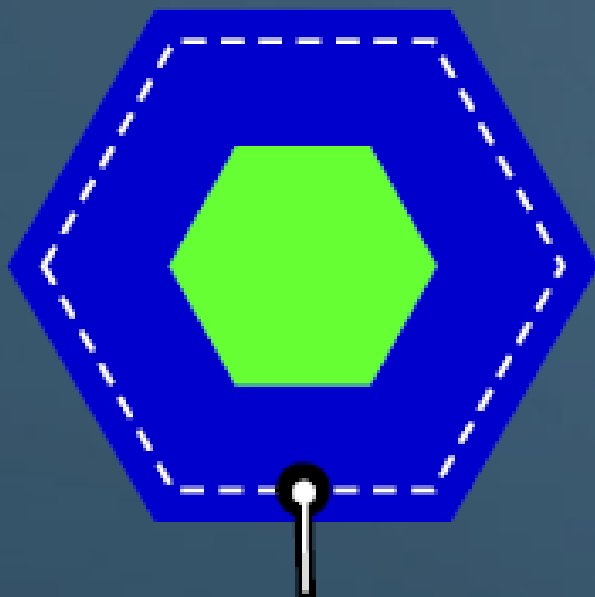
4G (LTE)

- **Increase cell capacity and user data rates**
- Objectives: 100 Mbps(D), 50 Mbps (U), reduced latency
- **Larger and flexible frequency bands** (e.g: Bw= 5, 10 or 20 MHz)
- OFDMA (DL), SC-FDMA (UL)
- Improve spectral efficiency:
 - **MIMO**
 - **Scheduler to optimize transmission**

HSPA

10 MHz, 20W

Sector Throughput
4.5 Mbps

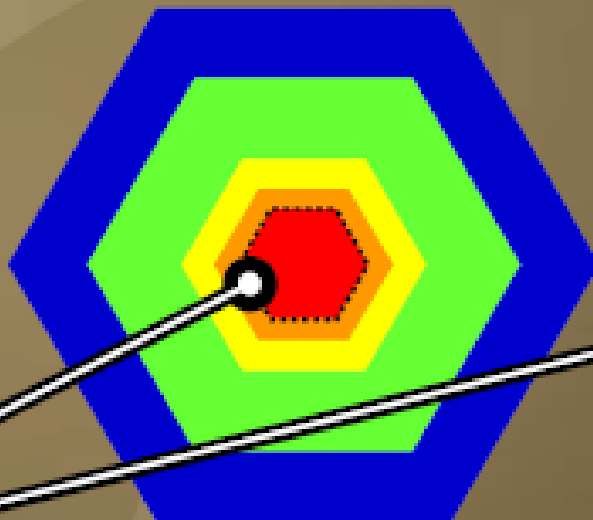


WCDMA cell breathing

LTE

10 MHz, 2 x 20W

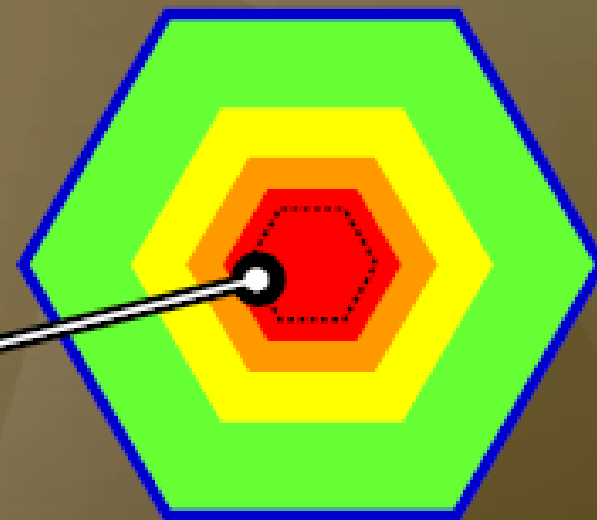
Sector Throughput
16.7 Mbps



LTE

20 MHz, 2 x 40W

Sector Throughput
33.4 Mbps



Aggregate
datarate



> 65 Mbps



> 30 Mbps



> 20 Mbps

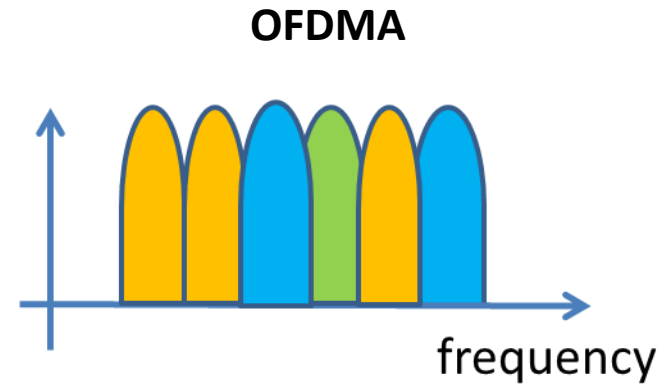


> 7 Mbps



> 1.2 Mbps

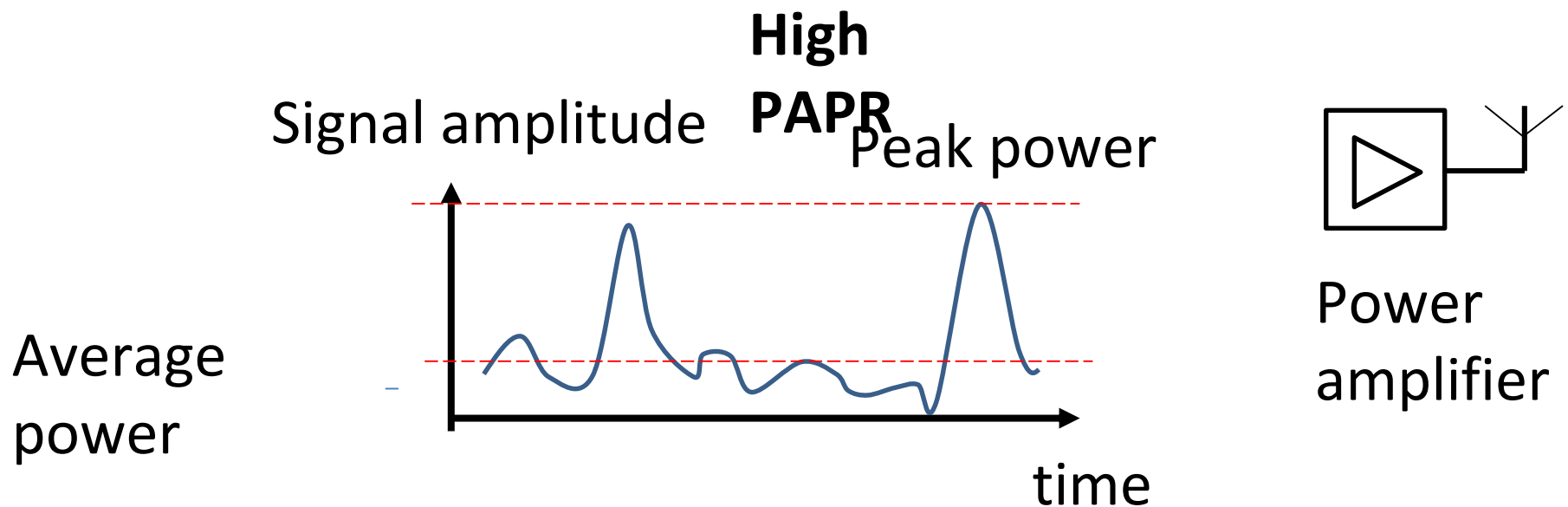
OFDMA



- Downlink channel in LTE uses OFDMA
- Long symbol times limits intersymbol interference due to multipath
- Allows flexible utilization of frequency spectrum
- Increases spectral efficiency due to orthogonality between sub-channels
- Allow optimization of data rates for users by transmitting on the sub-channels with lower fading

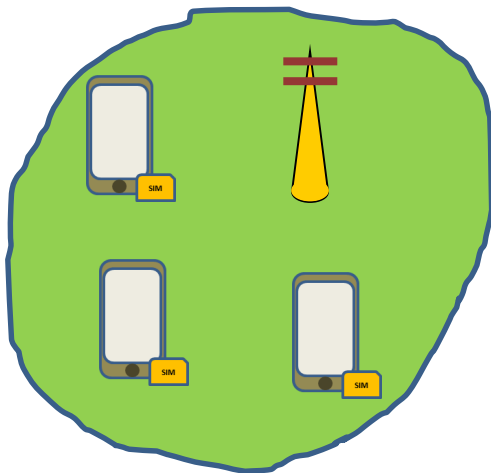
Problems with OFDMA

- The OFDMA signal has a high **Peak to Average Power Ratio (PAPR)**, meaning that the power amplifier needs a high power consumption as it must be able to amplify signals with high peaks

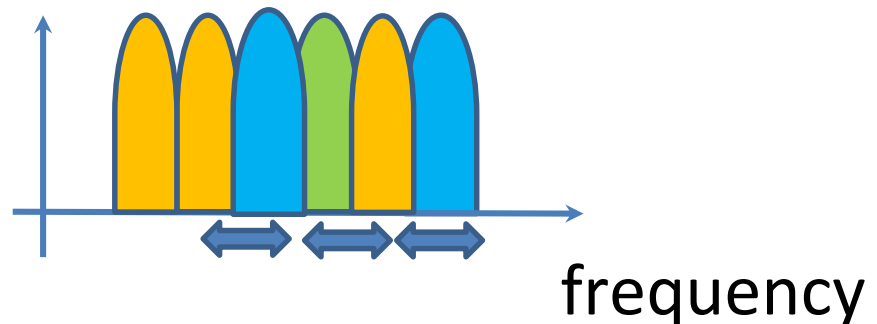


Problems with OFDMA

- In the **uplink**, the multiple channels produced by different phones are not perfectly aligned (instability in electronics, Doppler spread). Due to this offset in carriers, there is some interference among sub-channels.

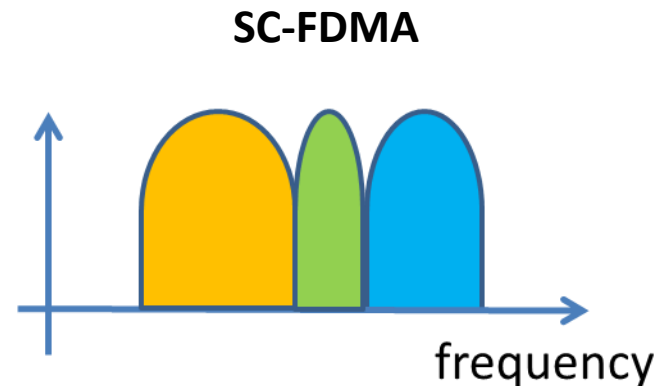


Multiple carriers

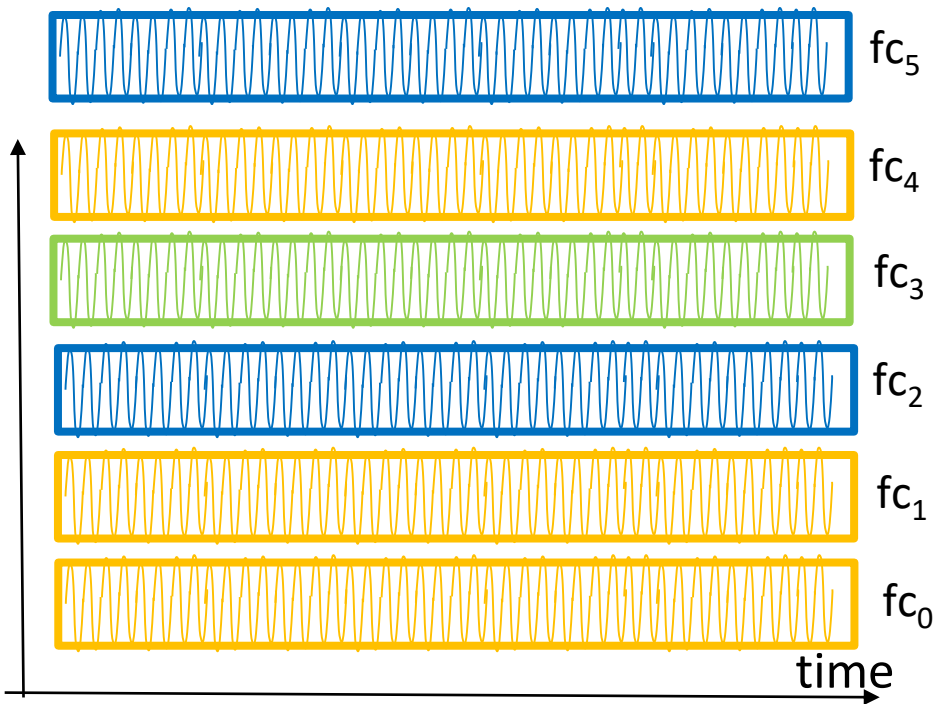
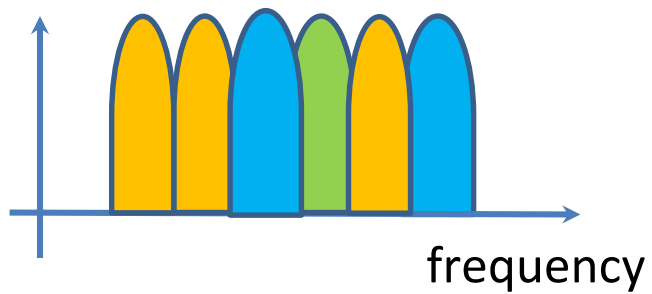


OFDMA and SC-FDMA

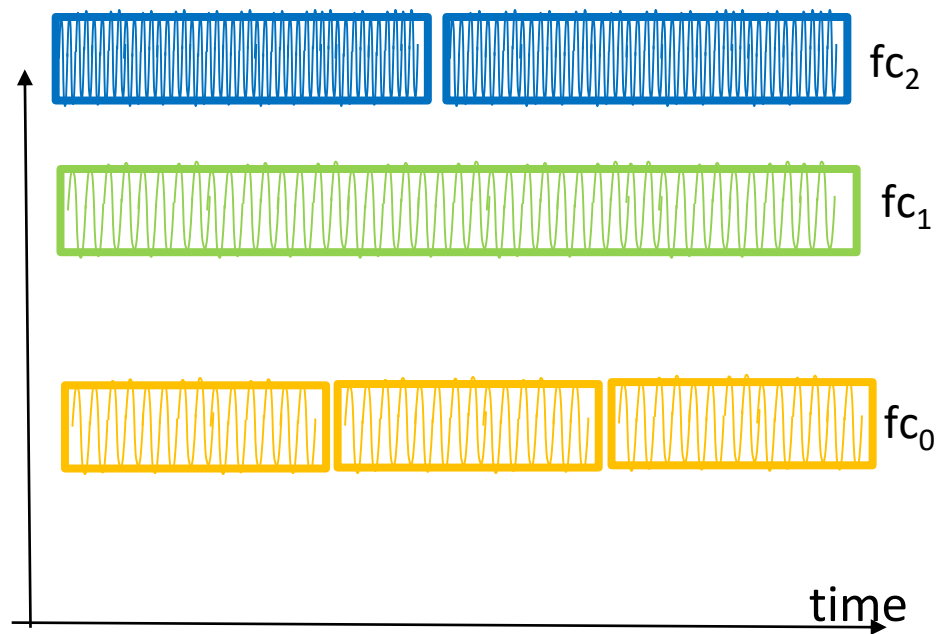
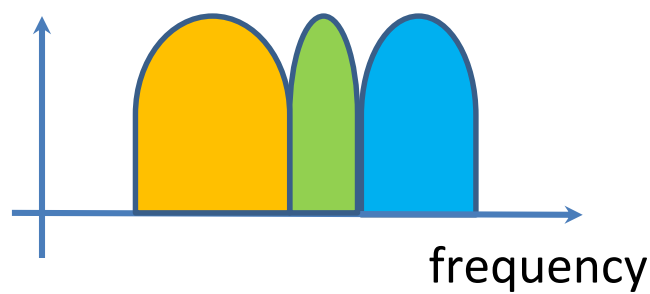
- Uplink transmission, and alternative technique called **Single Carrier- FDMA (SC-FDMA)** is used.
- A number of adjacent subchannels is allocated to each user. The **symbols are transmitted sequentially** using a single carrier per user
- The inter-symbol interference is solved by a more complex receiver at the BS.



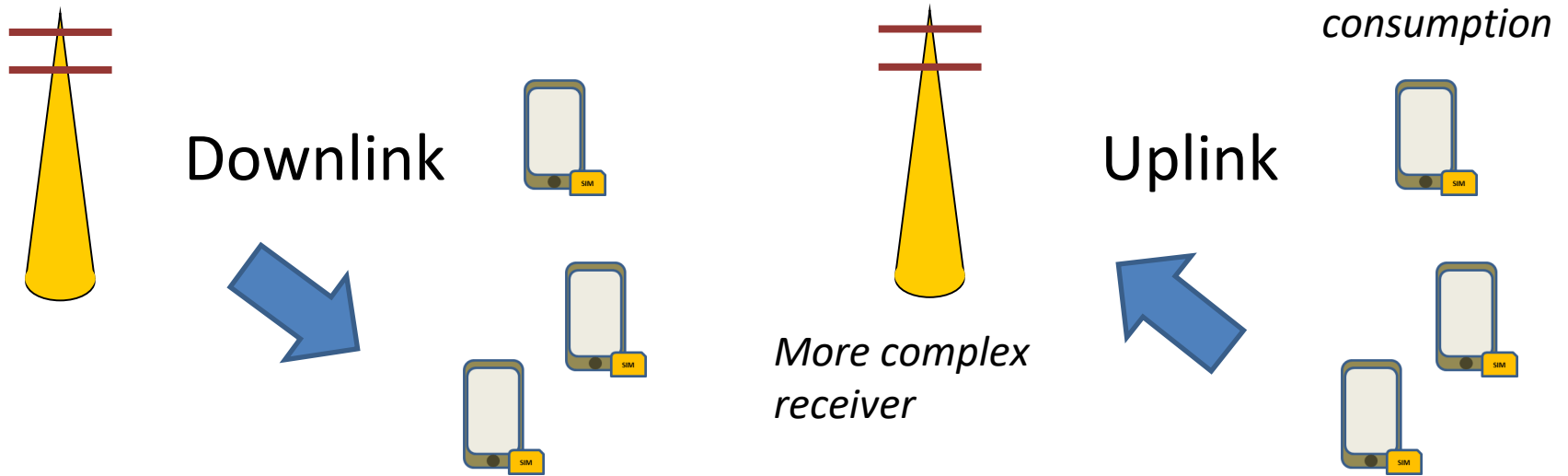
OFDMA



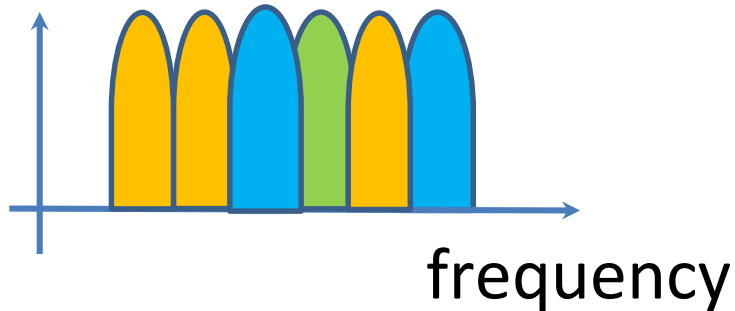
SC-FDMA



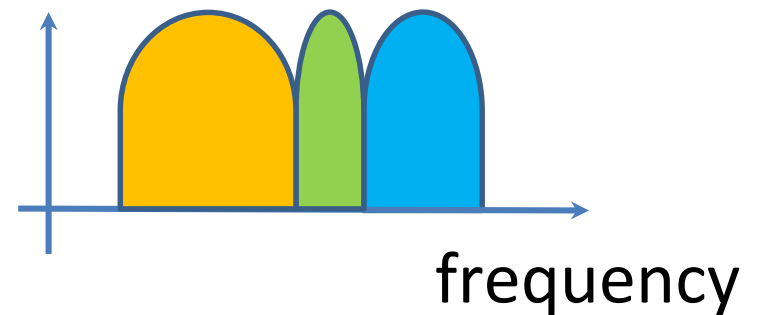
OFDMA and SC-FDMA



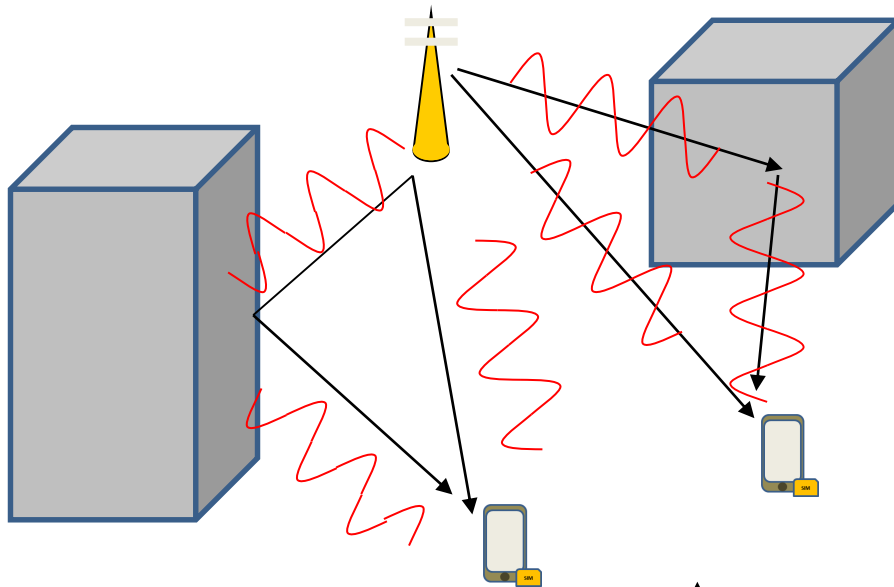
OFDMA



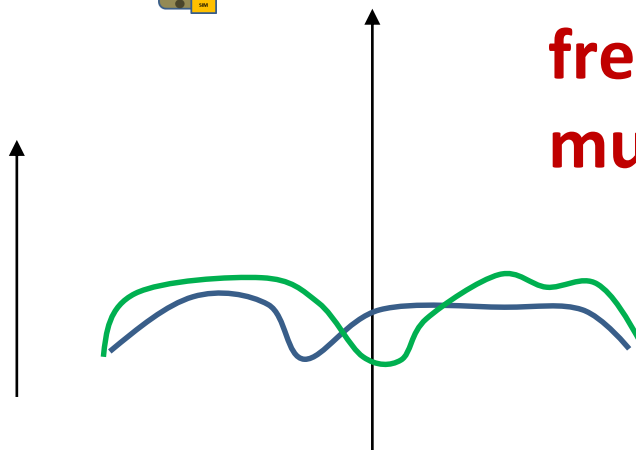
SC-FDMA



Different users in the cell suffer different multipath fading

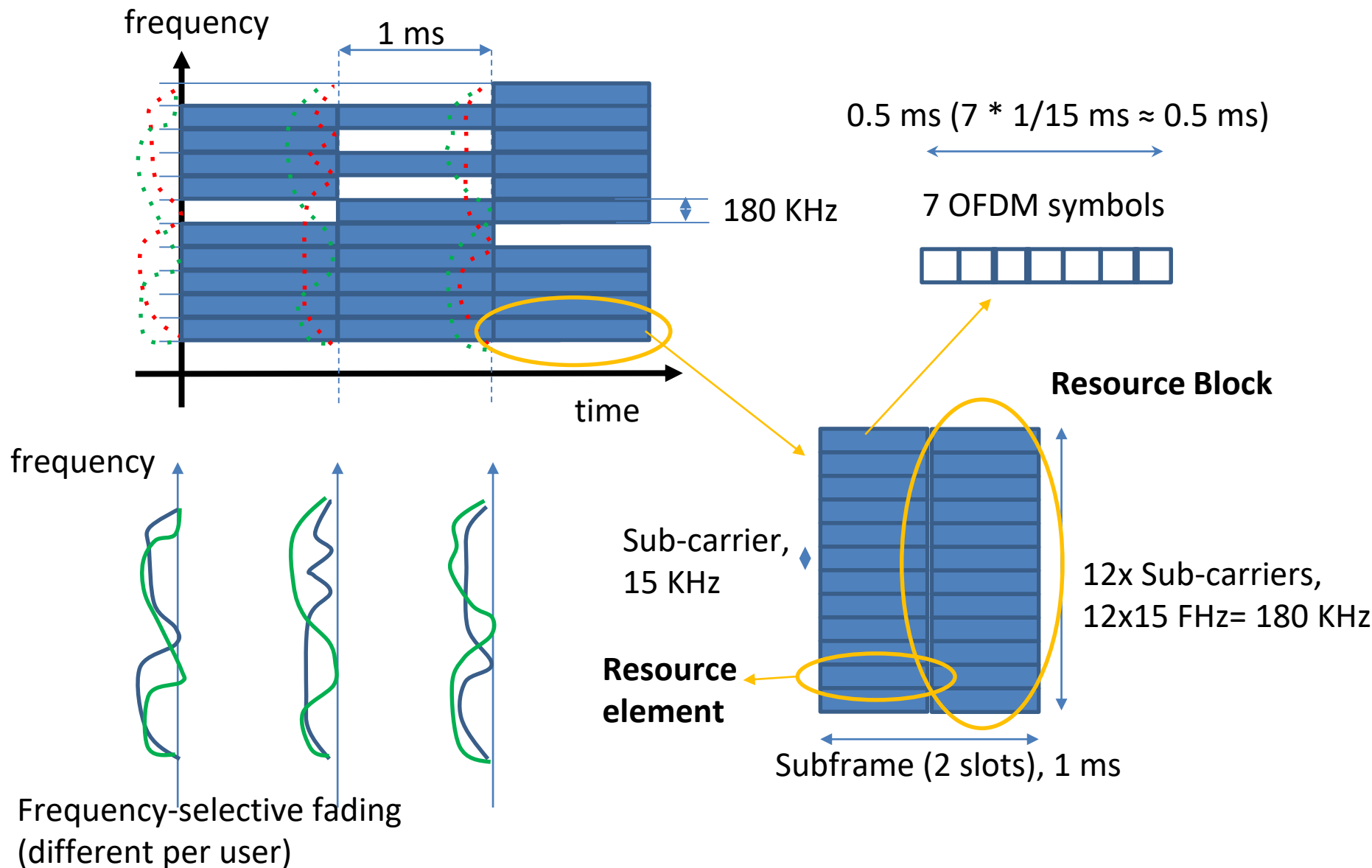


**frequency selective
multipath fading**



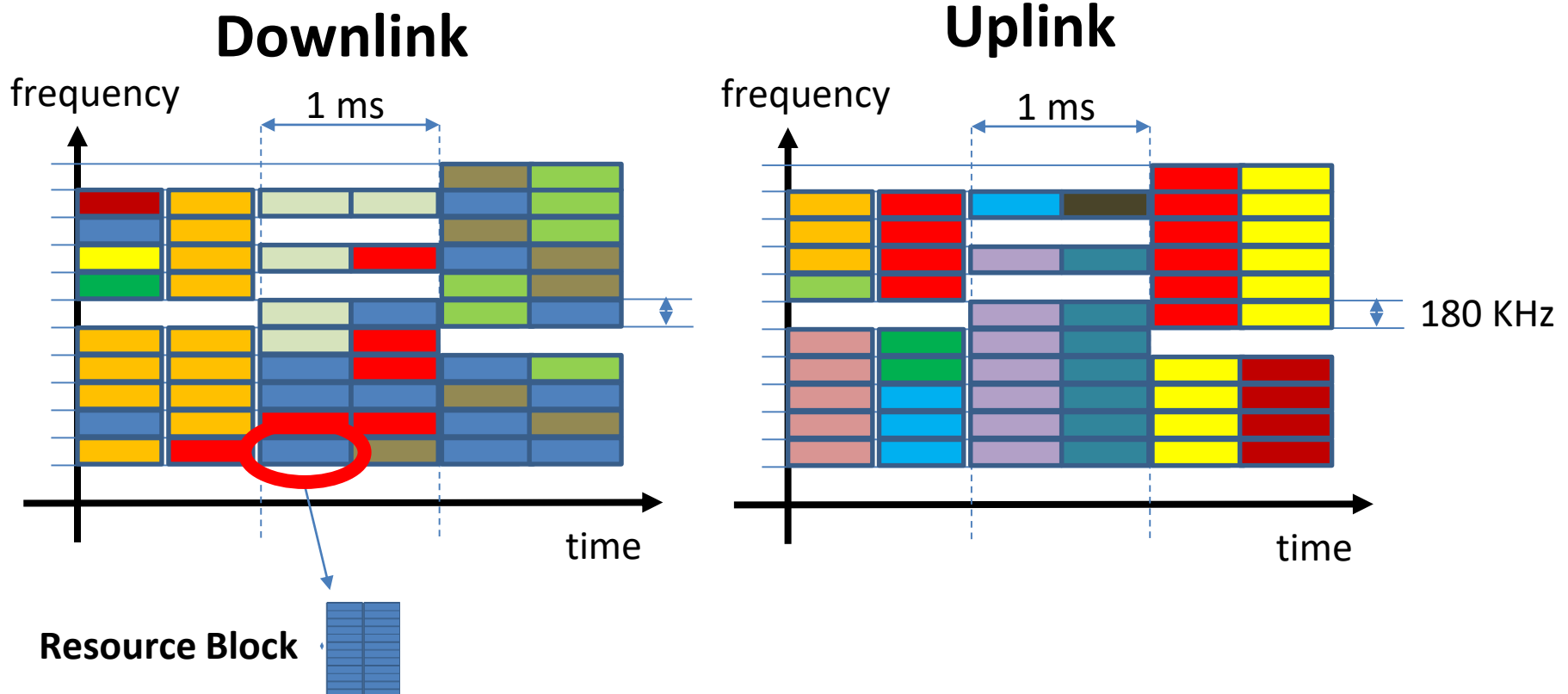
frequency

LTE PHY



RB scheduling

Scheduling of RBs is done on a subframe basis for both the uplink (consecutive RBs for same user) and uplink (non-consecutive RBs for user)



LTE channels

Downlink LTE Channels	
Name	Description
PBCH	Physical Broadcast Channel: Carries general information about the cell, like number of antennas on the base station and total bandwidth.
PDCCH	Physical Downlink Control Channel: Sends downlink control messages, e.g. for resource allocation on uplink/downlink.
PDSCH	Physical Downlink Shared Channel: Holds downlink data meant for users in resource blocks indicated by the PDCCH.
PHICH	Physical Hybrid-ARQ Channel: Contains positive or negative acknowledgments for uplink data.
Uplink LTE Channels	
PUCCH	Physical Uplink Control Channel: Holds control information from users to base stations, e.g. ACKs, channel reports and uplink scheduling requests.
PUSCH	Physical Uplink Shared Channel: Mainly carries uplink data in resource blocks indicated by the PDCCH.

Carries scheduling decisions
(which UEs tx/rx on which RBs, with what modulation)

The primary channel for delivering data to UEs. Shared among all users in the cell, the scheduler divides it into Resource Blocks allocated per subframe

The primary channel for UEs sending data to the network
Also time/frequency shared among users. When a UE has data to send, it piggybacks control info (CQI, ACK/NACK) onto PUSCH instead of using PUCCH

PRACH: Physical Random Access Channel (Uplink) Used for **initial access:** the UE's first contact with the network (random access preamble, chosen among 64 orthogonal preambles)

PCCH: Paging Control Channel (Downlink) carries paging messages over the PCH (Paging Channel) transport channel, **ultimately transmitted on a Physical channel**, the PDSCH.

Access Grant: The UE sends a **random access preamble** on the **PRACH** (Physical Random Access Channel), which is equivalent to GSM's RACH. The eNodeB responds with a **Random Access Response (RAR)** carried on the **PDSCH**

Control, broadcast and data channels

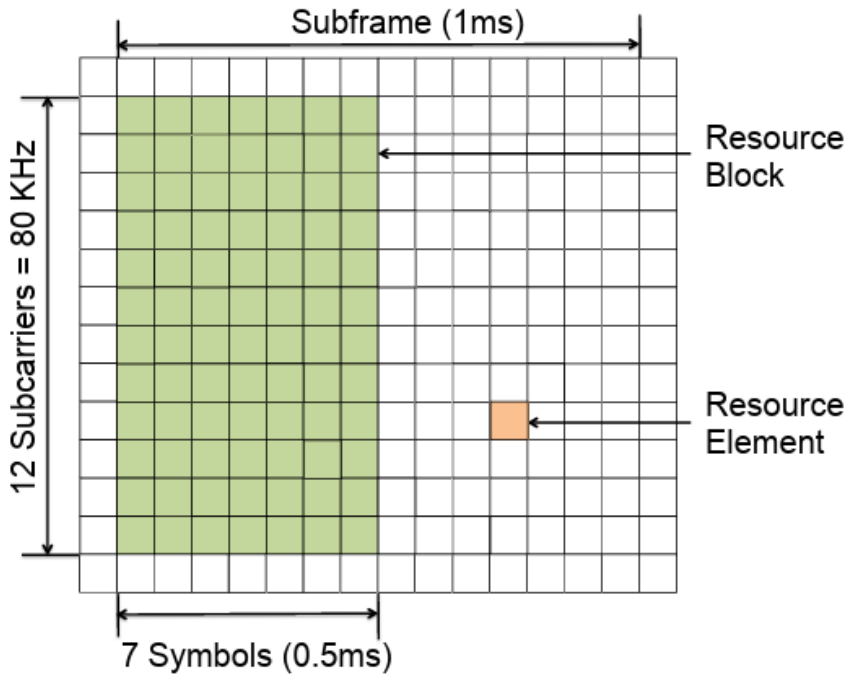


Figure 1: The Resource Grid is divided into multiple Resource Blocks.

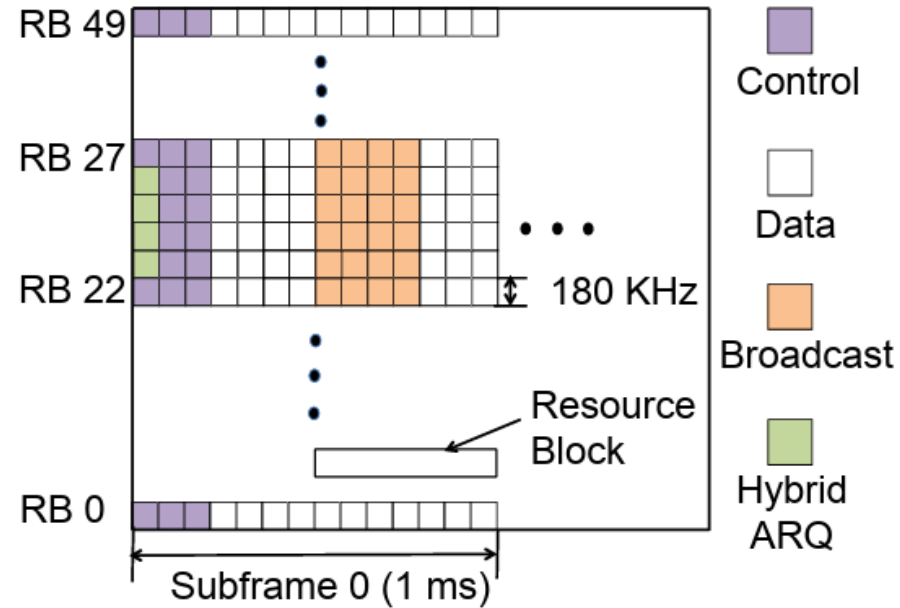


Figure 2: Physical Channels are mapped to well defined Resource Elements in the Grid.

5G

5G NR replaced SC-FDMA with DFT-s-OFDM in the uplink, though the two are closely related.

5G NR uplink uses **CP-OFDM** (Cyclic Prefix OFDM) as the primary waveform, the same as the downlink, but *also* supports **DFT-s-OFDM** (DFT-spread OFDM) as an alternative uplink waveform.

DFT-s-OFDM and SC-FDMA are functionally equivalent. DFT-s-OFDM is essentially SC-FDMA under a different name that reflects the signal processing chain (a DFT precoding step applied before the OFDM IFFT).

CP-OFDM offers more flexibility, higher throughput, and easier MIMO support.

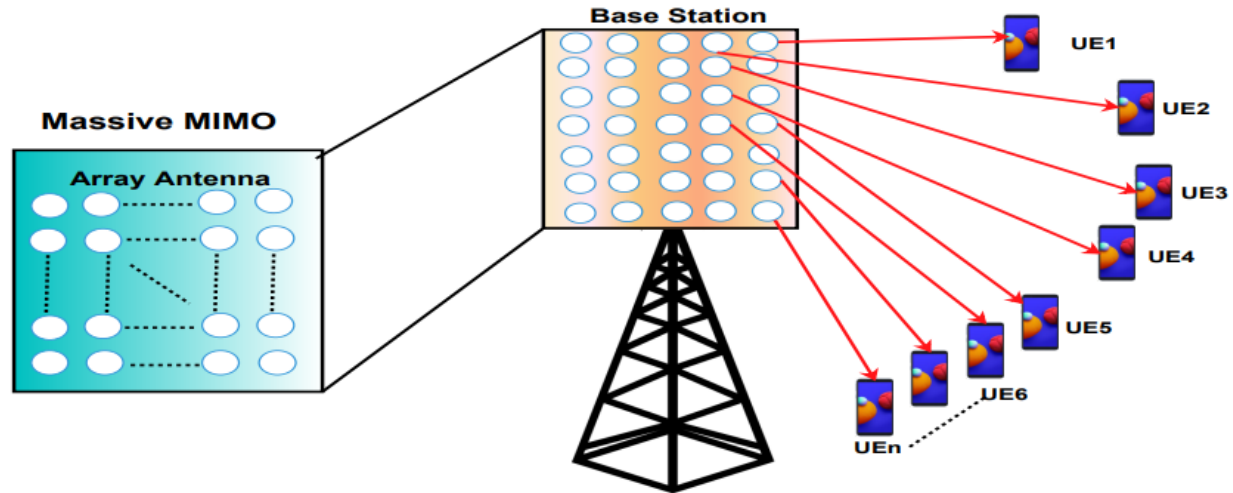
DFT-s-OFDM ($\approx$ SC-FDMA) is used when lower PAPR matters, e.g., for coverage-limited scenarios or devices that need to maximize transmit power efficiency (like IoT devices or cell-edge users).

So the choice in 5G NR uplink is context-dependent, which is a key design improvement over LTE's fixed SC-FDMA-only uplink.

5G

- Increased data rates, decreased latencies
 - Massive MIMO (More than 100 antennas in eNodeB, Array antenna)
 - Non-orthogonal Multiple Access (NOMA)
 - mmWave band (26.5-27.5 GHz) and lower frequency bands (700 MHz and 3.4-3.8 GHz)
 - Small cells (e.g. Use many small cells for mmWave to avoid obstacles)

Massive MIMO



Non-ortogonal Multiple Access (NOMA)

(in the study/evaluation phase, strong candidate for 6G). Increases efficiency of OFDMA by letting multiple users share the *same* time-frequency resource simultaneously, separated instead by **power levels**.

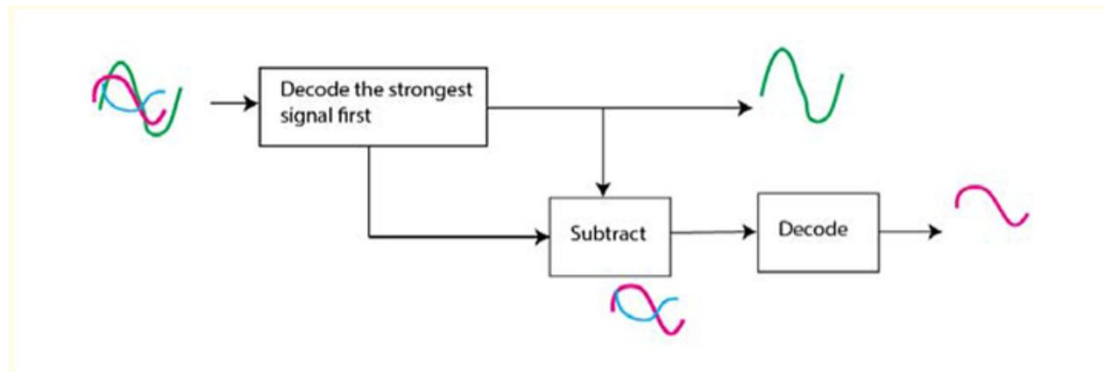
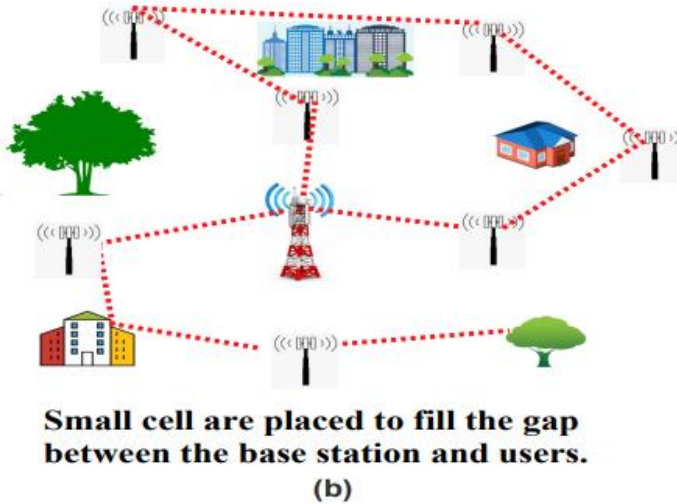
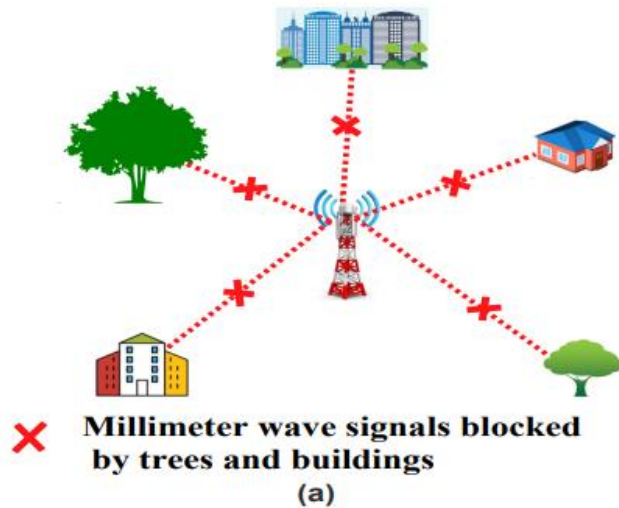
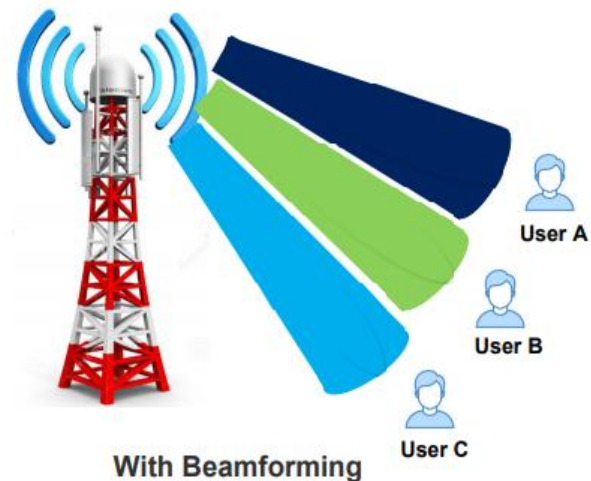
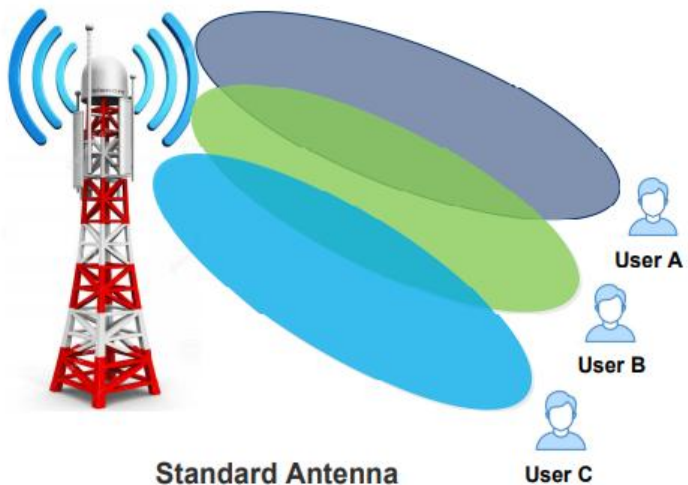


Figure 2.
Successive interference cancellation.

Small cells for mmWave

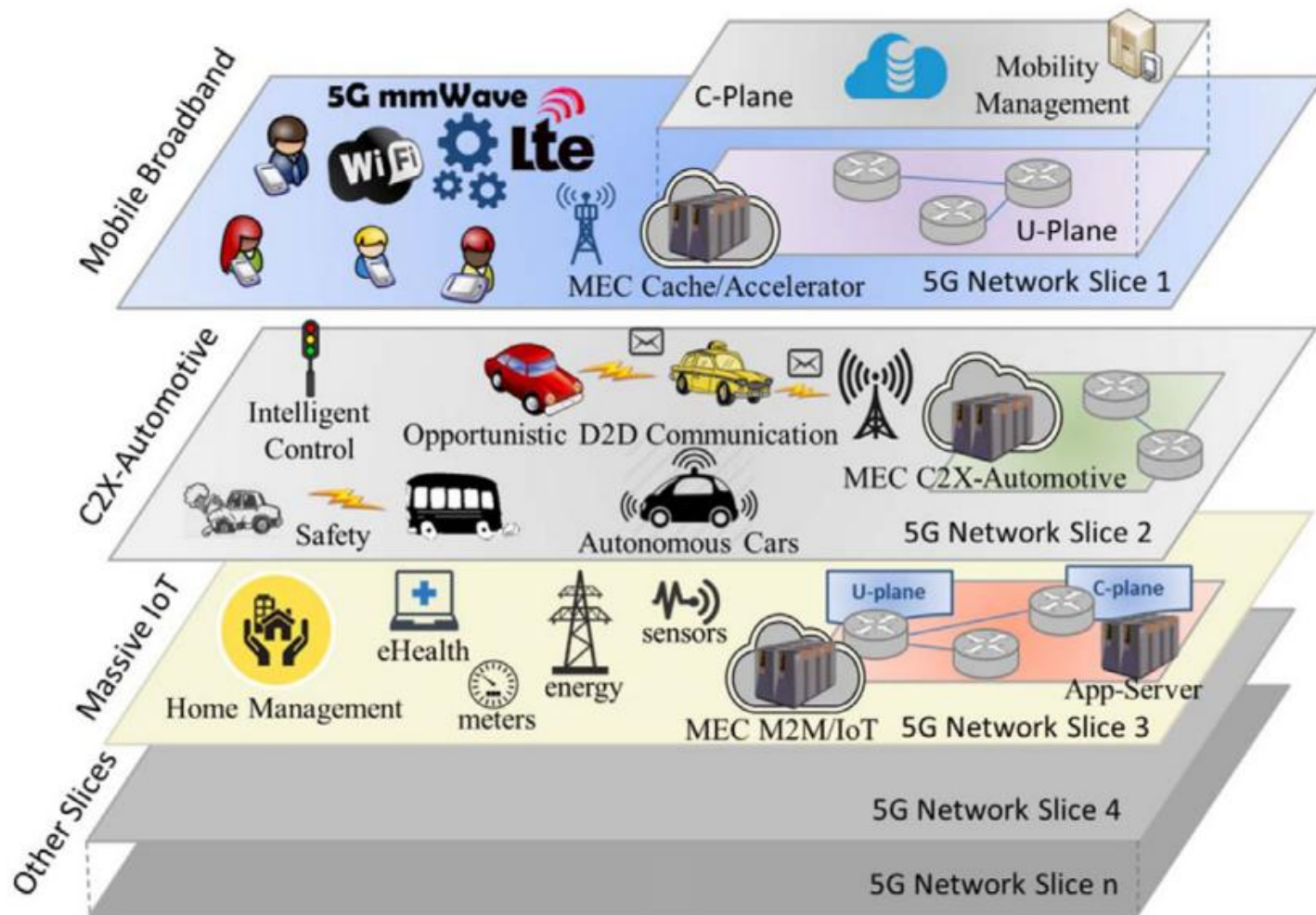


Beamforming



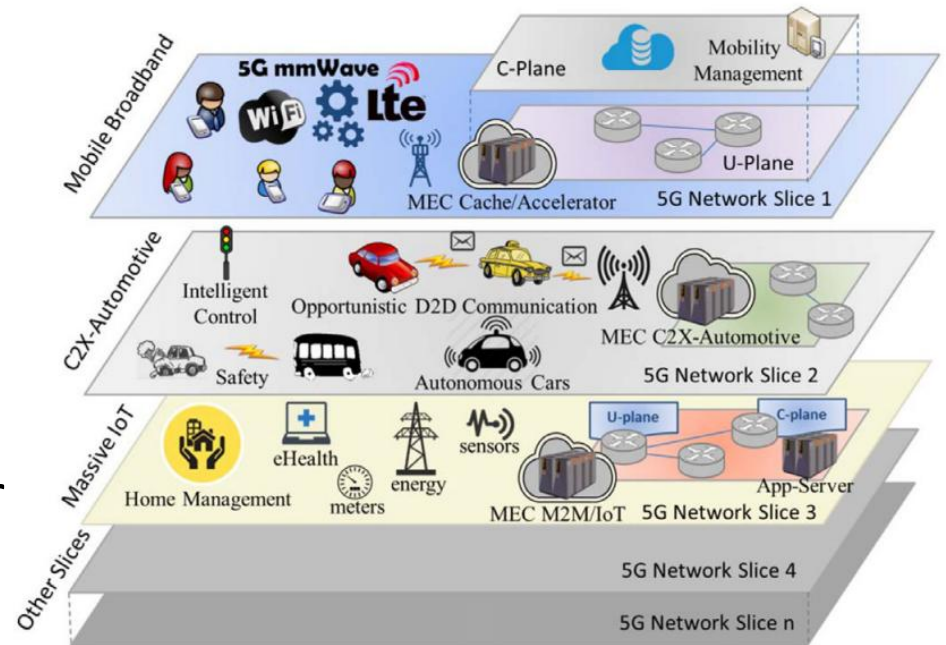
Mobile Edge Computing: reduced latency

Network slicing: create different virtual networks with specific performance



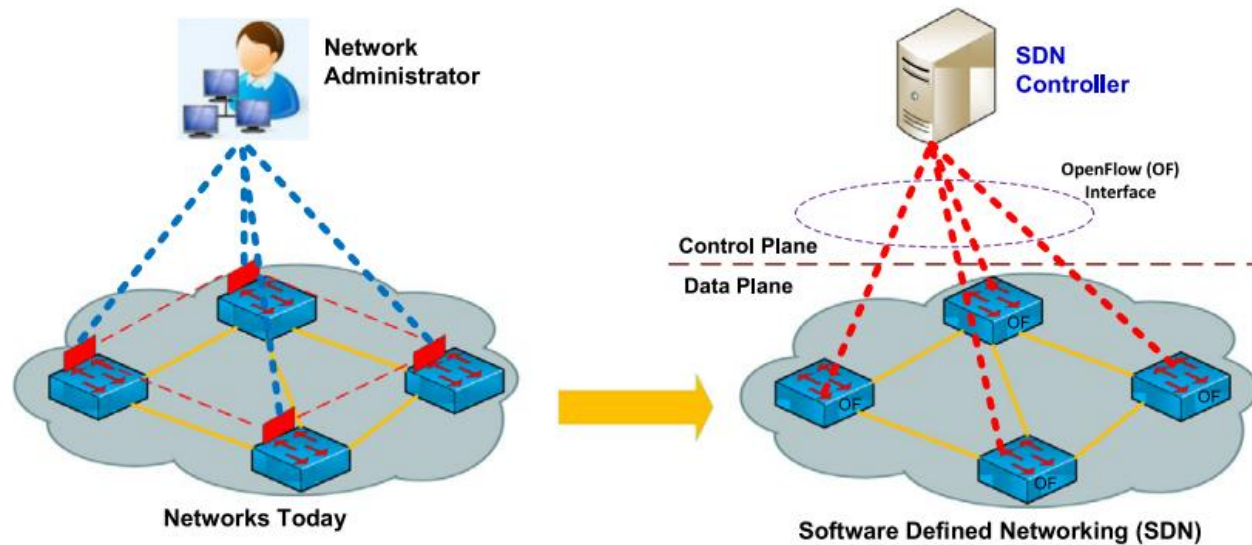
Network slicing

- The network slice is a logically separated, self-contained, independent and secured part of the network, targeting different services with different requirements on speed, latency and reliability.
- Network slice characteristics are for example low latency, high bandwidth and ultra-reliability for a critical IoT use case or higher latency and lower bandwidth for a massive IoT use case.



Software Defined Networks (SDN) and Network Function Virtualization (NFV)

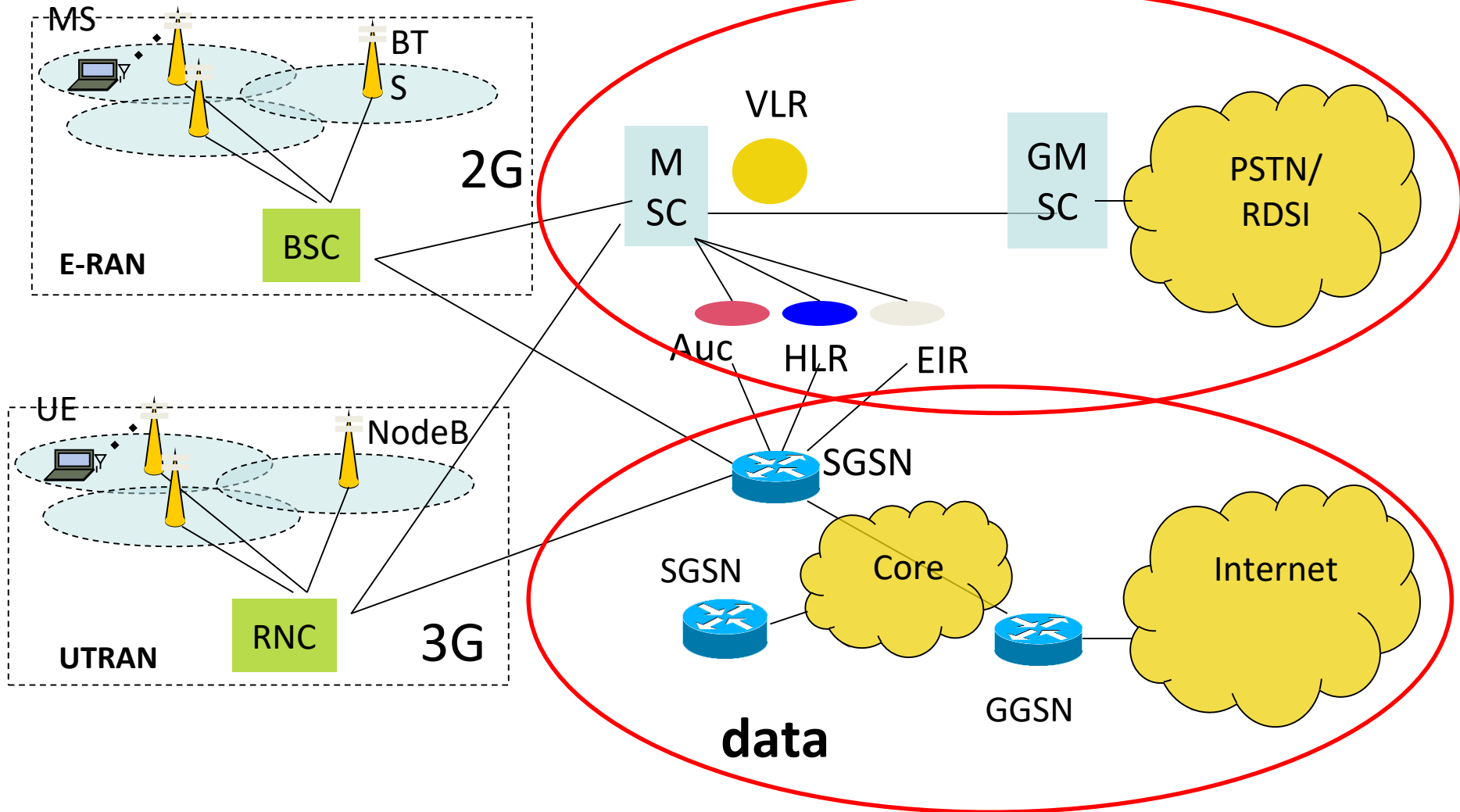
SDN: Separate forwarding (data) and control plane



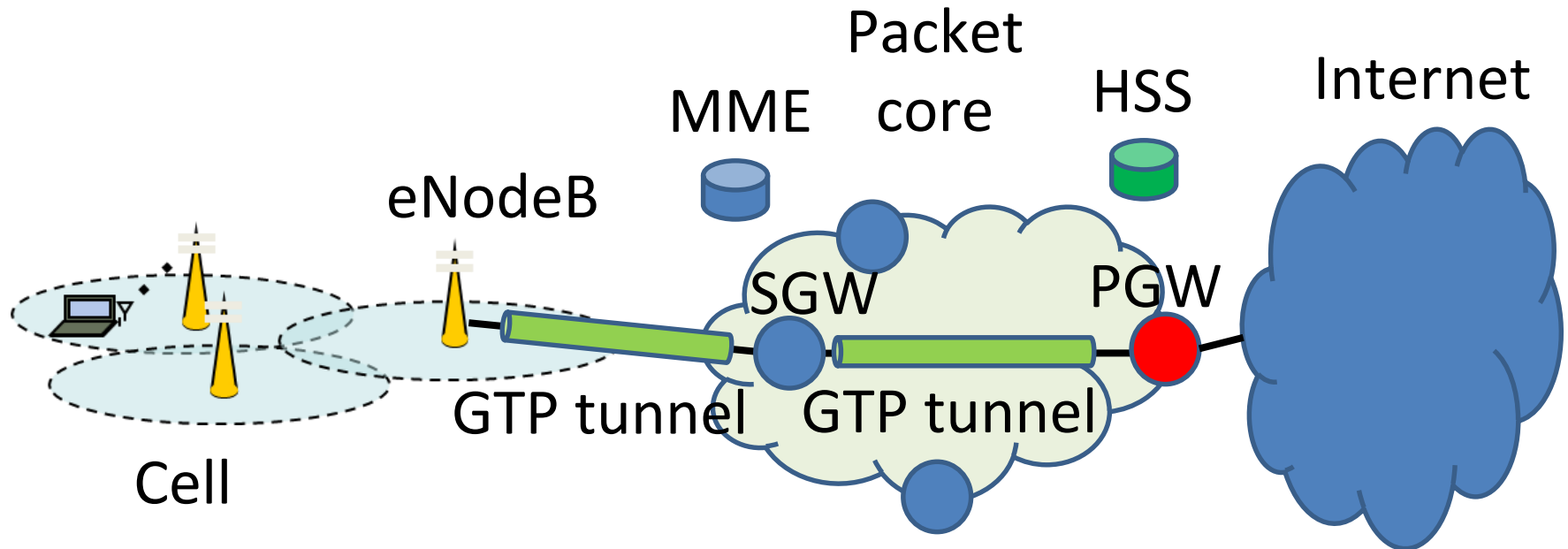
NFV: virtualize network functions (e.g., FWs, NAT, VPN, etc) on top of commodity Hw devices

Cellular networks: Mobile Packet Core

2G/3G architecture

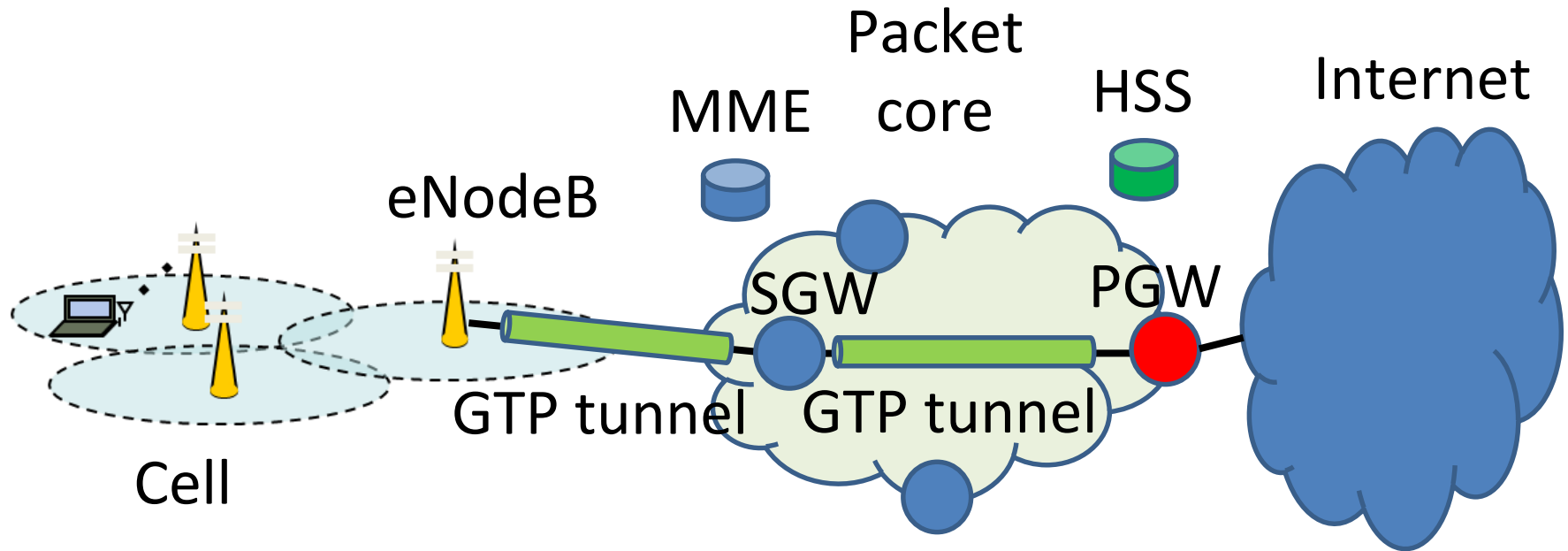


4G: Evolved Packet Core



Does not support a circuit-switching domain to create an end-to-end packet network, providing required QoS (VoIP).

LTE (Evolved Packet Core)



Serving Gateway (SGW) (3g SGSN)

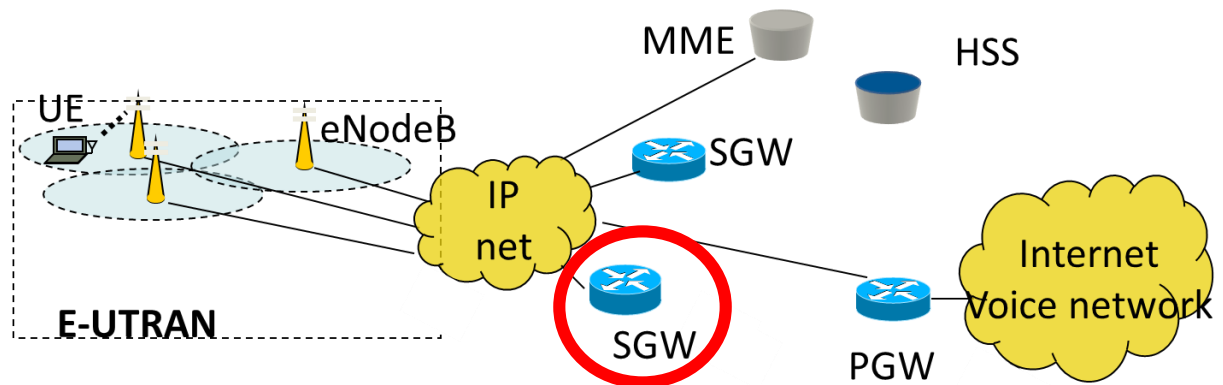
Packet Data Network Gateway (PGW), (3g GGSN)

Mobility Management Entity (MME) (3g VLR)

Home Subscriber Server (HSS) (3g HLR, AuC)

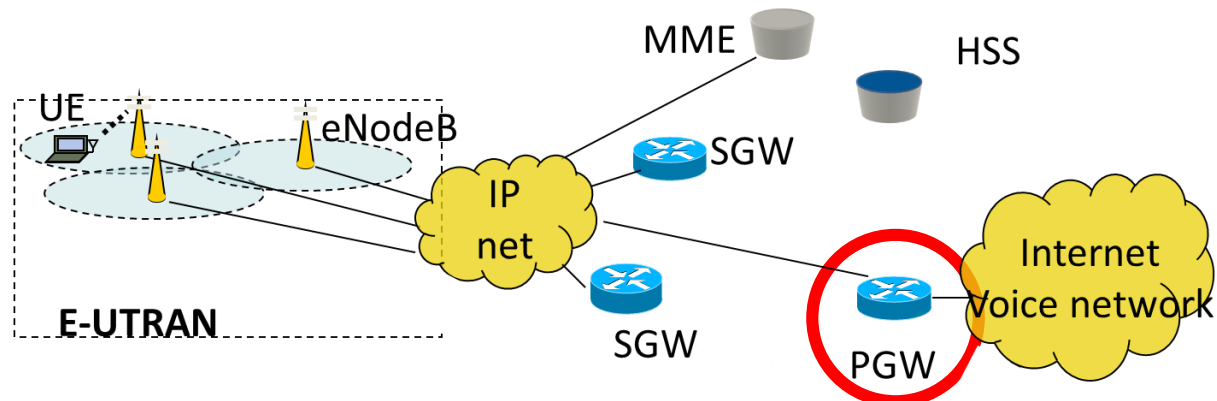
4G: Evolved Packet Core

- **Serving Gateway (SGW):**
 - Manage user-plane mobility
 - SGW maintains data paths between eNodeBs and the PDN Gateway (PGW).



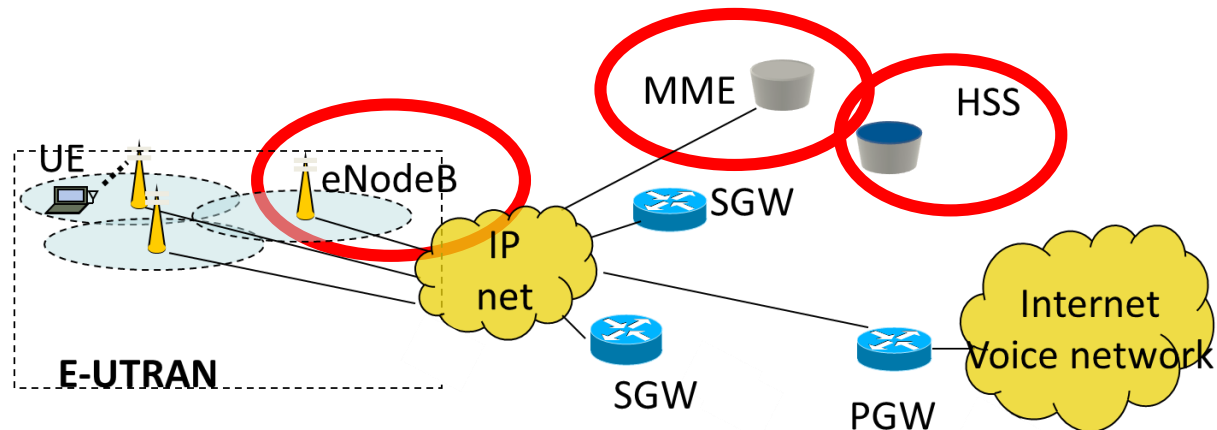
4G: Evolved Packet Core

- **Packet Data Network Gateway (PGW):**
 - Interface towards the Internet. Anchor point for sessions towards the external Internet.
 - Policy enforcement features (resource allocation and usage), Packet filtering (deep packet inspection), Charging support



4G: Evolved Packet Core

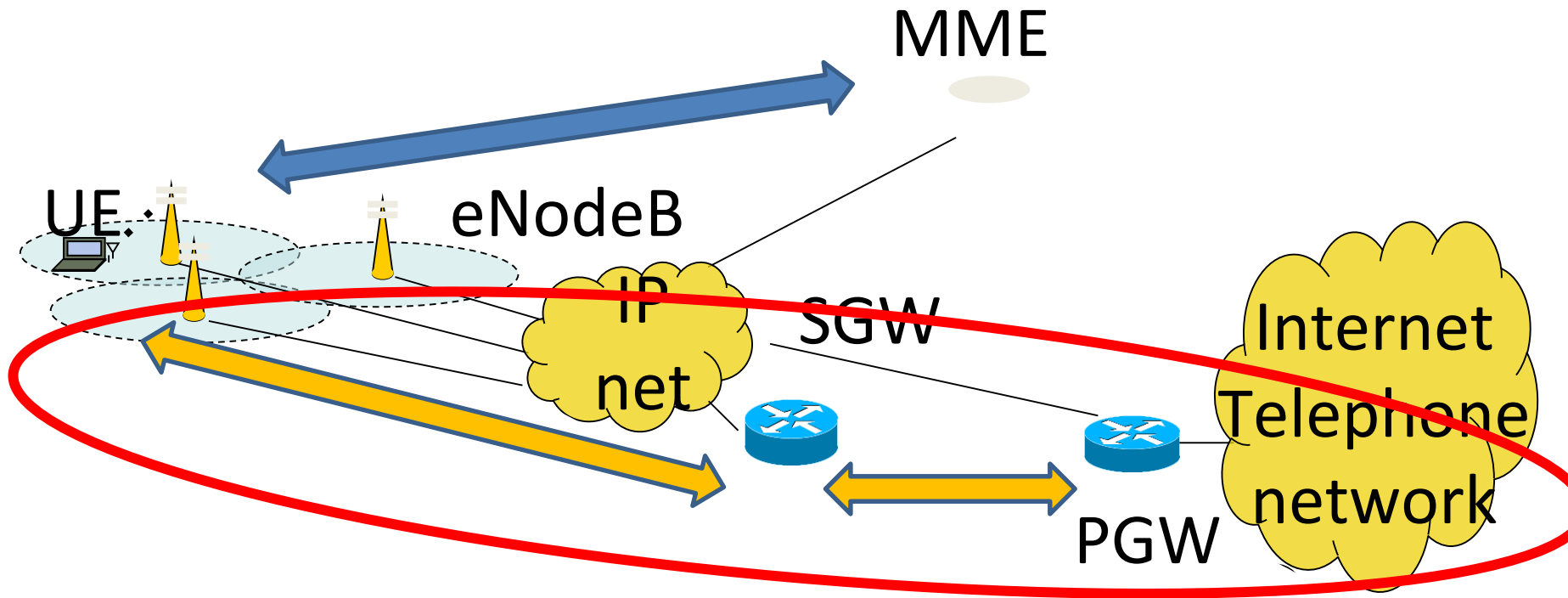
- **The Mobility Management Entity (MME):**
 - Performs the signaling and control functions to support tracking, paging, roaming and handovers.
- **The Home Subscriber Server (HSS):**
 - Performs the functions of the HLR (Home location register) and the AuC (Authentication center)



Cellular networks:
User-plane, EPS bearers

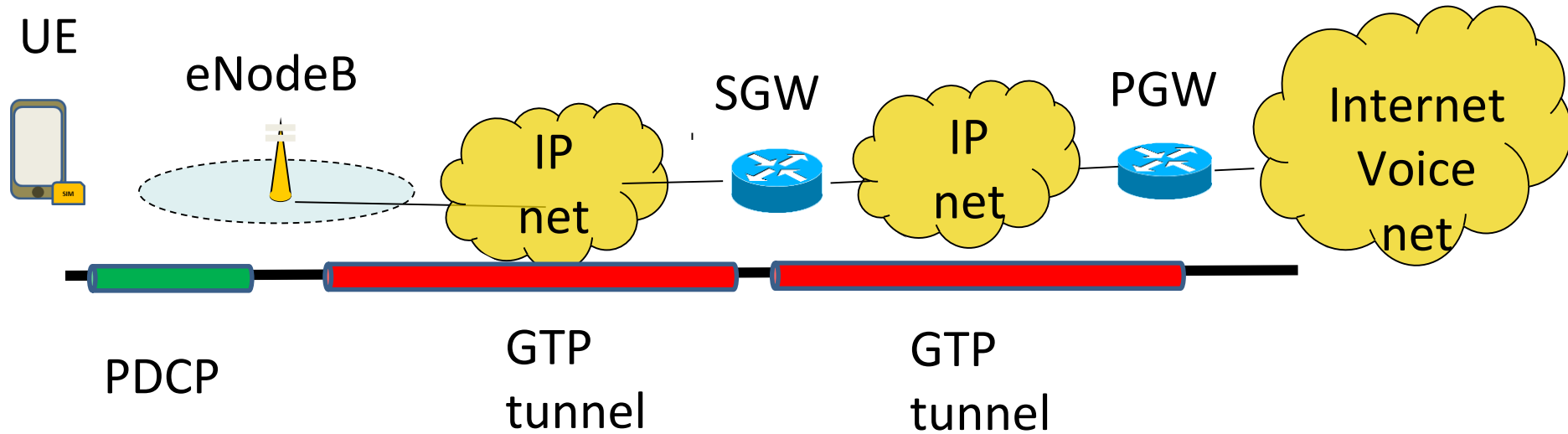
4G: Evolved Packet Core

Signalling (Control-plane)



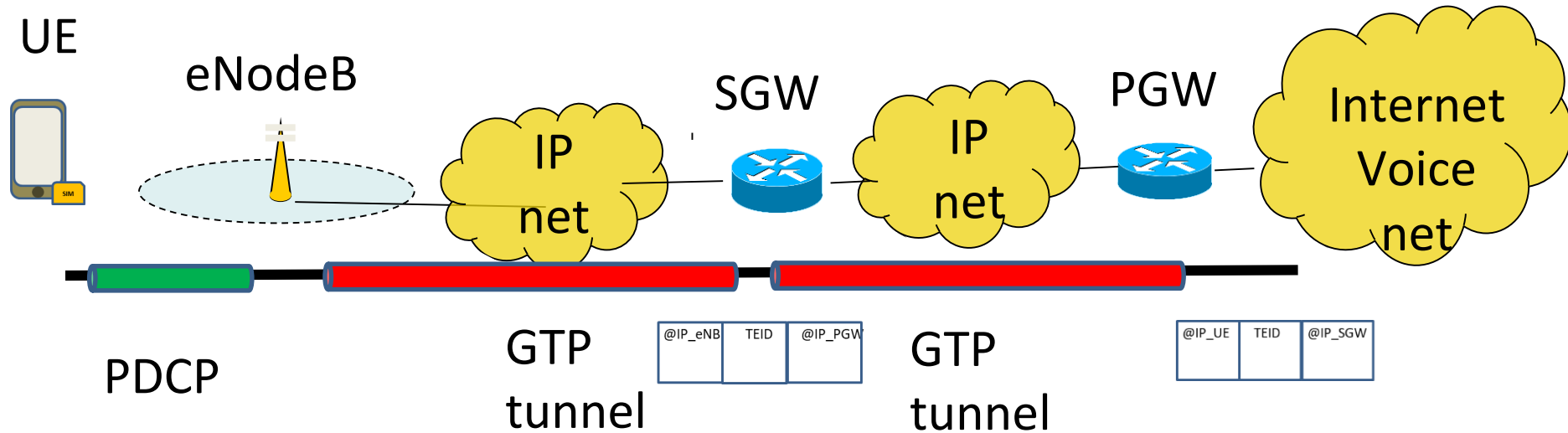
Voide+data (User-plane)

LTE EPS bearers



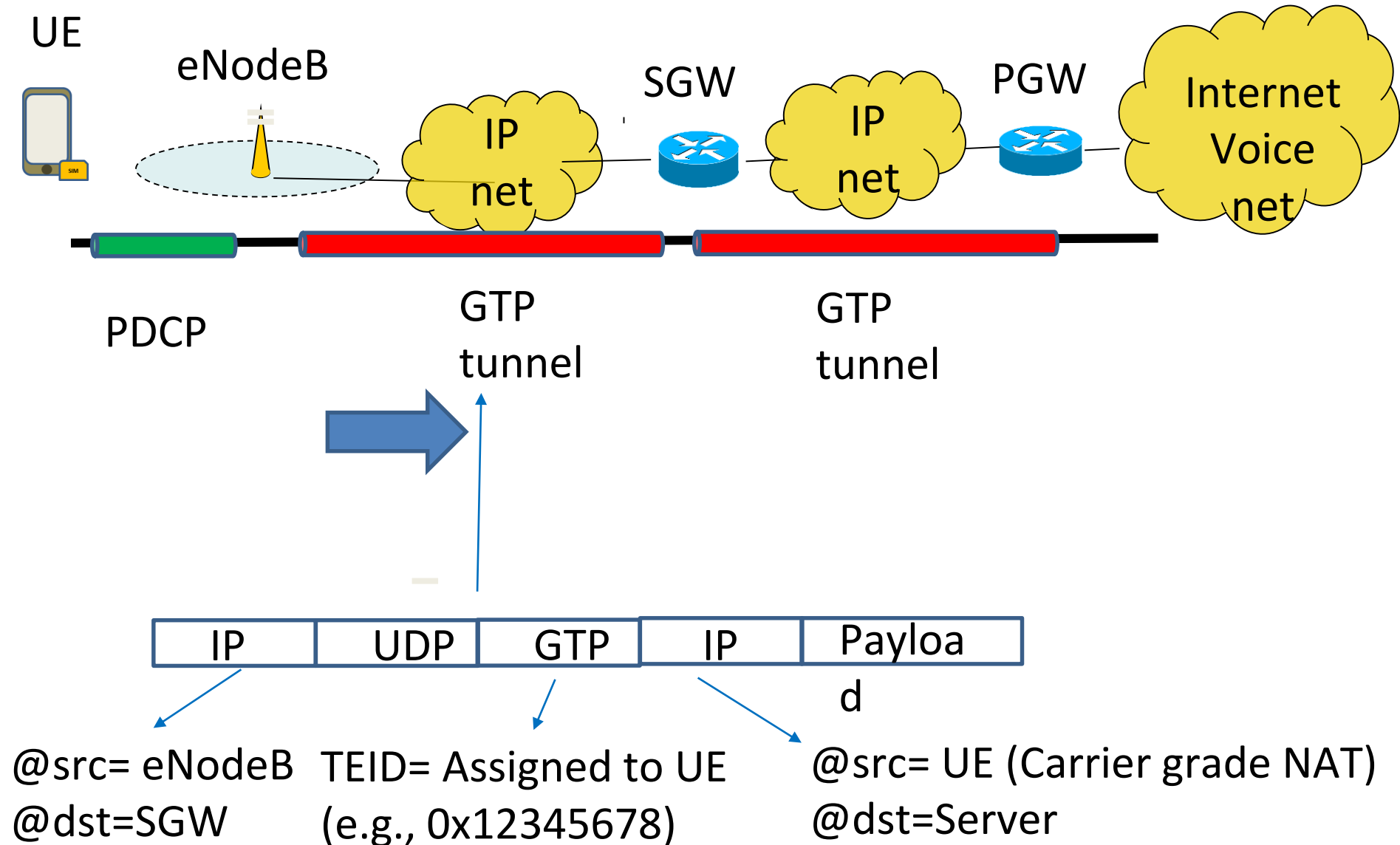
- Packet Data Convergence Protocol (PDCP): IP header compression, encryption
- GPRS tunneling protocol (GTP): tunneling of IP packets, tunnel management

LTE EPS bearers

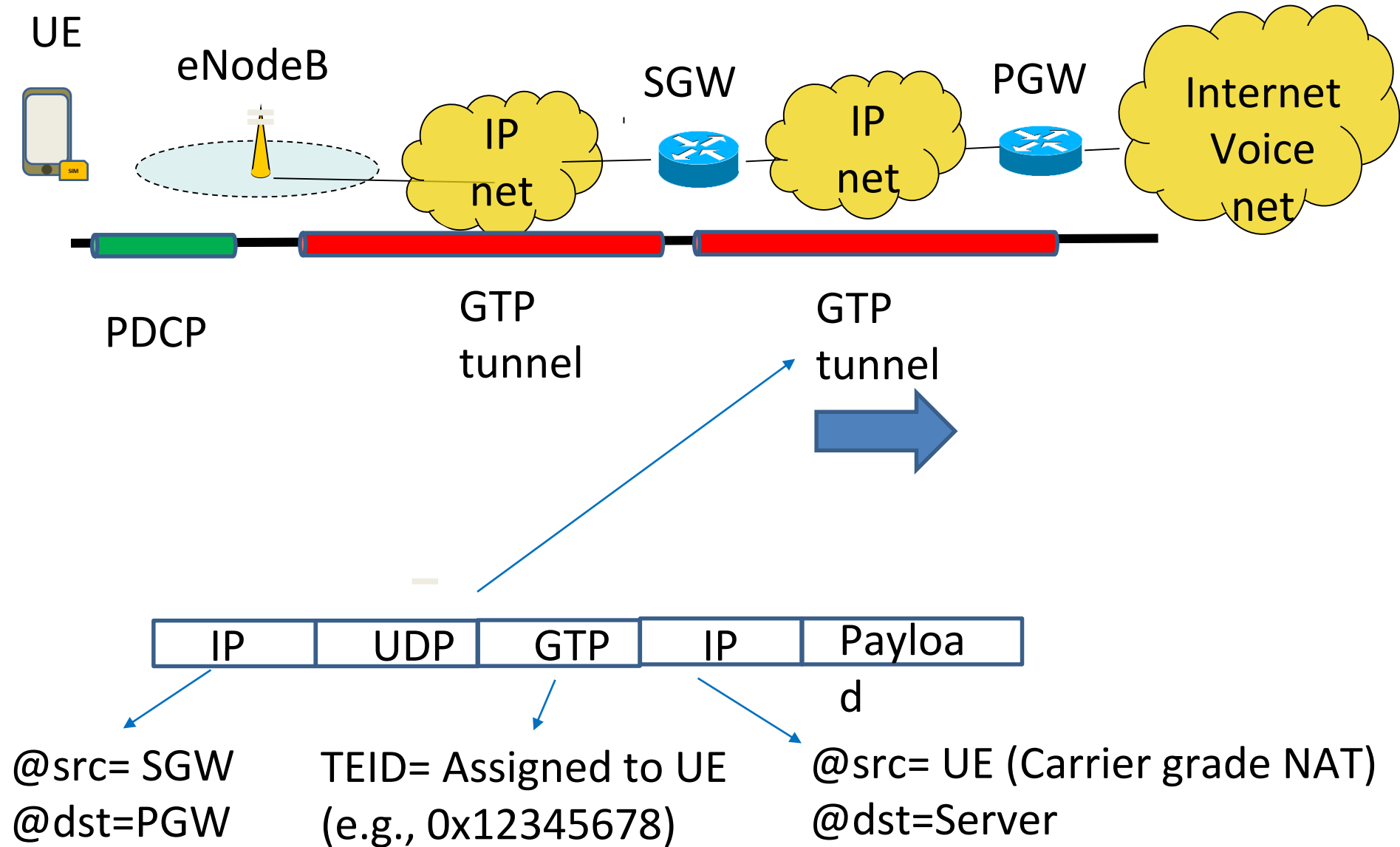


- EPS bearers need to be established via signalling (tunnel management) using GTP signalling packets
- Tunnels are reconfigured to support mobility (mobility management)
- **QoS** with different latency, PER, etc (conversational voice, video, RT gaming, TCP-data, etc,)

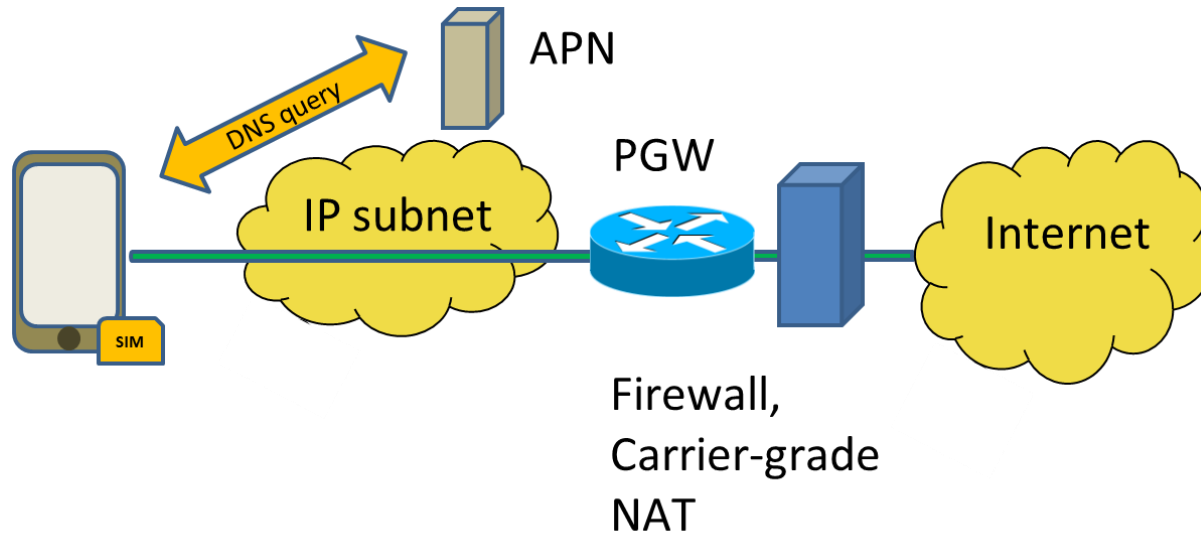
LTE EPS bearers



LTE EPS bearers



UE's view of the network



DNS in 4G: Once the PDN connection is established:

- UE gets an IP address + DNS server IP(s) from the PGW
- UE sends DNS queries to those servers **through the GTP tunnel** (over the LTE data plane)
- DNS server responds → UE resolves names → normal IP traffic flows.

Several APNs are possible: (**internet**: general browsing, **ims**: VoLTE, etc)

My phone...

Sistema/Acerca del teléfono/estado Red: Tuenti. Dirección IP: 100.67.42.10 (rango **Carrier Grade-NAT**)

```
c:\adb\platform-tools_r27.0.1-windows\platform-tools>adb shell "ip route show table 0"
```

```
default dev dummy0 table 1002 proto static scope link
```

```
default via 100.67.42.10 dev rmnet0 table 1003 proto static
```

```
local 100.67.42.10 dev rmnet0 table local proto kernel scope host src 100.67.42.10
```

```
broadcast 100.255.255.255 dev rmnet0 table local proto kernel scope link src 100.67.42.10
```

```
broadcast 127.0.0.0 dev lo table local proto kernel scope link src 127.0.0.1
```

```
local 127.0.0.0/8 dev lo table local proto kernel scope host src 127.0.0.1
```

```
local 127.0.0.1 dev lo table local proto kernel scope host src 127.0.0.1
```

rmnet0: the 4G mobile data interface (your LTE connection)

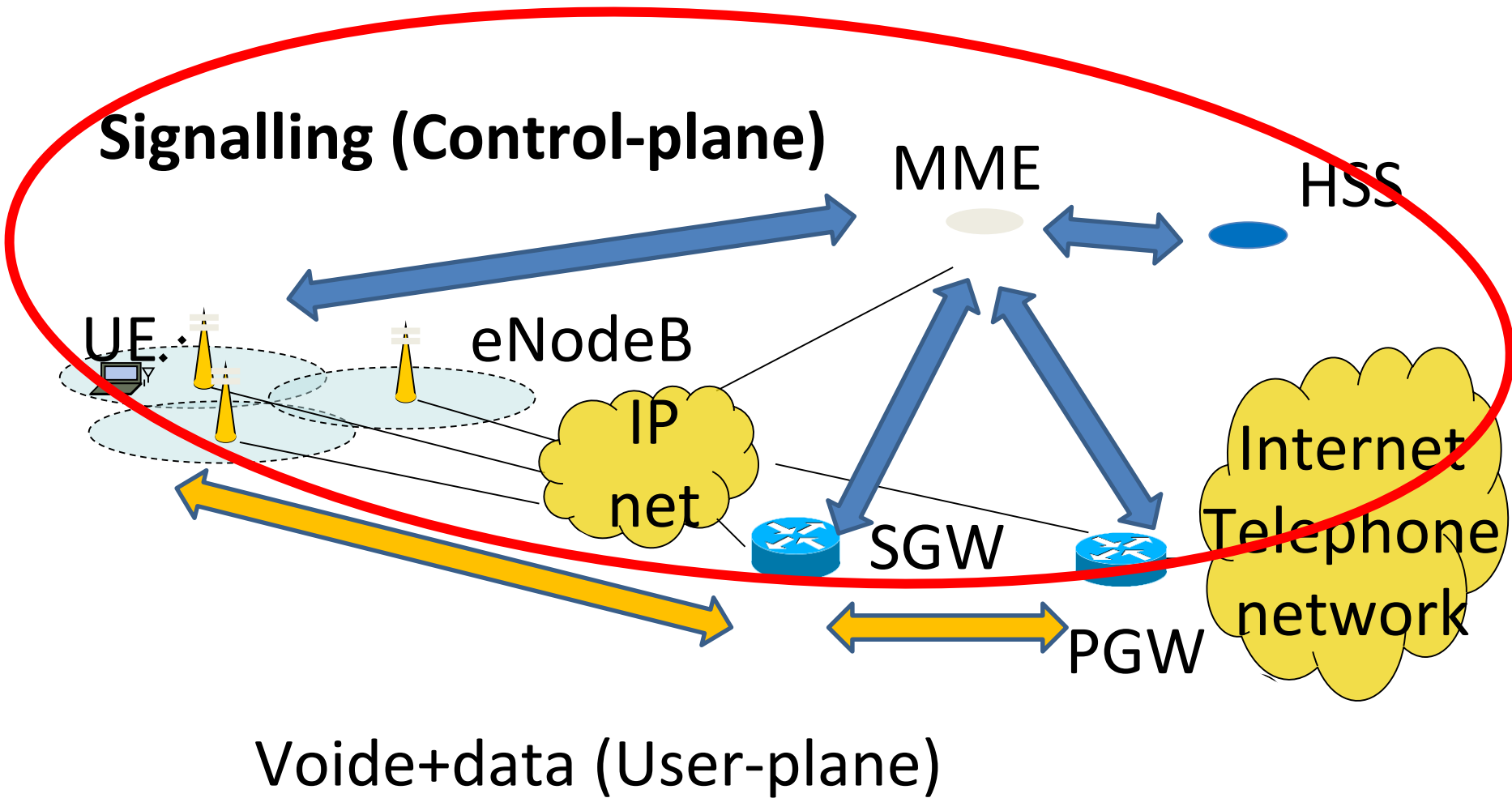
lo: the loopback interface (internal, always 127.0.0.1)

dummy0: a virtual placeholder interface (used internally by Android)

- **default via 100.67.42.10 dev rmnet0 table 1003**: All internet traffic exits through rmnet0. *“for any packet that policy directs to table 1003, send it out through rmnet0 via rmnet0 (this is a point to point link, no IP address of the default gw is needed)”*
- **local 100.67.42.10 dev rmnet0 table local**: My IP address, traffic destined to myself is handled locally
- **broadcast 100.255.255.255 dev rmnet0 table local**

Cellular networks:
Signalling procedures. Mobility and
location management

4G: Evolved Packet Core

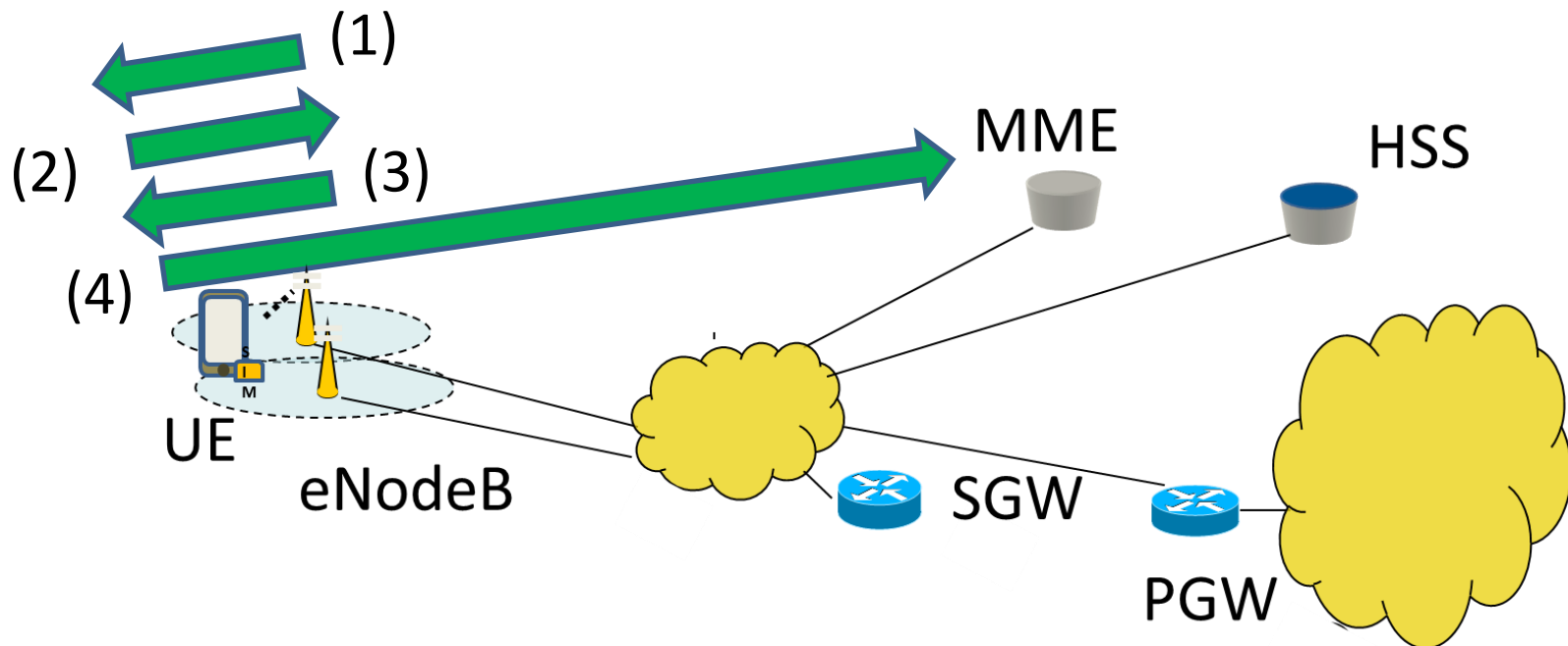


Signalling procedures

- **Radio Resources Management**
 - Establish, modify, close Radio Session
 - Handover (with/without change of SGW/MME)
- **Mobility Management**
 - Attach/Deattach to/from network
 - Tracking Area update
 - Service request/Paging
- **EPS bearer management**
 - Establish/modify/free EPS bearer

Attach (Register in network)

- (1) Listen to BCCH
- (2) Use RACH to request use of a channel
- (3) Use AGCH to grant use of a channel
- (4) Attach request (userid)



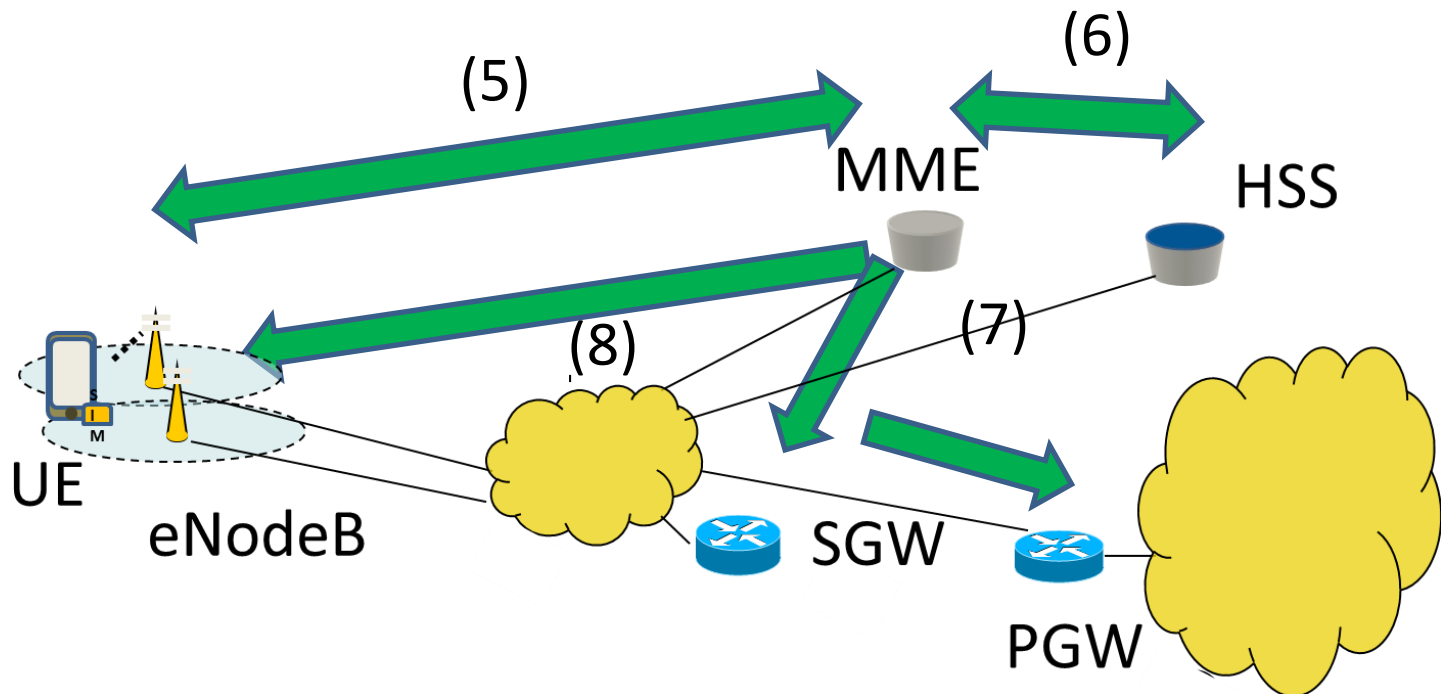
Attach (Register in network)

(5) Authentication and key negotiation

(6) Update IMSI location, insert subscriber data

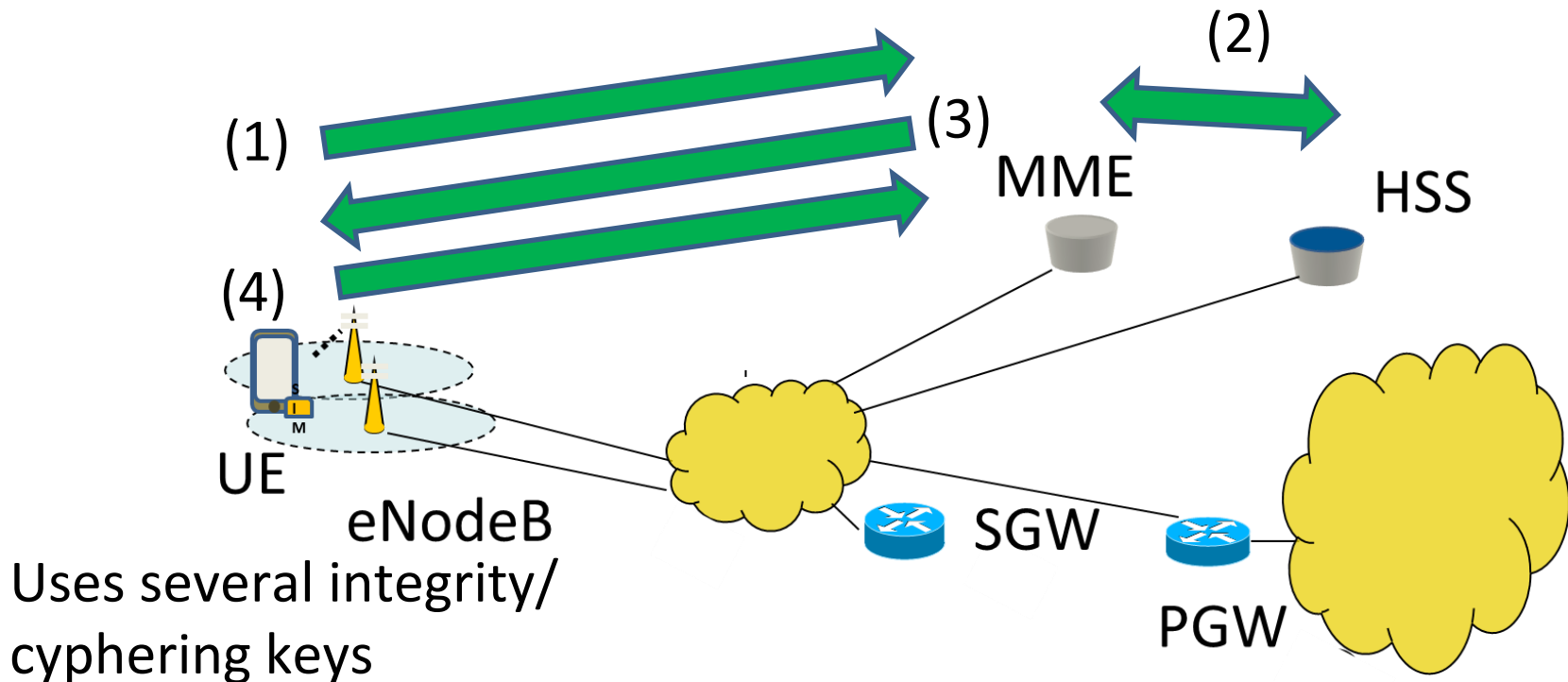
(7) Create default EPS bearer

(8) Attach accept



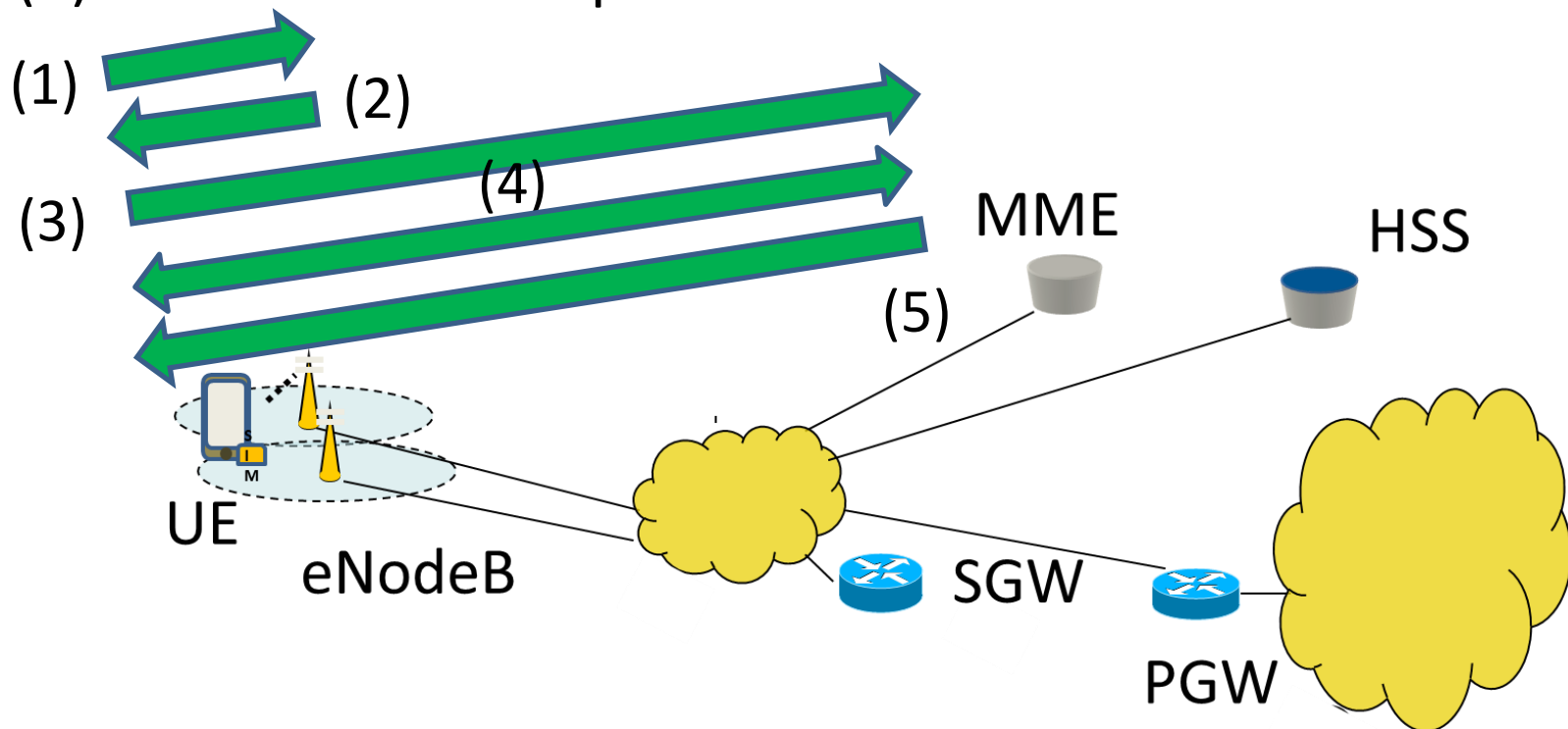
Authentication and key negotiation

- (1) Attach request
- (2) Obtain Authentication Vector from Subscriber auth key in SIM (AUTH, XRES)
- (3) Authentication request (AUTH). USIM computes RES. Derive keys. (also authenticates network with AUTH which encrypts using info in user's SIM).
- (4) Authentication response (RES) $\Rightarrow$ RES == XRES?



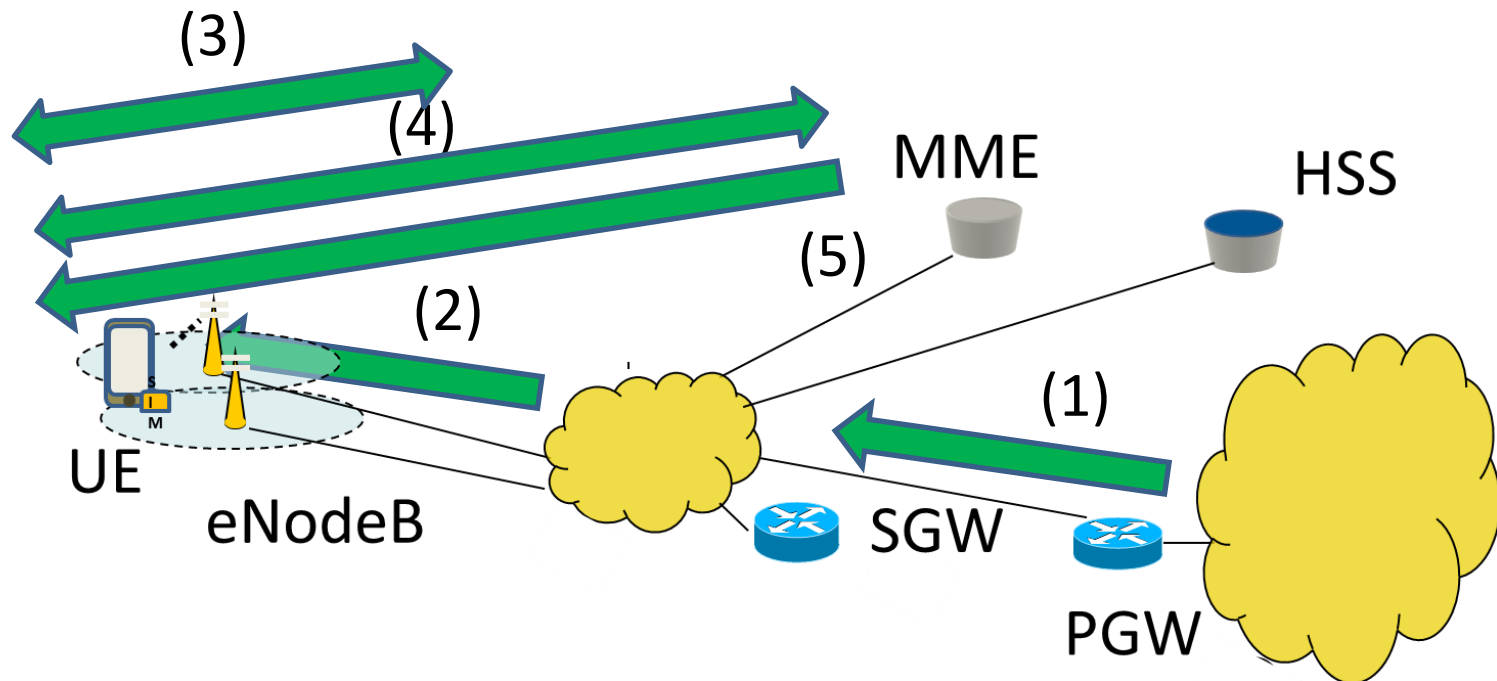
User initiated service request

- (1) Use RACH to request use of a channel
- (2) Use AGCH to grant use of a channel
- (3) Service request
- (4) Authentication and key negotiation
- (5) Establishment request



Paging

- (1) Data arrives from network
- (2) Use PCH for paging user
- (3) Random Access (RACH+AGCH)
- (4) Authentication and key negotiation
- (5) Establishment request + Data sent to UE



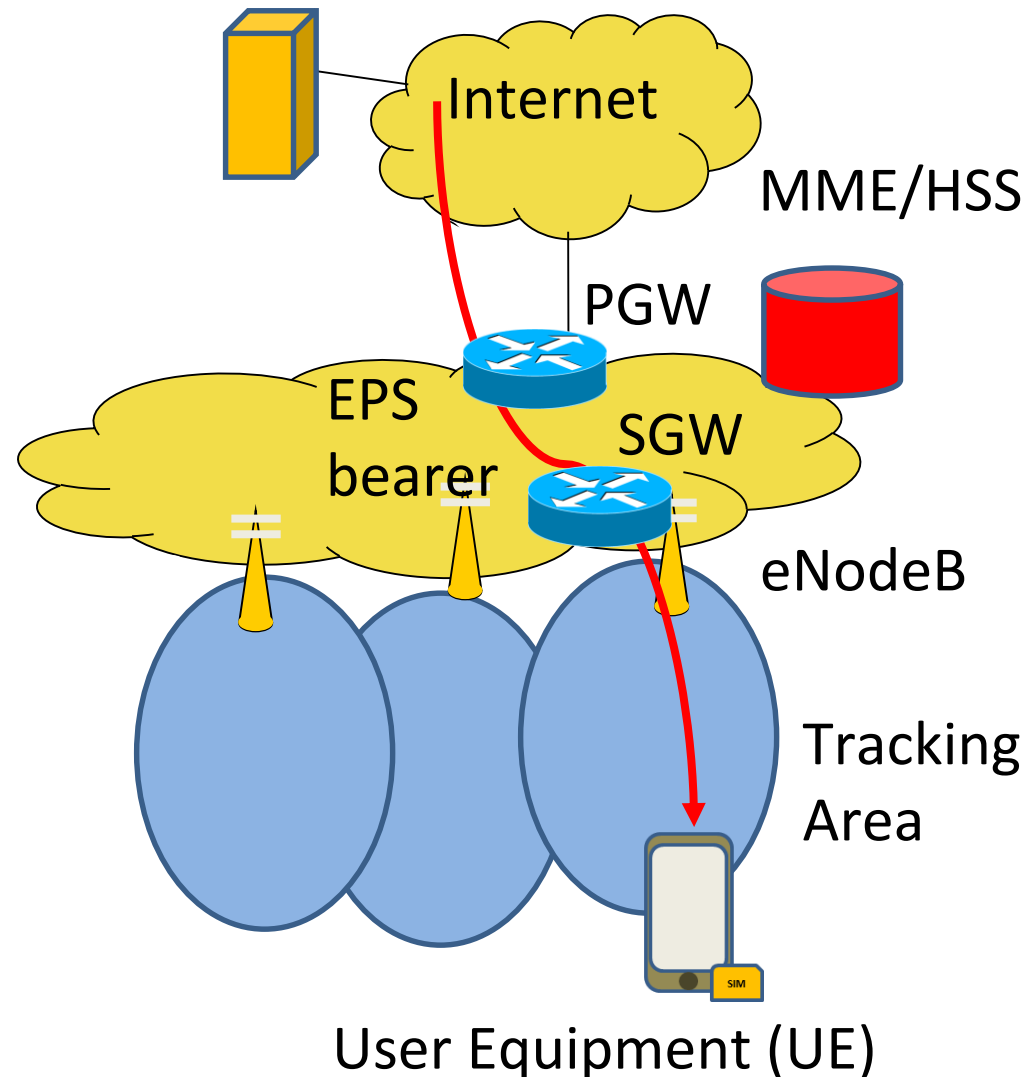
Roaming & Handovers

Roaming:

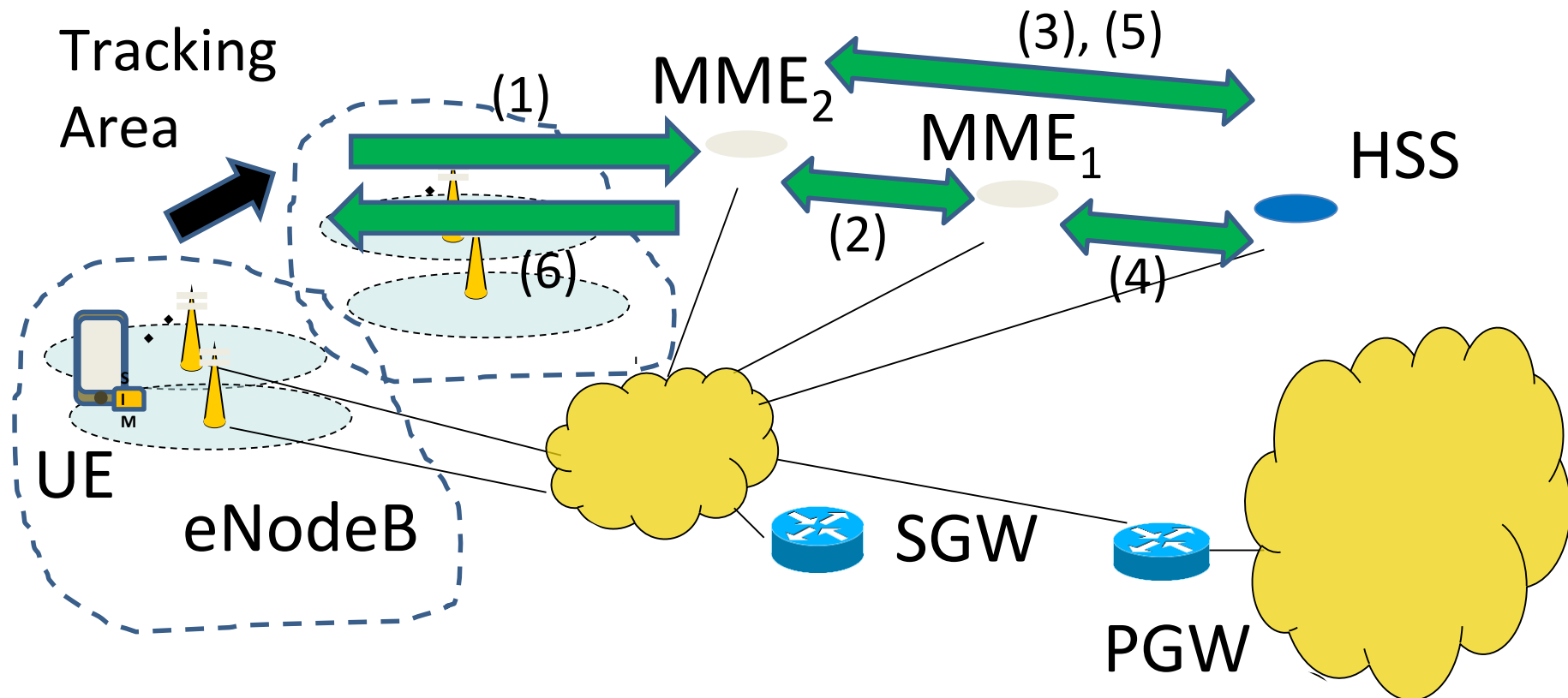
User changes its location
=> the system should be
informed of its current
location (**Tracking area**)

Handover:

User changes location
while engaged in a
conversation (change
eNodeB, SGW, and/or
MME)



Tracking Area Update



(1) TA Update request

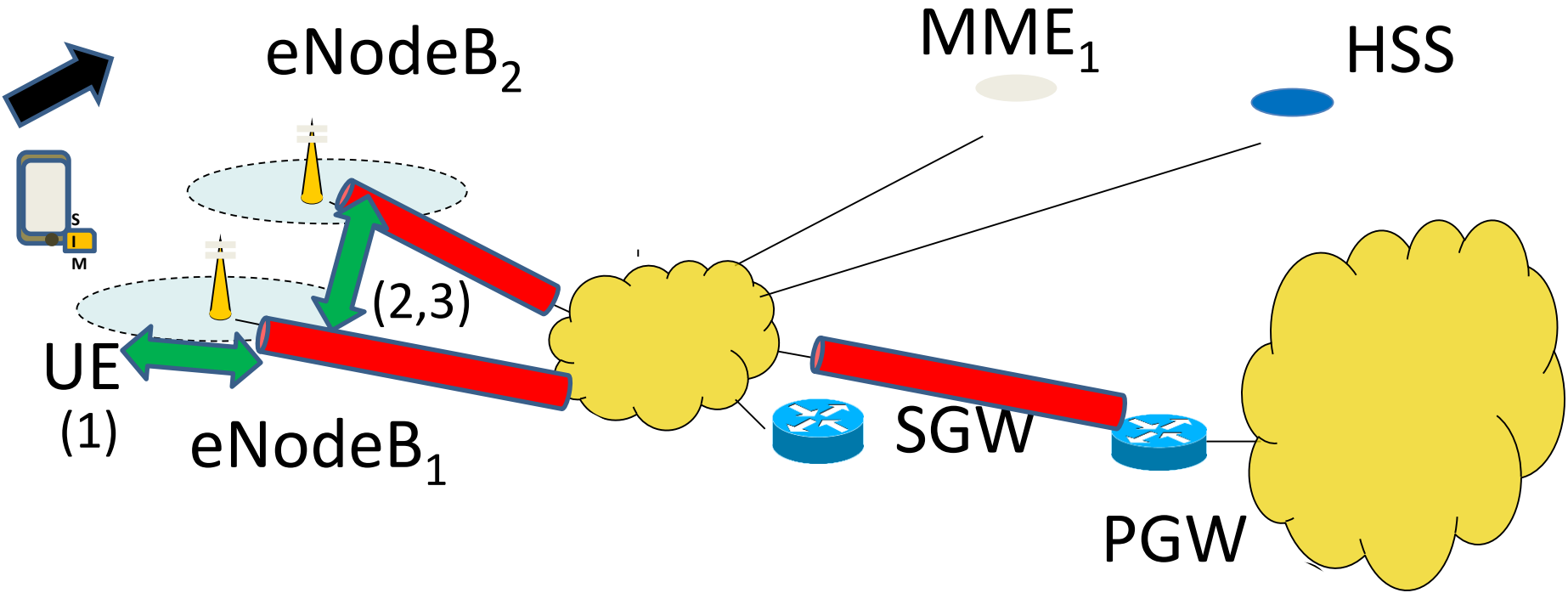
(2) Context request/response

(3) Update Location request

(4) Cancel Location, (5) Update Location ACK

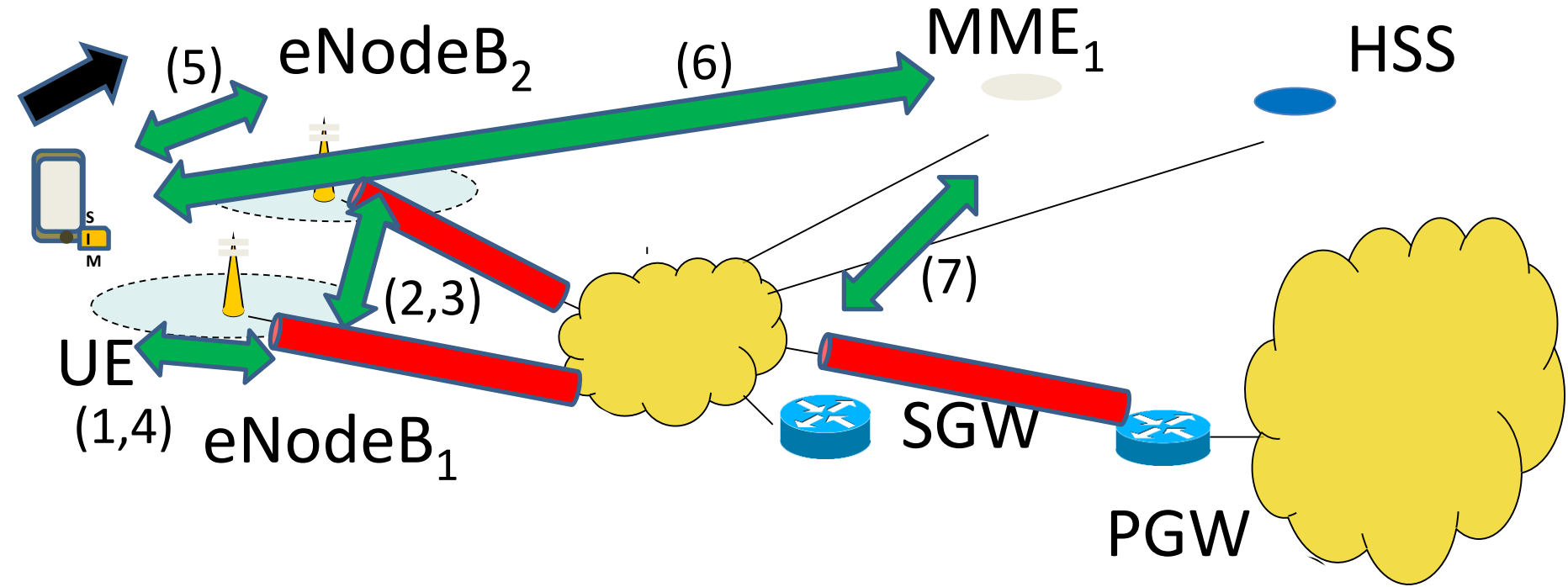
(6) TAU accept

Handover with same SGW



- (1) UE takes measurements, and sends them to eNodeB₁
- (2) eNodeB₁ requests HO to eNodeB₂
- (3) eNodeB₂ HO request ACK, eNodeB₁ sends packets waiting to be sent to UE

Handover with same SGW



(4) eNodeB₁ sends HO Command to UE

(5) UE sends HO confirm to eNodeB₂

(6) eNodeB₂ sends Path Request to MME

(7) MME modifies EPS bearer, eNodeB₁ frees radio resources

Mobile Virtual Operators

